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**Analisi degli isotopi stabili e struttura
trofica della comunità pelagica
nel Mar Mediterraneo occidentale**

**Stable Isotope Analysis and Trophic
Structure of the Pelagic Community
in the Western Mediterranean Sea**

Tesi di Laurea Magistrale di
Eleonora Carmen Savino

Relatore
Prof. Roberto Danovaro

Correlatrici:
Dr. Marta Coll
Dr. Maria Bas
Dr. Claudia Aparicio

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*"Non importa
chi ti abbia messo al mondo,
tutte le persone sono figli e figlie del mare"*

List of abbreviations and key definitions

- **Stable isotope analysis (SIA):** Quantification of natural abundance ratios of stable isotopes, commonly carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$), in consumer tissues to infer dietary sources, habitat use, and trophic position (Fry, 2006).
- **Stomach content analysis (SCA):** Identification and quantification of prey remain in stomachs to determine recent diet composition and feeding habits (Hyslop, 1980).
- **Trophic position (TP):** An organism's relative position in the food web, estimated from $\delta^{15}\text{N}$ enrichment between consumer and baseline representing primary production (Post, 2002).
- **Trophic enrichment factor (TEF):** The predictable increase in stable isotope ratios ($\Delta^5\text{N}$) between a consumer's tissues and its diet, typically assumed to be $\sim 3.4\text{‰}$ per trophic level for nitrogen (Post, 2002).
- **Isotopic niche:** A specific area in a multi-dimensional " δ -space" based on the stable isotope ratios (typically $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) representing the diversity of resources and habitats exploited by a consumer, acting as a proxy for the ecological niche (Newsome et al., 2007).
- **Tissue turnover rate:** The time required for isotopic incorporation to reflect the dietary inputs, influenced by tissue metabolism and growth (Kim et al., 2012; Nielsen et al., 2017).

Abstract

Pelagic fish communities are fundamental for Mediterranean marine ecosystems, transferring energy from planktonic production to top predators. However, understanding their feeding dynamics is often limited by the specific timeframe of the used method. This thesis investigated the trophic structure of pelagic communities in two contrasting regions of the western Mediterranean Sea, the productive Ebro Delta and the oligotrophic La Azohía, by integrating stomach content analysis (SCA) with multi-tissue stable isotope analysis (SIA). The comparison of the trophic positions' estimates, derived from both methodologies, revealed a distinct temporal divergence across both study areas. While stomach contents provided a snapshot of recent ingestion, often overestimating trophic positions due to digestive bias; stable isotope reflected the long-term assimilated diet. Notably, isotopic signatures suggested that the growth of medium and large pelagic species is largely sustained by lower-trophic resources than those visible in stomach contents. In La Azohía, the multi-tissue approach highlighted that isotopic variability can be related to species- and tissue-specific assimilation and turnover rates. Fast-turnover tissues may capture recent spatial and dietary heterogeneity; while inert tissues preserved lifetime trophic and habitat history, allowing for the reconstruction of ontogenetic shifts. Overall, this work highlights the limitations of relying on a single tissue to understand trophic dynamics. By integrating multiple temporal windows, the multi-tissue approach moves beyond static food webs, revealing how species interact with their environment across different timescales.

Riassunto

Le comunità di pesci pelagici rivestono un ruolo cruciale negli ecosistemi marini del Mar Mediterraneo, fungendo da collegamento tra la produzione planctonica e i livelli trofici superiori. Tuttavia, la comprensione e la ricostruzione delle loro dinamiche trofiche risultano difficili quando si impiegano singole metodologie con limitata risoluzione temporale. La presente tesi ha indagato la struttura trofica delle comunità pelagiche in due regioni del Mediterraneo occidentale, il Delta dell'Ebro e La Azohía, integrando l'analisi dei contenuti stomacali (SCA) con l'analisi degli isotopi stabili multi-tessuto (SIA). La comparazione delle stime delle posizioni trofiche, derivate da entrambe le metodologie, ha risaltato la divergenza temporale in entrambe le aree studio. Mentre il contenuto dello stomaco ha fornito un'istantanea dell'ingestione, spesso con stime di posizioni trofiche più elevate a causa del bias digestivo; gli isotopi stabili hanno riflesso l'assimilazione della dieta a lungo termine. In particolare, le firme isotopiche hanno indicato che la crescita delle specie pelagiche, di medie e grandi dimensioni, è in gran parte sostenuta da prede di livello trofico inferiore rispetto a quelle identificate nei contenuti stomacali. A La Azohía, l'approccio multi-tessuto ha evidenziato che la variabilità isotopica è determinata dai tassi di assimilazione e di turnover specifici per specie e tessuto. I tessuti con rapidi tassi di turnover rilevano la recente eterogeneità spaziale e alimentare; mentre i tessuti inerti hanno preservato le informazioni relative alla storia trofica e dell'habitat, permettendo la ricostruzione dei cambiamenti ontogenetici. In conclusione, la presente ricerca evidenzia le limitazioni dell'affidarsi ad analisi isotopiche svolte su un singolo tessuto per comprendere le dinamiche trofiche. L'approccio multi-tessuto, che integra diverse finestre temporali, consente di superare la visione delle reti trofiche statiche, rivelando come le specie interagiscono con il loro ambiente attraverso diverse scale temporali.

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Chapter 1: Introduction

This thesis investigates the trophic ecology of pelagic fish communities in the western Mediterranean Sea, combining the analysis of biomarkers and stomach contents in two study areas, the Ebro Delta region and the Azohía coast. In this first section, it outlines the ecological significance of the pelagic ecosystem, its socioeconomic importance and the environmental pressures shaping its ecological dynamics in the Mediterranean Sea. The complementary approaches used to study trophic structure, in this case, stomach content analysis (SCA) and stable isotope analysis (SIA), are presented, together with the rationale for considering tissue-specific isotopic variability of small, medium and large pelagic fish. Finally, the research questions, hypotheses and scope of the thesis are briefly summarised.

1.1 BACKGROUND

1.1.1 The pelagic ecosystem

The pelagic ecosystem includes the open-water domain of the ocean, where organisms inhabit and feed within the water column rather than on the seabed or coastal substrates (Angel, 1993; Coll et al., in press).

In the Mediterranean Sea, pelagic fish communities comprise diverse functional groups, defined as ontogenetic fractions of a species, single species or groups of species sharing common ecological traits that are linked through their trophic interactions (Corrales et al., 2017). These groups include: small planktivorous species such as European sardine *Sardina pilchardus* and European anchovy *Engraulis encrasicolus*; medium-sized zooplanktivorous and ichthyophagous species such as mackerels *Scomber spp.* and horse mackerels *Trachurus spp.*; and larger migratory top predators feeding on smaller fish and cephalopods, including various species of tunas like Bluefin *Thunnus thynnus*, swordfish *Xiphias gladius* and the blue shark *Prionace glauca* (Palomera et al., 2007; Coll et al., 2008). These organisms are organised into distinct trophic levels based on their position in the energy flow hierarchy, creating

intricate networks of feeding interactions known as food webs (Lindeman. 1942; Post. 2002). Consequently, these species contribute fundamentally to the energy transfer across ecosystem compartments and structure the dynamics of pelagic food webs (Albo-Puigserver et al., 2021).

By converting planktonic production into biomass accessible to higher predators, small pelagic fish form a biological bridge between lower and upper trophic levels (Cury et al., 2000. Young et al., 2017). Their position at intermediate trophic levels often generates a “wasp-waist” flow control, in which fluctuations in the abundance or size structure of small pelagic propagate both upward and downward through the food web (Cury et al., 2000; Bakun. 2006). This mechanism produces disproportionate ecological effects; for instance, a collapse in forage fish biomass can cause massive breeding failures in top predators like seabirds, while simultaneously allowing zooplankton blooms to persist unchecked due to the release of predation pressure (Cury et al., 2000). These wasp-waist dynamics are particularly significant in semi-enclosed systems like the Mediterranean Sea, where low productivity amplifies the cascading influence of small forage fish (Coll et al., 2019; Navarro et al., 2020; Ruzicka et al., 2024).

1.1.2 Ecological and socioeconomic relevance of small pelagic fishes

The ecological centrality of small pelagic species is mirrored by their socioeconomic importance. According to the 2025 Status Report of the General Fisheries Commission for the Mediterranean (FAO, 2025). European anchovy and sardine are among the species with the highest commercial value in the western Mediterranean. Specifically, European anchovy (7.8%), sardine (22.6%) and round sardinella *Sardinella aurita* (16.9%) together account for 47.3% of the total landings (tonnes). Consequently, small pelagic species remain among the most intensively exploited groups across the basin, supporting industrial and small-scale fisheries, regional food supply, and aquaculture through fishmeal production (Palomera et al., 2007; Pennino et al., 2020; Ortega et al., 2024; FAO 2025).

Furthermore, medium and large pelagic species, despite lower captured biomass compared to small pelagics, support economically critical fisheries due to their high

market value. Species such as the Atlantic bluefin tuna, swordfish, and various smaller tunas (e.g., Atlantic Bonito, *Sarda sarda*, and Bullet tuna, *Auxis rochei*) are vital for both industrial fleets and artisanal fisheries, driving significant export revenues and supporting local fishing communities (FAO, 2025). Collectively, these pelagic resources strongly influence local communities, many of which remain dependent on these fisheries for employment, income, and cultural identity (Lloret et al., 2015; Fernández-Corredor et al., 2021).

Because of this dual ecological and socioeconomic relevance, fluctuations in small pelagic populations can rapidly affect both marine communities and human societies. Shifts in stock size or demographic structure can alter its predator foraging behaviour, modify trophic link strengths and influence fisheries profitability (Coll et al., 2019; Lloret et al., 2015). These fluctuations often arise from interacting drivers such as climate variability, exploitation pressure and hydrographic anomalies, affecting resource availability, community composition and ecosystem resilience (Pennino et al., 2020; Ruzicka et al., 2024; Palomera et al., 2007). Particularly, even subtle shifts in recruitment or body condition of anchovies and sardines can propagate through the food web due to their pivotal role as prey for both commercial and protected predators (Fernández-Corredor et al., 2021; Navarro et al., 2020).

1.1.3 The Mediterranean Sea: a vulnerable and pressured basin

The Mediterranean Sea presents a unique environmental context for studying pelagic ecology. As a semi-enclosed basin with strong evaporative losses, elevated salinities and limited nutrient inputs, it is one of the most oligotrophic marine regions globally (Béthoux et al., 1998; Rodellas et al., 2015). Primary productivity is generally low and declines eastward, driven by restricted riverine inputs and intense seasonal stratification (Lazzari et al., 2012). These productivity regimes generate compact, efficiency-dependent food webs in which variations in plankton dynamics can strongly affect fish condition, recruitment and abundance (Thingstad et al., 1998; Coll et al., 2010).

Despite its high biodiversity, the Mediterranean Sea is highly vulnerable to environmental change due to its restricted water exchange with the Atlantic and intense

anthropogenic pressure (Lejeusne et al., 2010; Coll et al., 2012; Micheli et al., 2013). Sea-surface temperatures have increased at approximately 0.037- 0.041 °C per year, exceeding global ocean rates (Pisano et al., 2020; Martínez et al., 2023). Concurrent with this warming, the basin is experiencing progressive salinification and acidification (Schroeder et al., 2016). These combined trends have intensified marine heatwaves, strengthened summer stratification and altered productivity regimes, with particularly strong impacts on species characterised by short life cycles and high metabolic demands (Darmaraki et al., 2019; Le Grix et al., 2021). Furthermore, these environmental changes may favour shifts towards microbial or gelatinous-dominated pathways, altering the energy transfer to higher trophic levels (Danovaro et al., 2009).

In parallel, the Mediterranean remains one of the most heavily exploited marine regions worldwide (Cózar et al., 2015; UNEP/MAP, 2023; FAO 2025). The 2025 FAO-GFCM Status Report indicates that while the percentage of stocks fished above sustainable levels has decreased from 75% in recent years, exploitation remains critically high, with 52% of assessed stocks still classified as overfished (FAO, 2025). The basin also sustains extremely high densities of maritime traffic and coastal population, contributing to cumulative pressures, including habitat alteration, chemical contamination and widespread microplastic pollution, among the highest recorded globally (Cózar et al., 2015; UNEP/MAP, 2023).

These interacting stressors reshape pelagic fish populations and ecosystem functioning (Coll et al., 2019; 2024). Recent studies document declines in body condition, growth and reproductive output of anchovy and sardine under warming and reduced productivity (Caballero-Huertas et al., 2023; Fernández-Corredor et al., 2021), and changes in abundance and size structure of these species propagate through food webs, modifying trophic interactions and predator behaviour (Navarro et al., 2020; Ruzicka et al., 2024). Understanding these trophic relationships is therefore essential for predicting ecosystem responses to ongoing climatic and anthropogenic change (Post, 2002; Coll et al., 2019).

1.2 METHODOLOGICAL CONTEXT

Trophic ecology investigates feeding interactions and the pathways through which energy and nutrients flow within ecological communities, thereby identifying trophic positions and food-web structure (Post, 2002; Newsome et al., 2007). To achieve a comprehensive understanding of these dynamics, this thesis integrates stomach content analysis (SCA) and stable isotope analysis (SIA). By combining the snapshot of recently ingested prey provided by SCA with the time-integrated record of assimilated diet from SIA, it is possible to reconstruct both immediate and longer-term dietary pathways that structure of the food-web (Layman et al., 2012; Hussey et al., 2012).

1.2.1 Stomach content analysis

Stomach content analysis (SCA) is a classical method for reconstructing diets, relying on the visual and taxonomic identification of prey remains in the digestive tract (Hyslop, 1980). Because it can resolve prey at fine taxonomic levels, SCA is particularly valuable for describing predator-prey relationships, feeding strategies, and resource partitioning (Cortés, 1997; Amundsen et al., 1996). This method also allows quantification of useful prey-specific metrics such as frequency of occurrence, numerical abundance and weight contribution, each providing complementary insight into feeding behaviour (Amundsen et al., 2019).

Numerous studies have applied SCA to describe the trophic ecology of pelagic organisms in the Mediterranean Sea, providing critical baselines for understanding community dynamics. For instance, research on small pelagic fish has used stomach contents to assess trophic overlap and resource competition between resident species (e.g., *Sardina pilchardus*, *Engraulis encrasicolus*) and expanding species such as the *Sardinella aurita* (Albo-Puigserver et al., 2019; Bachiller et al., 2021). Similarly, for large pelagic predators, SCA has been used to reconstruct the long-term feeding history of Atlantic bluefin tuna (Pardo et al., 2025) and to validate juvenile swordfish diet as a sentinel indicator of ecosystem change (Fernández-Corredor et al., 2023). Furthermore, recent integrative efforts have combined SCA with other methodologies

to reveal large-scale spatial patterns, such as latitudinal gradients in the feeding habits of forage fish across the western basin (Bachiller et al., 2020).

However, SCA provides only a short temporal snapshot, typically hours to a few days, because prey detectability declines rapidly through digestion (Rindorf & Lewy, 2004; Baker et al., 2014). Soft-bodied or gelatinous prey are often underestimated or entirely lost, generating biases that can misrepresent predator-prey links, especially in small pelagic fishes whose diets often include fragile zooplankton prey (Buckland et al., 2017; Ibáñez et al., 2021). Additional constraints, including regurgitation during capture, diel feeding rhythms, and the prevalence of empty stomachs, and potential observer biases arising from insufficient taxonomic expertise or misidentification, further limit the interpretability of SCA datasets (Hyslop, 1980; Baker et al., 2014). Despite these limitations, SCA remains indispensable for identifying prey taxa that would otherwise be isotopically indistinguishable, making it a critical complement to isotopic methods (Layman et al., 2012; Davis et al., 2012).

1.2.2 Stable isotope analysis (SIA)

Stable isotopes are non-radioactive forms of elements that differ in neutron number and atomic mass and remain stable over ecological timescales (Peterson & Fry, 1987; Fry, 2006). Biochemical and physical processes cause predictable fractionation between heavy and light isotopes, producing characteristic isotopic ratios that can be used to infer ecological processes such as energy flows, nutrient cycling, and dietary sources within ecosystems (Post, 2002; Boecklen et al., 2011). These ratios are expressed in delta (δ) notation relative to international standards, Vienna Pee Dee Belemnite (VPDB) for carbon and atmospheric nitrogen (AIR) for nitrogen (Coplen, 1994), enabling comparison across tissues, species, and ecosystems (Fry, 2006).

Carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) are the most widely used stable isotopes in marine trophic studies because they elucidate the two primary axes of food web topology: the habitat and the vertical trophic levels (Layman et al., 2012; Peterson & Fry, 1987). $\delta^{13}\text{C}$ exhibits relatively minor trophic enrichment but varies substantially among primary production sources, distinguishing benthic from pelagic, coastal from offshore, and terrestrial from marine carbon pathways (France, 1995; Post, 2002).

Conversely, $\delta^{15}\text{N}$ exhibits substantial fractionation during assimilation, as consumers preferentially retain the heavier isotope while excreting the lighter one (DeNiro & Epstein, 1981). Consequently, $\delta^{15}\text{N}$ increases, typically 2-4‰, with each trophic transfer, providing a robust estimator of trophic position (Minagawa & Wada, 1984; Vander Zanden & Rasmussen, 1999; McCutchan et al., 2003). These isotopic metrics have been successfully applied to the Mediterranean pelagic ecosystem, for instance, to describe the trophic structure of pelagic assemblages, including seasonality (Albo-Puigserver et al., 2016) and to identify the trophic strategies of coexisting predators across different temporal scales (Navarro et al., 2020).

Stable isotope analysis (SIA) has become central to trophic ecology because it provides a time-integrated record of assimilated diet over periods defined by tissue turnover (Post, 2002; Jackson et al., 2011; Hobson, 1999). Compared with SCA, which reflects recent ingestion, SIA integrates dietary information over weeks, months, or even years, depending on tissue type, physiological rate, and growth rate (Hobson, 1999; Hussey et al., 2014) (*see section 1.2.3. below*). SIA also requires relatively small tissue samples and can be conducted non-lethally in some cases, expanding its applicability to protected or data-limited species (Henderson et al., 2016; Larocque et al., 2021). Nonetheless, SIA entails higher analytical costs compared to traditional methods and it is sensitive to physiological processes such as metabolic routing, isotopic discrimination, tissue biochemical composition, regional and temporal baseline of the marine ecosystem and analytical precision, all of which must be accounted for to avoid misinterpretation (Gaye-Siessegger et al., 2004; Barnes et al., 2008; Jardine & Cunjak, 2005).

Beyond standard trophic reconstruction, recent studies have demonstrated that multi-tissue isotopic datasets can reveal ontogenetic dietary shifts and movement across isotopically heterogeneous habitats, providing insights that surpass the temporal scope of single-tissue approaches (Hussey et al., 2012; Nielsen et al., 2017). While SIA lacks the fine taxonomic resolution of preys compared to stomach content analysis, it resolves the temporal limitations of the latter (Layman et al., 2012). Therefore, integrating both methods mitigate their respective constraints, showing a comprehensive and temporally resolved reconstruction of trophic relationships that

neither method could achieve in isolation (Boecklen et al., 2011; Davis et al., 2012; Bachiller et al., 2020).

1.2.3 Tissue-specific isotopic turnover

An important consideration when interpreting SIA is that different tissues within the same organism do not incorporate the stable isotopes from diet at the same rate (Post, 2002). Because isotopic turnover depends on metabolic activity, growth dynamics, and biochemical composition, each tissue effectively integrates dietary information over a specific temporal window (Post, 2002; Nielsen et al., 2017; Kim et al., 2012). The kinetics of this process are commonly described by a turnover rate (λ), often expressed as an isotopic half-life (t_{50}): the time required for 50% of a tissue's atoms to equilibrate with a new dietary isotopic baseline (Hesslein et al., 1993).

Two biological mechanisms drive isotopic turnover: growth dilution and metabolic replacement (Hesslein et al., 1993). Growth dilution occurs when new tissue is synthesised from the recent diet and dilutes the isotopic signature of older biomass (Madigan et al., 2012). This mechanism tends to dominate during life stages of rapid growth, such as juveniles of many pelagic fishes (Herzka & Holt, 2000). Metabolic replacement involves the catabolism and resynthesis of existing molecules without a net increase in tissue mass (Hesslein et al., 1993). As somatic growth slows, e.g. in adult fishes, metabolic replacement becomes the primary driver of isotopic change (Buchheister & Latour, 2010).

The relative contribution of growth dilution and metabolic replacement is strongly influenced by organism physiology (Carleton & Martínez del Río, 2010). In general, turnover scales allometrically with body size, with smaller fishes generally incorporating isotopes more rapidly than large, slow-growing predators (Martínez del Río et al., 2009; Vander Zanden et al., 2015). Thermal physiology further modulates turnover: for instance, regionally endothermic taxa (e.g. tunas, lamnid sharks) exhibit elevated protein synthesis and replacement rates relative to ectothermic species of comparable size, potentially accelerating isotopic turnover across tissues (Madigan et al., 2018). These physiological principles provide a predictive framework, yet real-

world turnover remains primarily determined by tissue-specific biochemical and metabolic properties (Carleton & Martínez del Río, 2010; Mohan et al., 2016).

Functionally, these physiological dynamics allow tissues to be broadly grouped into two categories based on their isotopic integration properties: **metabolically active tissues**, such as liver and blood, which provide a moving average of recent assimilated diet; and **metabolically inert or archival tissues**, such as otoliths and eye lenses, which retain the isotopic composition present at the time of deposition and thus preserve longer-term ecological histories (Campana, 1999; Martínez del Río et al., 2009).

Building on this understanding, the analysis in this thesis is to take advantage of distinct temporal windows of information. By systematically comparing marine animal tissues, we aim to disentangle general physiological patterns from the specific ecological differences driven by species-specific behaviours. The specific integration properties and temporal resolutions of the key tissues are synthesised in **Table 1.2**.

BOX 1. METABOLICALLY ACTIVE TISSUES

Fast turnover tissues (days to weeks)

Metabolically active tissues include the splanchnic organs (e.g. liver, stomach, and stomach), where rapid biochemical turnover drives fast isotopic incorporation, typically characterised by isotopic half-lives ranging from **days to weeks** (Tieszen et al., 1983; Buchheister & Latour, 2010). The **liver**, among the most dynamic tissues in fishes, exhibits high rates of protein synthesis and lipid cycling and therefore reflects dietary inputs over timescales of **days to weeks**, depending on species, temperature and feeding conditions (Carleton & Martínez del Río, 2010; Suzuki et al., 2005; Buchheister & Latour, 2010).

Reproductive tissues, such as **gonads**, also fall within this category, but their turnover is uniquely governed by the gametogenic cycle rather than continuous metabolic maintenance. In income breeders, such as the European and Japanese

anchovies, gonadal $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values mirror the diet consumed immediately before spawning, reflecting rapid deposition during oocyte growth (Yoneda et al., 2025).

Other metabolically active tissues with rapid or intermediate turnover include **blood plasma** and **epidermal mucus**, which incorporate dietary isotopes within days to weeks and are widely used in non-lethal monitoring studies (Kim et al., 2012; Church et al., 2009; Winter et al., 2019), as well as **red blood cells**, which renew more slowly and therefore integrate diet over several weeks to months (Kim et al., 2012; Vander Zanden et al., 2015).

Slow turnover tissues (months to years)

Somatic tissues with lower metabolic demands integrate diet over extended periods, ranging from **several months to over a year** depending on the species' life history and tissue type (Buchheister & Latour, 2010). **White muscle** is the standard tissue for fish isotope ecology due to its relatively homogeneous biochemical composition and low metabolic turnover (Pinnegar & Polunin, 1999; Carleton & Martínez del Río, 2010). Its isotopic values generally represent diet over medium to long terms, though turnover rates vary significantly with body size, temperature, and species-specific metabolic rates (Herzka, 2005; Buchheister & Latour, 2010; Mohan et al., 2016). For example, in juvenile Pacific bluefin tuna (*Thunnus orientalis*), muscle tissue requires approximately 167 days to equilibrate with a new diet, illustrating the extended integration window of large pelagic species (Madigan et al., 2012). Turnover in muscle is allometric; that is, it scales with body size, such that larger fish exhibit slower incorporation rates than smaller individuals (Vander Zanden et al., 2015).

Fin tissues, commonly used as non-lethal surrogates for muscle, show mixed turnover behaviour: **the soft inter-ray membrane** incorporates isotopes at a rate comparable to muscle, whereas the **mineralised fin rays** grow by accretion and may preserve a partial chronological record, although vascularisation and resorption in some pelagic taxa can attenuate early-life layers (Willis et al., 2013; Tzadik et al., 2017).

Metabolically inert (Archival) tissues

In contrast to active tissues, archival tissues grow through accretion and remain metabolically inert after deposition, preserving a chronological record of the fish's life history (Campana, 1999).

Vertebral structures serve as long-term isotopic archives, although their reliability depends on the distinct physiological properties of the taxa involved (Andrews et al., 2023). In elasmobranchs, vertebral cartilage grows by continuous accretion, depositing discrete, non-remodelling bands (annuli) that reflect the entire life history of the individual (Estrada et al., 2006; Andrews et al., 2023). This calcified cartilage is largely metabolically inert, allowing for the chronological reconstruction of dietary trajectories across the individual's lifetime via micro sampling (Kim et al., 2012; Carlisle et al., 2012). However, caution is required regarding spatial resolution in older specimens, as band compression can blur the temporal distinctness of isotopic signatures (Andrews et al., 2023).

In contrast, the use of teleost vertebrae as chronological archives has evolved with our understanding of bone physiology (Witten & Huysseune, 2009; Bas & Cardona, 2018). While early views considered the fish skeleton to be metabolically inactive, it is now clear that teleost bone is subject to metabolic resorption and remodelling to maintain mineral homeostasis, particularly during periods of nutritional or reproductive stress (Witten & Huysseune, 2009). Consequently, teleost vertebral collagen acts as a long-term integrated average of the fish's diet rather than a precise chronological archive (Tzadik et al., 2017; Andrews et al., 2023). This implies that early-life isotopic signatures may be partially overwritten by later metabolic activity (Tzadik et al., 2017; Andrews et al., 2023).

Otoliths are carbonate structures precipitated on an organic matrix within the inner ear of teleosts (Campana, 1999). Once deposited, their carbonate material is metabolically inert; the organic fraction likewise becomes fixed, preserving the isotopic composition at the time of deposition (Campana, 1999). This property makes otoliths ideal for reconstructing long-term ontogenetic shifts in diet or habitat (Campana, 1999; Grønkjær et al., 2013).

Furthermore, **eye lenses** grow by sequential addition of concentric protein layers (crystallins), which remain chemically stable after deposition (Wallace et al., 2014). Because of their nitrogen-rich composition and lack of metabolic turnover, each layer of the lens can record the isotopic signature at the time of formation, offering a high-resolution archive of dietary or habitat changes throughout an individual's life (Wallace et al., 2014; Tzadik et al., 2017). Recent studies have successfully used this archival property by analysing sequential protein laminations to reconstruct the complete lifetime trophic histories of complex migrants, establishing the method as a robust tool for identifying precise ontogenetic shifts in trophic position and habitat use (Quaeck-Davies et al., 2018; Curtis et al., 2020).

Understanding tissue-specific turnover is particularly critical for SIA studies and especially when dealing with pelagic fishes, which often undergo ontogenetic dietary shifts and migrate among isotopically heterogeneous habitats (Hussey et al., 2012; Kim et al., 2012). By sampling different tissues simultaneously, a multi-tissue SIA approach can reconstruct dietary histories across temporal scales: metabolically active tissues reflect recent feeding conditions and nutritional state; archival tissues reveal long-term or lifetime resource use, trophic shifts, and habitat transitions (Campana, 1999; Grønkjær et al., 2013). Integrating these multiple temporal windows offers a powerful mechanism to resolve complex life-history dynamics that single-tissue approaches would not reveal (Nielsen et al., 2017).

Table 1.2. Functional classification of tissues used in stable isotope analysis. The table summarises the distinct physiological turnover mechanisms, approximate temporal integration windows, and the specific ecological utility of each tissue for reconstructing the trophic history of pelagic fishes.

Tissue category	Tissue type	Turnover	Temporal range	Ecological utility	References
Metabolically active	Blood plasma / Mucus	Very fast	Days - Weeks	Recent diet; short-term feeding events; non-lethal sampling	Church et al. (2009); Kim et al. (2012)
	Liver	Fast	Days - Weeks	Recent diet; seasonal pulses	Tieszen et al. (1983); Buchheister & Latour (2010)
	Gonads (in income breeders)	Fast-variable	Days - Weeks (pre-spawning)	Diet just before reproduction; resource allocation studies	Yoneda et al. (2025)
	Red blood cells (erythrocytes)	Intermediate	Weeks - Months	Intermediate-term diet; migratory species monitoring	Kim et al. (2012); Vander Zanden et al. (2015)
	White muscle	Slow	Months ~Years	Seasonal to annual diet integration; inter-individual comparisons	Madigan et al. (2012); Pinnegar & Polunin (1999)
	Fin membrane (inter-ray soft tissue)	Slow	Months	Non-lethal surrogate for muscle in long-term studies	Willis et al. (2013)

	Fin rays (lepidotrichia)	Very slow / accretion	Months- Years	Potential long- term record (limited)	Tzadik et al. (2017)
Metabolically inert (archival)	Teleost vertebrae (bone/collagen)	Very slow / remodelled	Lifetime (integrated average)	Long-term trophic baseline; limited chronological resolution	Witten & Huysseune (2009); Andrews et al. (2023)
	Elasmobranch vertebrae (cartilage bands)	Accretion / non- remodelling	Lifetime (chronolog ical)	Lifetime dietary trajectories; ontogenetic shifts; habitat migration	Estrada et al. (2006); Carlisle et al. (2012)
	Otoliths	Accretion (aragonite)	Lifetime (chronolog ical)	High-resolution lifetime dietary or habitat history	Campana (1999); Grønkjær et al. (2013)
	Eye lens	Accretion (crystallins)	Lifetime (chronolog ical)	Lifetime dietary / habitat shifts; sequential record from embryonic to adult stages	Quaeck-Davies et al. (2018); Wallace et al. (2014)

1.3 CONTEXT

Despite the established utility of multi-method approaches, the simultaneous application of SIA and SCA remains methodologically complex due to their differing temporal resolutions and analytical sensitivities (Post, 2002; Layman et al., 2012). While studies in freshwater systems have successfully demonstrated how combining these methods reveals ontogenetic shifts undetectable to either method alone (Davis et al., 2012), such integrative frameworks have been applied less frequently to Mediterranean pelagic communities.

Indeed, research in the northwestern Mediterranean has largely examined food-web structure using single methods, relying either on isotopic (Albo-Puigserver et al., 2016; Navarro et al., 2020) or stomach content data (Bernal et al., 2015; Costalago & Palomera, 2014). Although recent studies have begun to bridge this gap by conducting direct quantitative comparisons for specific species (Bachiller et al., 2020, 2021; Fernández-Corredor et al., 2023; Pardo et al., 2025), systematic evaluations of the consistency between stomach contents and multi-tissue isotopic signatures remain limited. This lack of comprehensive validation hinders our ability to robustly calibrate isotopic metrics against taxonomic data, limiting the full resolution of temporal feeding dynamics in this highly pressured marine basin (Hussey et al., 2012).

Furthermore, while tissue-specific turnover is a known physiological phenomenon (Mohan et al., 2016), its application in Mediterranean pelagic ecology has been largely restricted to specific case studies. For instance, Navarro et al. (2020) successfully used a multi-tissue comparison (e.g., muscle vs. liver vs. fin) to differentiate trophic strategies in local predators, however standard isotopic assessments in the region still predominantly focus on single-tissue analysis (typically white muscle). By relying on a single temporal window, these assessments frequently neglect the "isotopic clock" potential of multi-tissue analysis to resolve temporal discrepancies (Fry, 2006; Hussey et al., 2012). Consequently, the capacity to reconstruct ontogenetic shifts or seasonal dietary changes within individuals remains largely untapped in current regional assessments (Nielsen et al., 2017).

Finally, the ecological interpretation of isotopic data has been significantly advanced by the formalisation of the isotopic niche concept and the development of quantitative metrics to measure it (Newsome et al., 2007; Layman et al., 2007). These indices use dispersion in $\delta^{13}\text{C}$ - $\delta^{15}\text{N}$ bi-plot space to estimate trophic diversity and resource use, providing a standardised method to compare community structure (Layman et al., 2012). Among the modern tools available, SIBER (Stable Isotope Bayesian Ellipses in R), a specialised package within the R statistical software environment, allows for the robust estimation of isotopic niche width and overlap even with small sample sizes (Jackson et al., 2011). However, these advanced Bayesian frameworks are rarely applied to multi-tissue datasets in the Mediterranean context

(Navarro et al., 2020), where most isotopic studies remain focused on single-tissue assessments (Albo-Puigserver et al., 2016; Cardona et al., 2015).

Therefore, addressing these gaps, specifically the cross-validation of SIA and SCA, the exploration of tissue-specific isotopic variability, and the application of Bayesian niche metrics, is crucial for improving the ecological interpretation of isotopic data and drawing robust inferences about trophic structure in pelagic marine ecosystems (Hussey et al., 2012; Nielsen et al., 2017).

1.4 OBJECTIVES

Building on this background, the fundamental purpose of this thesis is to elucidate the trophic structure and isotopic variability of two pelagic fish communities in the western Mediterranean Sea by combining different methodologies. Two complementary objectives were developed:

1. **To compare trophic information derived from SIA and SCA** for pelagic fish species collected in the *Ebro Delta* region (*Case study 1*) system associated with the purse seine fishery during 2022-2023, assessing whether both SCA and SIA methods showed consistent or divergent insights into trophic relationships and ecosystem structure.

2. **To examine isotopic variability among tissues** (muscle, gonads, liver, bone, fin and eye) from small, medium and large pelagic fish sampled in *La Azohía* Tuna trap during spring 2025 (*Case study 2*). This objective integrates stomach content analysis as a temporal baseline to interpret tissue-specific differences in relation to turnover rates and extends the methodological comparison of dietary metrics to an oligotrophic environment.

Research questions and hypotheses

1. *To what extent do SIA and SCA provide complementary or divergent information on trophic relationships among pelagic fish species in the western Mediterranean Sea?*

H1: SIA and SCA are expected to produce broadly consistent trophic position estimates, with potential discrepancies arising from differences in temporal integration, digestive selectivity, and short-term variability in prey availability, regardless of the study area

2. *How do $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values vary among tissues within individuals and species, and what does this reveal about temporal dietary integration and life-history information?*

H2a: Metabolically active tissues (e.g., liver, gonad, muscle, dorsal fin) will reflect recent dietary inputs and show greater inter-individual variability.

H2b: Archive tissues (e.g., vertebrae, eyes) will integrate longer-term dietary signals and exhibit $\delta^{15}\text{N}$ enrichment correlated with body size, reflecting ontogenetic trophic shifts.

1.5 SIGNIFICANCE AND SCOPE

This research contributes to advancing the knowledge about the trophic ecology of the pelagic community in the Mediterranean Sea by integrating complementary dietary approaches and evaluating isotopic variability across tissues. From a methodological perspective, it refines the interpretation of SIA results and contextualises SCA findings by identifying potential biases through cross-validation, thereby linking tissue-specific isotopic signatures with ecological and physiological processes.

Ecologically, *Case study 1* focuses on the *Ebro Delta* pelagic system (Catalonia), where the combination of SIA in white muscle and SCA is used to describe the trophic structure and quantify relationships among pelagic fish species in a productive, river-influenced ecosystem. *Case study 2*, conducted in *La Azohía* tuna trap system (Murcia), integrates SIA across multiple tissues and SCA to investigate intra-individual isotopic variability. This approach provides insights into how different tissues integrate dietary information over distinct temporal scales and allows for the characterisation of recently ingested prey and evaluating the consistency of trophic estimates across contrasting environmental systems.

The overall study's scope is limited to pelagic fish species collected during discrete sampling periods (2022–2025) and focuses on stable isotopes of nitrogen ($\delta^{15}\text{N}$) and carbon ($\delta^{13}\text{C}$). Nevertheless, its findings have broader implications for stable isotope ecology, trophic modelling, and ecosystem-based management by reducing uncertainty in trophic estimates and improving the temporal resolution of dietary studies.

This is particularly relevant for the Mediterranean Sea, where complex food webs and rapid environmental changes require robust, multi-method monitoring tools, even though isotopic baselines are generally well-established in the region.

1.6 THESIS OUTLINE

The next chapters of this thesis are structured as follows:

- **Chapter 2 - Materials and Methods:** Describes the study areas, sampling design, laboratory procedures, and analytical methods used for both SIA and SCA, including isotopic corrections and statistical analyses.
- **Chapter 3 - Results:** Presents the main findings, including isotopic and stomach content data for the *Ebro Delta (Case study 1)* and inter-tissue isotopic variability and stomach content data for *La Azohía (Case study 2)*.
- **Chapter 4 - Discussion:** Interprets the results considering existing literature, evaluates the hypotheses, and explores the ecological and methodological implications of integrating SIA and SCA in the current research using these methods. While exploring how tissue turnover and life history traits shape the isotopic signals of pelagic species.
- **Chapter 5 - Conclusions:** Synthesises the key findings, highlights methodological and ecological insights, and provides recommendations for future research and ecosystem-based fisheries management in the western Mediterranean Sea.

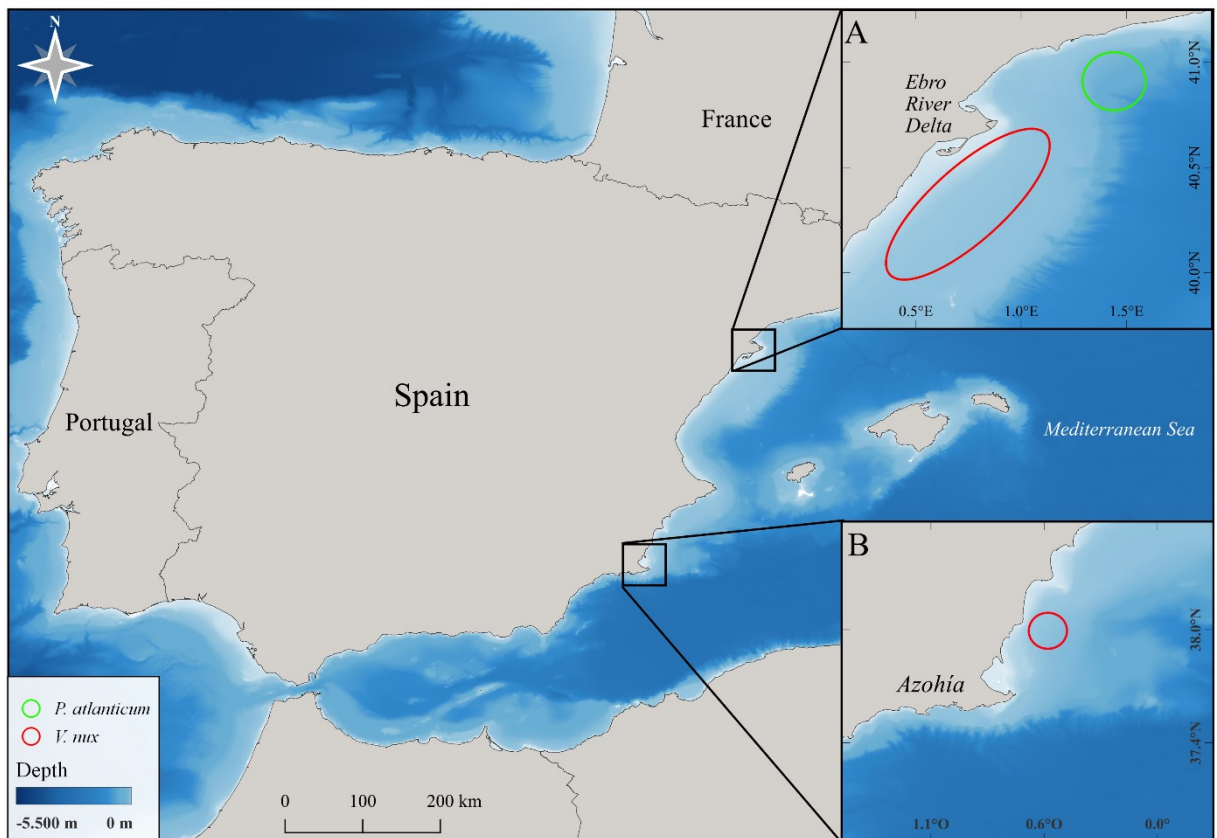
Chapter 2: Materials and Methods

2.1 STUDY AREAS AND SAMPLE COLLECTION

2.1.1 Study areas

This study was conducted in two distinct regions of the western Mediterranean Sea: the *Ebro Delta* (Catalonia, Spain) and *La Azohía* (Murcia, Spain) (**Figure 2.1**). These areas represent contrasting pelagic environments influenced by different oceanographic, climatic, and anthropogenic conditions within a larger region of the western Mediterranean Sea.

Figure 2.1 Location of the two study areas in the western Mediterranean Sea: (a) Ebro Delta (Catalonia, Spain) and (b) La Azohía (Murcia, Spain). Circles represent where the baseline organisms were collected: *P. atlanticum* (green) and *V. nux* (red). Maps generated with QGIS



The **Ebro Delta** is located in the northwestern Mediterranean, where the Ebro River discharges into the sea, forming one of the largest coastal wetlands in the region (Rodríguez-Santalla & Navarro, 2021). The area is characterised by a wide continental shelf, extending up to 70 km offshore with depths reaching approximately 100 m at the shelf break, and high primary productivity driven by nutrient-rich freshwater inputs and strong seasonal variability (Salat, 1996; Coll et al., 2008). This nutrient enrichment sustains a productive pelagic ecosystem dominated by small pelagic species such as European pilchard and European anchovy, which serve as key prey for larger predators, including mackerels, Atlantic horse mackerels, and small tunas (Atlantic Bonito, Bullet tuna and little tunny) (Palomera et al., 2007; Ouled-Cheikh et al., 2022). However, the system is increasingly impacted by anthropogenic pressures, such as reduced river flow due to damming and irrigation, eutrophication, and intense fishing activity, which collectively influence trophic interactions and food-web stability (Coll et al., 2008; Llebot et al., 2011). These characteristics establish the Ebro Delta as a high-productivity reference system for examining pelagic trophic dynamics (Navarro et al., 2020).

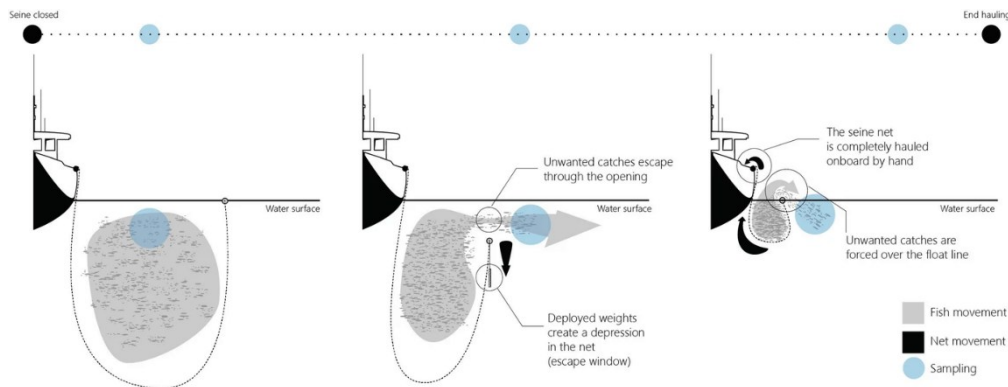
La Azohía, located along the southeastern coast of Spain within the Alboran Sea (Murcia region), represents an oligotrophic, open coastal ecosystem influenced by Atlantic inflow through the Strait of Gibraltar (García-Lafuente et al., 2002). In contrast to the Ebro, this area is distinguished by a constricted continental shelf with steep bathymetry, where depths rapidly exceed 200 m close to shore, and by the presence of tuna traps, known as *almadrabas* (Di Natale et al., 2009). These are stationary, labyrinth-like nets anchored proximate to the shoreline to intercept migrating pelagic species, including Atlantic bonito (*Sarda sarda*), bullet tuna (*Auxis rochei*) and longfin tuna (*Thunnus alalunga*). La Azohía offers a lower-productivity environment where isotopic variability across tissues can be explored in relation to oligotrophic baseline conditions (Bosc et al., 2004; Estrada, 1996).

2.1.2 Sample collection

Fish samples were collected during two independent sampling campaigns conducted between 2022 and 2025, corresponding to the two study areas described above.

For **Case study 1 (Ebro Delta)**, samples were collected from 22 August 2022 to 19 September 2023 using purse-seine fisheries operating in the vicinity of the delta. This active netting technique is designed to encircle pelagic schools with a large vertical net, allowing efficient capture of small- to medium-sized pelagic fishes while preserving their integrity for subsequent laboratory processing (Marçalo et al., 2018) (**Figure 2.2**). In total, **165 individuals** representing 16 species were collected in this area, ranging from small forage fish like the European anchovy to large predators such as the swordfish (**Table 3.1**).

Figure 2.2. Diagram of sampling at sea for purse-seine operations (adapted from Marçalo et al. 2018).



For **Case study 2 (La Azohía)**, sampling was conducted from 4 May 2025 to 7 September 2025 in collaboration with local fisheries employing the almadraba technique (**Figure 2.3**).

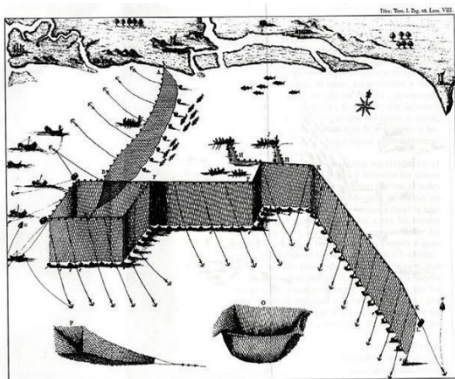


Figure 2.3. Representation of a tuna trap in profile. Sáñez Reguart, Volumen I, lámina VIII, p. 68 (reproduced from Cambrils Pueblo de Pescadores, 2024).

This traditional fixed-trap system intercepts migratory pelagic species using anchored net chambers (Di Natale et al., 2009), ensuring the minimal capture stress and optimal specimen preservation required for high-precision multi-tissue analysis. A total of **55 individuals** belonging to 9 species were sampled directly from the trap. Additionally, to ensure robust sample sizes for specific elasmobranchs, this dataset was supplemented with isotopic data for 56 blue sharks (*Prionace glauca*) collected in the region (Fernández-Corredor et al., 2025) (**Table 3.4**). Across both study areas, target species were selected to represent multiple trophic levels and ecological roles within the pelagic food web, ensuring a representative coverage of the entire pelagic community (**Table 2.1**).

To estimate trophic positions, isotopic baseline samples were collected from a scientific bottom trawl survey as part of the **Mediterranean International Trawl Survey (MEDITS)** campaign¹. These samples consisted of obligate filter-feeding organisms selected to represent the primary consumer level. Specifically, for the Ebro Delta, the pelagic tunicate *Pyrosoma atlanticum* was collected from deep offshore waters (339-5757 m depth), while the demersal bivalve *Venus nux* was sampled from shallower continental shelf (36-77 m depth). For the La Azohía system, *Venus nux* samples were collected at depths ranging from 75 to 95 m. These primary consumers provide a proxy for the base of the food web, as they effectively integrate high-frequency isotopic fluctuations of basal resources (Post, 2002).

¹ Funded by the EU through the European Maritime and Fisheries Fund (EMFF) within the National Programme of collection, management and use of data in the fisheries sector and support for scientific advice regarding the Common Fisheries Policy.

Table 2.1. List of pelagic fish species analysed in each study area, grouped by functional category, with common names, IUCN conservation status, and tissues analysed for SIA. Species are visually separated by functional group (Small, Medium, Large pelagic fishes and Baseline) to facilitate interpretation. IUCN categories: LC = Least Concern; NT = Near Threatened; VU = Vulnerable; EN = Endangered; CR = Critically Endangered; DD = Data Deficient; NE = Not Evaluated. Tissues analysed: MU = Muscle; ST = Stomach; LI = Liver; GO = Gonads; DF = Dorsal fin; VE = Vertebrae; EY = Eye. Studies: 1 (Ebro Delta), 2 (Azohía).

Species	Common name	IUCN status	Tissue analysed	Study
Small pelagic				
<i>Boops boops</i>	Bogue	LC	MU	1
<i>Cheilopogon exsiliens</i>	Exocoetid flyingfish	LC	MU	1
<i>Engraulis encrasicolus</i>	European anchovy	LC	MU, ST, LI, GO, DF, VE, EY	both
<i>Sardina pilchardus</i>	European sardine	LC	MU, ST, LI, GO, DF, VE, EY	both
Medium pelagic				
<i>Auxis rochei</i>	Bullet tuna	LC	MU	1
<i>Auxis thazard</i>	Frigate tuna	LC	MU, ST, LI, GO, DF, VE, EY	2
<i>Euthynnus alletteratus</i>	Little tunny	LC	MU, ST, LI, GO, DF, VE, EY	2
<i>Sarda sarda</i>	Atlantic bonito	LC	MU, ST, LI, GO, DF, VE, EY	both
<i>Scomber colias</i>	Atlantic chub mackerel	LC	MU, ST, LI, GO, DF, VE, EY	both
<i>Scomber scombrus</i>	Atlantic mackerel	LC	MU	1
<i>Seriola dumerili</i>	Greater amberjack	LC	MU	1
<i>Trachurus trachurus</i>	Atlantic horse mackerel	LC	MU	1
Large pelagic				
<i>Mola mola</i>	Ocean sunfish	VU	MU, ST, LI, GO, DF, VE, EY	2
<i>Prionace glauca</i>	Blue shark	NT	MU, ST, LI, GO, DF, VE, EY	2
<i>Tetrapturus belone</i>	Mediterranean spearfish	DD	MU, ST, LI, GO, DF, VE, EY	2
<i>Thunnus thynnus</i>	Atlantic bluefin tuna	EN	MU	1
<i>Tylosurus acus</i>	Keel-jawed needlefish	LC	MU	1
<i>Xiphias gladius</i>	Swordfish	VU	MU	1
Baseline				

Species	Common name	IUCN status	Tissue analysed	Study
<i>Pyrosoma atlanticum</i>	Sea pickle	NE	Whole body	1
<i>Venus nux</i>	Saltwater clam	NE	Whole body	1
Other				
<i>Ariosoma balearicum</i>	Bandtooth conger	LC	MU	1
<i>Illex coindetii</i>	Broadtail shortfin squid	LC	MU	1
<i>Remora remora</i>	Common remora	LC	MU	1

2.2 LABORATORY ANALYSES

2.2.1 Fish measurement and dissection

Laboratory processing was conducted at the **Institut de Ciències del Mar (ICM-CSIC, Barcelona)** directed by the Trophic Ecology and Connectivity scientific-technical service. Morphometric measurements were selected according to the specific body morphology of each taxon to ensure standardised comparisons, following established fisheries guidelines (Holden & Raitt. 1974).

Total length (TL) was recorded for species with truncate or rounded caudal fins, including clupeoids (*Sardina pilchardus*, *Engraulis encrasicolus*), sparids (*Boops boops*), and the ocean sunfish (*Mola mola*), as well as for elasmobranchs (*Prionace glauca*). **Fork length (FL)** was used for scombrids and carangids (*Scomber colias*, *Trachurus trachurus*, *Thunnus thynnus*, *Auxis rochei*, *Sarda sarda*, *Seriola dumerili*) to avoid bias caused by potential damage to the caudal fin tips. For billfishes and needlefishes (*Xiphias gladius*, *Tetrapturus belone*, *Tylosurus acus*), **lower-jaw fork length (LJFL)** was recorded. Additionally, **dorsal mantle length (ML)** was measured for cephalopods (*Illex coindetii*).

All specimens were weighed (**total weight, TW**) to the nearest 0.01 g. Sex and gonadal maturity were determined macroscopically upon dissection. The full biometric dataset for each study area is summarised alongside the isotopic results in **Table 3.1** and **Table 3.3** (see Chapter 3).

Following measurement, all individuals were dissected to obtain the biological tissues required for SIA and SCA. The dissection protocol followed standardised procedures developed and routinely applied at the ICM, ensuring methodological consistency and preventing contamination.

The sequence of dissection proceeded as follows:

1. The **dorsal fin** was first excised and stored in labelled plastic bags.
2. A small incision was made above the lateral line, in the posterior third of the body, to extract a sample of dorsal white **muscle**.

3. A ventral incision from anus to mouth was made to expose the abdominal cavity, taking care not to puncture internal organs. **Gonads, liver, and stomach** were separated to identify them. Once the gonads were weighed, sex and the mature stage were identified. While the stomach was preserved entirely, for the liver and gonads only a slice was needed.
4. Both **eyes** were removed and stored separately in plastic bags.
5. The **otoliths** were extracted by opening the cranial cavity behind the eyes, removing the brain tissue, and retrieving the sagittae using fine forceps.
6. **Vertebral column** was removed and cleaned of adhering muscle and connective tissue using distilled water. To ensure anatomical standardisation across all specimens, the first thoracic vertebra was used (Panfili et al., 2002). However, to yield sufficient mass for isotopic analysis, the number of consecutive vertebrae retained was adjusted allometrically: the first two vertebrae were excised for large pelagics (*Thunnus thynnus* and *Mola mola*), the first six for medium-sized species (*Scomber* spp.). and the first ten for small pelagic fish (*Sardina pilchardus*, *Engraulis encrasicolus*). Of this pooled sample, the anterior half was designated for processing, while the posterior section was archived as a replicate.

Following dissection, white muscle samples were immediately oven-dried at 50°C, whereas all other tissues were individually labelled and preserved frozen at -20°C to prevent degradation until subsequent processing. Each sample was catalogued in a digital database, linking tissue type, specimen ID, species, and collection metadata.

2.2.2 Stomach content processing and laboratory identification

Stomach content processing was performed on all individuals collected in both study areas. Laboratory procedures followed standard protocols (Hyslop, 1980), adapted from the internal guidelines of the Institut de Ciències del Mar (ICM-

CSIC) to accommodate species-specific morphology and sample condition. To prevent cross-contamination, all operations were conducted using sterilised dissection tools and decontaminated work surfaces.

Stomach extraction and fullness assessment

Upon dissection, the digestive tract was ligated at the oesophagus and pylorus to prevent content loss. The stomach was then excised, incised longitudinally, and its contents emptied into a glass Petri dish. Stomachs were visually classified according to their fullness state: empty, containing prey, exhibiting evidence of net feeding (codend ingestion), or regurgitated. To distinguish between true physiological fasting and stress-induced regurgitation, the colouration and distension of the gall bladder were examined following Robb (1992) and Weitkamp & Sturdevant (2008):

- Pale / natural colour: Indicates a meal has been eaten recently.
- Pale green to emerald. green: Indicates a recently eaten meal is currently being digested.
- Dark green or dark blue: Suggests that at least 40 hours have passed since the last meal.

Prey identification and quantification

Prey items were sorted manually under a stereomicroscope and identified to the lowest feasible taxonomic level. Identification followed the COCAD (IEO-Cádiz) guide, supplemented by regional ichthyological and invertebrate keys (Lloris, 2015).

To characterise the state of decomposition, prey items were assigned to one of three digestion categories:

1. **Conditioned prey:** Identifiable items showing minimal digestion with morphology largely preserved.
2. **Partially digested prey:** Items with external features moderately degraded but still measurable and taxonomically informative.
3. **Highly digested prey:** Amorphous remains, fragments, or slurry-like material lacking diagnostic features.

For quantification, individuals of the same prey taxon found within a single stomach were pooled to record numerical abundance (N) and aggregate wet weight (W) using a high-precision balance (± 0.001 g). This standardised approach was applied across all predator size classes. Where morphological integrity permitted, the average length of prey items was also measured. Highly digested material that could not be taxonomically resolved was weighed in aggregate as "highly digested material".

2.2.3 Stable isotope sample preparation

Soft tissue processing

Frozen tissue samples (muscle, liver, gonad, dorsal fin) were thawed and subsampled into microcentrifuge tubes. For the shark species (*Prionace glauca*), tissues underwent a specialised urea extraction protocol before drying to prevent bias in $\delta^{15}\text{N}$ values associated with ureosmotic regulation (Vidal et al., 2023). Each muscle subsample was rinsed three times with deionised water to leach soluble urea. Subsequently, all soft tissue samples were dehydrated in a laboratory oven (Memmert ULE 400) at 50°C for 48 hours to eliminate moisture content.

Once dried, samples were homogenised into a fine powder to ensure isotopic uniformity using a high-speed bead mill (Retsch MM 400). Three stainless steel beads were added to each microcentrifuge tube, and samples were agitated at maximum frequency (30 Hz) for 15 minutes. Samples were visually inspected for granularity, and the grinding cycle was repeated if necessary to achieve a consistent texture. Metal beads were retrieved using a magnetic extractor.

Eyes processing

Eye lenses were dissected under a stereomicroscope (Leica M205 C) using a dual approach to differentiate between bulk isotopic signatures and potential chronological records.

- **Lens extraction:** For all specimens, the lens was extracted by incising the cornea, removing the vitreous humor and connective tissue, and rinsing with deionised water.

- **Bulk analysis:** The left lens was dried whole at 50°C for 48 hours and subsequently powdered using the bead mill protocol described above.
- **Chronological delamination:** The right lens was delaminated following standard protocols to separate concentric growth layers (Wallace et al., 2014). These samples were processed and archived for future high-resolution chronological reconstruction studies.

For *P. glauca*, lens samples were rinsed three times with deionised water before drying to remove retained urea, ensuring accurate C:N ratio measurements.

Vertebra processing

Vertebral samples underwent a rigorous multi-step cleaning protocol to isolate bone collagen.

1. **Cleaning and isolation:** Adherent organic matter was mechanically removed using a scalpel and stiff brush. To isolate the vertebral centra, the neural and hemal arches, which may exhibit different remodelling rates, were excised. The isolated centra were then subjected to an ultrasonic bath in deionised water to remove microscopic tissue residues. Sonication duration was standardised to specimen size: 30 seconds at low power for small species and 30 seconds at high power for larger species. Cleaned centra were oven-dried at 50°C for 24 hours and pulverised into a fine, homogeneous powder using an agate mortar and pestle.

2. **Chemical treatment:** Because bone tissue contains both inorganic carbonates (which enrich $\delta^{13}\text{C}$) and lipids (which deplete $\delta^{13}\text{C}$; DeNiro & Epstein, 1977), a dual-aliquot approach was employed to minimise isotopic bias (Guiry et al., 2016). The homogenised powder was divided into two subsamples following Bas & Cardona (2018):

- **Subsample 1** (target: $\delta^{15}\text{N}$): This aliquot was analysed untreated. Chemical demineralisation was avoided for nitrogen analysis as acid treatment can alter collagen $\delta^{15}\text{N}$ values (Bas & Cardona, 2018).
- **Subsample 2** (target: $\delta^{13}\text{C}$): This aliquot underwent a two-step chemical purification to isolate the organic carbon fraction:

2.1 Lipid removal: Lipids were removed using a 2:1 Chloroform: Methanol solution following Folch et al. (1957). The solvent was added to the powder and replaced daily until the solution remained clear, indicating complete lipid removal. Samples were subsequently dried at 50 °C for 24 hours.

2.2 Demineralisation: To remove inorganic carbonates, the lipid-free powder was treated with 0.5 N Hydrochloric Acid (HCl) for 24 hours or until bubbles ceased, following standard protocols for marine vertebrates (Newsome et al., 2006; Bas & Cardona. 2018).

Following treatment, the residue was rinsed with distilled water to neutral pH and dried again at 50 °C for 24 hours. This chemically treated fraction was used for $\delta^{13}\text{C}$ determination.

2.2.4 Isotope analysis and lipid correction

Mass spectrometry

Stable isotope measurements were conducted at the Laboratorio de Isótopos Estables (LIE) at the Estación Biológica de Doñana - CSIC. (Seville, Spain). All prepared samples were analysed using Continuous Flow Isotope Ratio Mass Spectrometry (CF-IRMS) coupled with a Flash High-Temperature Elemental Analyser (Fry, 2006). Samples were combusted at 900 °C to convert organic material into CO_2 and N_2 gases for isotopic determination (Brenna et al., 1997).

Data expression and standards

Isotopic results are expressed in standard δ -notation (parts per thousand ‰) according to the equation:

$$\delta X = \left(\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1 \right) \times 1000$$

where X is ^{13}C or ^{15}N , and R is the corresponding ratio of heavy to light isotopes ($^{13}\text{C}/^{12}\text{C}$ or $^{15}\text{N}/^{14}\text{N}$) (Peterson & Fry, 1987). Isotopic values were normalised relative to international standard scales: Vienna Pee Dee Belemnite (VPDB) for carbon and Atmospheric Nitrogen (AIR) for nitrogen (Coplen, 1994). Analytical precision,

determined by the replicate analysis of internal laboratory standards, was approximately $\pm 0.1\text{‰}$ for $\delta^{13}\text{C}$ and $\pm 0.2\text{‰}$ for $\delta^{15}\text{N}$ (Jardine & Cunjak, 2005).

Lipid correction and quality control

Because lipids are significantly depleted in ^{13}C relative to proteins, variations in tissue lipid content can bias $\delta^{13}\text{C}$ values and can bias the dietary signals (DeNiro & Epstein, 1977). A multi-tiered correction strategy was applied based on tissue-specific biochemical properties and atomic C:N ratios, following the recommendations of et al. (2007) and Logan et al. (2008).

1. **Chemical extraction** (vertebrae): As detailed in Section 2.2.3, vertebral samples, which typically exhibit C:N ratios between 3.0 and 3.3 for pure collagen (Guiry et al., 2020; Kim et al., 2012), underwent direct chemical lipid extraction. This procedure was necessary to ensure accurate $\delta^{13}\text{C}$ measurement from the organic collagen fraction, as mathematical models often fail to account for the inorganic carbonate phase in bone (Guiry et al., 2020). The quality of the chemical extraction was verified by monitoring atomic C:N ratios. Post-treatment values were assessed against the theoretical range for unaltered collagen (2.9 - 3.6) to confirm the successful isolation of the organic fraction (DeNiro, 1985; Guiry et al., 2016).
2. **Mathematical normalisation**: For soft tissues, chemical extraction was avoided to prevent the alteration of ^{15}N signatures (Post et al., 2007). Instead, the atomic C:N ratio was used as a proxy for lipid content. Samples exceeding a threshold of C: N > 3.5 were normalised using tissue-specific models:
 - **Muscle and other tissues**: White muscle typically exhibits C:N ratios between 3.0 and 4.5 (Kiljunen et al., 2006; Logan et al., 2008). For these tissues, as well as gonads, dorsal fins, and eyes, we applied the non-linear model proposed by Kiljunen et al. (2006), which is robust for tissues with low-to-intermediate lipid content. This method first estimates the percentage of lipids (L) based on the C:N ratio:

$$L(\%) = \frac{93}{1 + (0.246 \times C:N - 0.775)^{-1}}$$

The estimated lipid content was then used to normalise the $\delta^{13}\text{C}$ values:

$$\delta^{13}\text{C}_{\text{normalized}} = \delta^{13}\text{C}_{\text{bulk}} + D \left(\frac{L}{100} \right)$$

Where the protein-lipid discrimination factor was set to $D=7.018$ (Kiljunen et al., 2006). Note: While dorsal fins and eye lenses are primarily structural tissues with negligible lipids (typical C:N ~ 2.8 -3.6) (Hussey et al., 2012; Wallace et al., 2014), this correction was applied to any individual outlier exceeding the 3.5 threshold to ensure methodological consistency.

- **Liver:** Hepatic tissue functions as an energy store and often exhibits C:N ratios ranging from 4.0 to 10.0 (Logan et al., 2008; Boecklen et al., 2011). Since general models can perform poorly at high lipid concentrations, liver $\delta^{13}\text{C}$ values were corrected using the tissue-specific logistic model described by Logan et al. (2008).

2.3 DATA ANALYSIS

2.3.1 Stomach content data analysis

Descriptive metrics

To measure diet composition, prey items were grouped into major functional categories, including **Chordata (Actinopterygii)**, **Mollusca (Cephalopoda)**, **Cnidaria (Schyphozoa)** and **Crustacea (Malacostraca)**, to facilitate comparison. The contribution of each prey category i to the diet was quantified using three standard indices (Hyslop, 1980):

- **Percentage number (%N):** The number of individuals of prey type i relative to the total number of identifiable prey items.
- **Percentage weight (%W):** The wet mass of prey type i relative to the total wet mass of stomach contents.
- **Percentage frequency of occurrence (%FO):** The percentage of non-empty stomachs containing at least one individual of prey type i .

To minimise the biases inherent in single metrics, such as the overestimation of small, numerous preys by %N or heavy, rare prey by %W, the compound **Index of relative importance (%IRI)** was calculated (Pinkas et al., 1971). To facilitate comparisons between species, the standardised percentage IRI (%IRI) was computed following Cortés (1997):

$$IRI_i = (\%N_i + \%W_i) \times \%FO_i$$

$$\%IRI_i = \left(\frac{IRI_i}{\sum IRI} \right) \times 100$$

This metric provides a robust, integrated measure of the trophic importance of each prey category i within the predator's diet (Cortés, 1997).

Diet diversity and overlap

Trophic diversity (niche breadth) was estimated using the **Shannon-Wiener diversity index (H')**. Calculated based on prey weight (%W) to emphasise the energetic contribution of different prey taxa (Shannon & Weaver, 1949). To assess the potential for resource competition between species, dietary overlap was quantified using **Schoener's Index (O_{jk})** (Schoener, 1970):

$$O_{jk} = 1 - 0.5 \sum |p_{ij} - p_{ik}|$$

Where p_{ij} and p_{ik} represent the proportional contribution (by weight, %W) of prey category i to the diet of species j and k , respectively (Schoener, 1970). Index values range from 0 (no overlap) to 1 (complete overlap), with values exceeding 0.6 traditionally considered to indicate biologically significant overlap (Wallace, 1981).

2.3.2 Stable isotope data analysis

Statistical framework and software

All statistical analyses were performed within the R computing environment (version 4.5.1) (R Core Team, 2024). Data cleaning and visualisation were conducted using the *dplyr* and *ggplot2* R packages. respectively (Wickham, 2016). Prior to hypothesis testing, the atomic **C:N ratio** was calculated for all samples to identify lipid

content and to validate the efficacy of chemical lipid extraction in bone samples (post-extraction C:N targets: 3.0-3.3).

Normality of isotopic distributions was assessed using the **Shapiro-Wilk test**, and homogeneity of variance was checked using **Levene's test** (Zuur et al., 2010). Data were visualised using **biplots** ($\delta^{13}\text{C}$ vs $\delta^{15}\text{N}$) to identify clustering patterns and **boxplots** to detect outliers. Intra-individual variability was initially explored using **paired-line plots**.

Isotopic niche metrics

To quantify trophic diversity and habitat use, we applied the **isotopic niche** concept (Newsome et al., 2007) using the **SIBER** package (Stable Isotope Bayesian Ellipses in R) (Jackson et al., 2011). Unlike convex hulls (Total Area, TA), which are sensitive to outliers (Layman et al., 2007), SIBER models standard ellipse areas (SEA) to represent the core niche occupied by roughly 40% of the population. This Bayesian framework allows for robust comparisons of community structure even with small sample sizes, addressing a specific methodological gap in Mediterranean multi-tissue studies (Navarro et al., 2020). Two specific metrics were calculated for analysis:

1. **SEAc (Sample-size corrected standard ellipse area)**: This metric was used for descriptive tables and unbiased graphical representation in isotopic biplots, allowing for direct magnitude comparisons of niche area (Jackson et al., 2011).
2. **SEAB (Stable isotope bayesian ellipses area)**: A metric calculated using Markov Chain Monte Carlo (MCMC) simulation (10,000 iterations) to generate a posterior distribution of niche sizes. This metric was used for probabilistic statistical comparisons of niche width between groups.

Isotopic niche overlap between species or tissues was quantified using **maximum likelihood (ML)** estimates as the proportion of the intersection between two SEAc ellipses, providing a metric of trophic segregation (Jackson et al., 2011).

Trophic position calculations

Trophic Position (TP_{SIA}) was estimated for each individual using the standard one-baseline model proposed by Post (2002):

$$TP_{SIA} = \lambda + \frac{(\delta^{15}N_{consumer} - \delta^{15}N_{baseline})}{\Delta^{15}N}$$

where λ represents the trophic level of the baseline organism. $\delta^{15}N_{baseline}$ is the mean nitrogen signature of that baseline, and $\Delta^{15}N$ is the trophic enrichment factor (TEF) (Post, 2002). A TEF of 3.4‰ was applied, which is the widely accepted mean fractionation value for marine consumers (Post, 2002). For the Ebro Delta, baseline $\delta^{15}N$ values were derived from whole-body samples of *Pyrosoma atlanticum* and *Venus nux*. For the La Azohía system, whole-body samples of *Venus nux* were used exclusively as the isotopic baseline.

Inter-tissue comparisons

(testing H2a) To test the hypothesis that metabolically active tissues differ from archive tissues in both isotopic signal and niche width; a multi-step analytical approach was employed:

1. **Multiple comparisons across tissues (location):** To determine if tissues record different dietary information (e.g., recent vs. long-term diet), we tested for significant differences in median $\delta^{13}C$ and $\delta^{15}N$ values. Statistical test selection was determined by the assessment of data distribution (**Normality**) and variance (**Homogeneity**).
 - **Parametric Approach:** When assumptions of normality and homogeneity were met, a one-way Analysis of Variance (**ANOVA**) was performed (Fisher, 1925), followed by **Tukey's HSD** post-hoc tests to identify specific pairwise differences (Tukey, 1949).
 - **Non-Parametric Approach:** For datasets revealing deviations from normality or homoscedasticity, **Kruskal-Wallis** test was applied (Kruskal & Wallis, 1952). Significant results were followed by **Dunn's**

tests (Dunn, 1964) adjusted with the **Holm-Bonferroni correction** (Holm, 1979) to identify specific tissue discrepancies.

2. **Differences in isotopic variability (niche width):** To quantify and compare isotopic variability, we modelled isotopic niches using **SIBER metrics** as previously mentioned. We formally tested for differences in niche width between tissues using a Kruskal-Wallis test on the ellipse area values (SEA_B). If significant, differences were tested using **Dunn's tests** adjusted with the **Holm-Bonferroni correction** to confirm whether metabolically active tissues occupy a significantly broader isotopic niche space than archive tissues.

Testing ontogenetic shifts

(testing $H2a$) To test whether archive tissues record cumulative ontogenetic trophic shifts, we modelled the relationship between body size and $\delta^{15}N$ (or Trophic Position).

1. **Regression modelling:** Linear regressions were fitted for $\delta^{15}N$ against body size (Length) for each tissue type:

$$\delta^{15}N_{tissue} \sim \alpha + \beta \times Size + \epsilon$$

We assessed the slope (β) and coefficient of determination (R^2), which represents the proportion of isotopic variance explained by body size. A stronger correlation (R^2) and steeper positive slope in archive tissues compared to somatic tissues were interpreted as evidence of the preservation of early-life trophic history (Hussey et al., 2014).

2.3.3 SIA-SCA integration and comparison

To evaluate the complementarity of the two dietary methods, trophic estimates derived from stomach contents were explicitly compared with those derived from stable isotopes, following the comparative framework established for Mediterranean food webs by Navarro et al., (2011).

Trophic position from stomach contents (TP_{SCA})

The trophic position of each species was estimated using the fractional trophic level method. This approach integrates the specific diet composition of the predator with the known trophic levels of its prey, following the standard equation for standardised diet analysis (Cortés, 1999):

$$TP_{SCA} = 1 + \sum_{j=1}^n (TL_j \times p_i)$$

where TL_j is the trophic level of prey taxon j . and p_i is the proportion by weight of prey j in the diet (where $\sum p_i = 1$). The addition of 1 account for the trophic step between the predator and its prey (Christensen & Pauly, 1992).

Instead of relying on generic global databases, prey trophic levels (TL_j) were obtained from the functional groups defined in the calibrated Ecopath model for the western Mediterranean (GSA 06 and GSA 07) (Coll et al., 2024). The complete assignment of functional groups and corresponding trophic levels for all identified prey taxa is detailed in **Appendix A (Tables A.1 and A.2)**. For example, prey identified as "Phytoplankton" were assigned $TL=1$, and "Micro- and mesozooplankton" were assigned $TL=2.02$, ensuring that TP_{SCA} estimates reflect the specific energy flow structure of the study region.

Methodological cross-validation

The agreement between stable isotope analysis (SIA) and stomach content analysis (SCA) was assessed using **Spearman's rank correlation** between mean TP_{SCA} and mean TP_{SIA} values across species (Zar, 2010). To visualise discrepancies, species were plotted relative to a 1:1 unity line ($TP_{SIA}=TP_{SCA}$). Deviations from this line were interpreted in the context of temporal integration windows:

- **SCA > SIA:** Suggests recent feeding on higher-trophic prey not yet assimilated into tissues (Hyslop, 1980) or TL from the Ecopath model representing larger sizes of prey than what has been eaten.

- **SCA < SIA:** Suggests assimilation of soft-bodied, high-trophic prey (e.g., gelatinous zooplankton) that are rapidly digested and undetected in SCA or feeding on lower-trophic prey during the sampling event compared to the long-term average (Hussey et al., 2012).

Tissue-specific correlations

To explicitly test whether metabolically active tissues better reflect recent diet, we compared the strength of the correlation, for example, between TP_{SCA} and Liver (TP_{Liver}) versus TP_{SCA} and Muscle (TP_{Muscle}). A stronger correlation with the fast-turnover tissue (liver) is expected if the population has recently shifted diet, supporting the utility of multi-tissue sampling for resolving temporal feeding dynamics (Nielsen et al., 2017).

Chapter 3: Results

3.1 DATA QUALITY AND STATISTICAL ASSUMPTIONS

3.1.1 Lipid content and extraction efficiency

The atomic C:N ratios were assessed in all tissues used in all objectives to determine the need for lipid normalisation. In the **Ebro Delta (Case study 1)**, pelagic muscle tissues exhibited generally high lipid content (mean C:N = 3.95 ± 0.40), with **98.9%** of individuals exceeding the 3.5 threshold. Consequently, mathematical correction was applied to nearly the entire dataset to standardize $\delta^{13}\text{C}$ values.

In **La Azohía (Case study 2)**, C:N ratios varied widely among tissues. Values ranged from a mean of 3.89 ± 0.44 in muscle to 8.22 ± 5.17 in liver. Tissues exceeding the threshold, including 100% of liver samples and >90% of gonads and dorsal fins, were mathematically corrected using the tissue-specific models described in the methodology.

For bone samples (vertebrae), the chemical lipid extraction successfully reduced C:N ratios to a mean of 3.38 ± 0.17 , falling within the theoretical target range. This may confirm the efficacy of the chemical treatment in removing inorganic carbonates and lipids.

3.1.2 Normality and homoscedasticity

Statistical assumptions were rigorously tested for both datasets prior to hypothesis testing (**Appendix B**). Shapiro-Wilk tests revealed significant deviations from normality in several key groups, necessitating a statistical framework adapted to each objective.

- **Case study 1 (Ebro):** The assumption of normality for $\delta^{15}\text{N}$ was met for the majority of species (11 out of 15). Significant deviations were detected primarily in groups with the largest sample sizes (*Scomber colias*, n=24; *Scomber scombrus*, n=29; *Thunnus thynnus*, n=12). However, Levene's test

revealed significant heterogeneity of variance among species ($p=0.002$). Given this heteroscedasticity, comparisons of trophic position ($\delta^{15}\text{N}$) were conducted using the non-parametric **Kruskal-Wallis test**, followed by **Dunn's post-hoc test** with **Holm-Bonferroni** correction. In contrast, $\delta^{13}\text{C}$ values satisfied normality assumptions (12/15 species passed) and exhibited homogeneity of variances (Levene's test. $p=0.07$). Therefore, spatial and inter-specific comparisons of carbon signatures were performed using One-way ANOVA, followed by **Tukey's HSD post-hoc tests**.

- **Case study 2 (Azohía):** Normality and homoscedasticity varied between the nine analysed species. For $\delta^{15}\text{N}$, normality was confirmed in the majority of species (7/9), and variances remained homogeneous across tissue types ($p>0.05$). Tissue-specific differences in nitrogen signatures were assessed using One-way ANOVA and **Tukey's HSD**. Conversely, corrected $\delta^{13}\text{C}$ values showed deviations from normality. The test failed for one or more tissue types in 7/9 species ($p<0.001$). Although the assumption of homogeneity of variance was satisfied, the violation of normality meant the non-parametric **Kruskal-Wallis** and **Dunn's post-hoc tests (Holm-Bonferroni corrected)** were used for tissue-based carbon comparisons.

3.2 CASE STUDY 1: THE EBRO DELTA PELAGIC COMMUNITY

A summary of the biometric data, maturity status, and mean isotopic signatures ($\delta^{13}\text{C}$, $\delta^{13}\text{C}_{\text{corr}}$, $\delta^{15}\text{N}$) for the 16 marine species and the isotopic baselines collected in the Ebro Delta are presented in **Table 3.2**.

Table 3.2. Summary of biological and isotopic metrics for the pelagic fish community of the **Ebro Delta** (Case study 1). Data are grouped by functional category and include sample size (n), biometric measurements (body length (BL) and weight), maturity status, mean ($\pm$ standard deviation), isotopic signatures ($\delta^{13}\text{C}$, lipid-corrected $\delta^{13}\text{C}_{\text{corr}}$, $\delta^{15}\text{N}$) and Trophic Position (TP_{SIA}) estimated from stable isotopes from muscle tissue.

Species	n	BL ± SD ¹	Weight ± SD	Maturity	Tissue	δ ¹³ C ± SD	δ ¹³ C _{corr} ± SD ²	δ ¹⁵ N ± SD	TP _{SIA} ± SD
Small pelagic									
<i>Boops boops</i>	5	17.44 ± 3.33	56.05 ± 26.68	Juveniles (2), Adults (3)	Muscle	-19.21 ± 0.29	-17.68 ± 0.33	9.85 ± 0.28	3.09 ± 0.08
<i>Cheilopogon exsiliens</i>	4	17.03 ± 0.81	37.42 ± 6.06	Juveniles (4)	Muscle	-18.34 ± 0.15	-17.00 ± 0.14	5.97 ± 0.51	1,83 ± 0.15
<i>Engraulis encrasicolus</i>	5	13.20 ± 0.64	15.38 ± 2.38	Adults (5)	Muscle	-18.91 ± 0.31	-17.48 ± 0.28	8.75 ± 0.30	2,70 ± 0.09
<i>Sardina pilchardus</i>	5	13.88 ± 0.55	22.60 ± 3.74	Juveniles (3), Adults (2)	Muscle	-18.70 ± 1.17	-17.02 ± 1.14	9.27 ± 0.40	2,96 ± 0.12
Medium pelagic									
<i>Auxis rochei</i>	8	180.25 ± 48.56	18.03 ± 4.86	Juveniles	Muscle	-18.11 ± 0.27	-16.84 ± 0.21	6.81 ± 0.68	2,58 ± 0.20
<i>Sarda sarda</i>	7	110.14 ± 7.43	11.01 ± 0.74	Juveniles (2), Adults (5)	Muscle	-18.47 ± 0.26	-16.89 ± 0.09	6.68 ± 0.95	2.16 ± 0.28
<i>Scomber colias</i>	24	263.50 ± 39.95	26.35 ± 4.00	Juveniles (2), Adults (22)	Muscle	-18.72 ± 0.26	-17.32 ± 0.23	9.39 ± 0.48	2.96 ± 0.14
<i>Scomber scombrus</i>	29	181.72 ± 39.73	18.17 ± 3.97	Juveniles (25), Adults (4)	Muscle	-19.02 ± 0.29	-17.55 ± 0.27	9.36 ± 0.85	2,89 ± 0.25
<i>Seriola dumerili</i>	5	101.20 ± 39.01	10.12 ± 3.90	Juveniles (4), Adults (1)	Muscle	-18.34 ± 0.25	-16.84 ± 0.23	6.99 ± 0.63	2,41± 0.19
<i>Trachurus trachurus</i>	29	108.62 ± 21.68	10.86 ± 2.17	Juveniles	Muscle	-19.99 ± 0.41	-18.28 ± 0.31	9.52 ± 0.32	3.00 ± 0.10
Large pelagic									

Species	n	BL $\pm$ SD ¹	Weight $\pm$ SD	Maturity	Tissue	$\delta^{13}\text{C} \pm$ SD	$\delta^{13}\text{C}_{\text{corr}} \pm$ SD ²	$\delta^{15}\text{N} \pm$ SD	TP _{SIA} $\pm$ SD
<i>Thunnus thynnus</i>	12	28.51 $\pm$ 35.20	4249.07 $\pm$ 14407.85	Juveniles	Muscle	-18.30 $\pm$ 0.27	-16.93 $\pm$ 0.14	7.09 $\pm$ 1.19	2.28 $\pm$ 0.35
<i>Tylosurus acus</i>	1	143.00	1750.00	Juveniles	Muscle	-17.57	-16.20	9.61	3.02
<i>Xiphias gladius</i>	3	36.95 $\pm$ 35.99	502.99	Juveniles	Muscle	-17.82 $\pm$ 0.45	-16.07 $\pm$ 0.60	8.78 $\pm$ 1.28	2.49 $\pm$ 0.38
Other									
<i>Ariosoma balearicum</i>	2	21.65 $\pm$ 4.03	16.85 $\pm$ 8.36	Juveniles	Muscle	-16.84 $\pm$ 0.04	-15.15 $\pm$ 0.07	10.88 $\pm$ 1.21	3.40 $\pm$ 0.36
<i>Illex coindetii</i>	25	8.67 $\pm$ 1.45	18.68 $\pm$ 7.36	Juveniles	Muscle	-19.23 $\pm$ 0.20	-17.25 $\pm$ 0.13	8.92 $\pm$ 0.25	2.82 $\pm$ 0.07
<i>Remora remora</i>	1	12.50	11.63	Adults	Muscle	-17.28	-15.60	12.56	3.89
Baseline									
<i>Pyrosoma atlanticum</i>	6	48.67 $\pm$ 18.59	- -	-	Whole body	-22.45 $\pm$ 0.41	-19.18 $\pm$ 0.40	5.72 $\pm$ 0.87	1.88 $\pm$ 0.25
<i>Venus nux</i>	6	34.88 $\pm$ 5.30	- -	-	Whole body	-20.16 $\pm$ 0.28	-17.63 $\pm$ 0.32	6.54 $\pm$ 0.49	2.12 $\pm$ 0.15

Note: ¹Body length (BL) represents the specific morphometric standard for each species (TL, FL, LJFL, or ML) as detailed in **Section 2.2.1**. ² $\delta^{13}\text{C}_{\text{corr}}$ refers to lipid-corrected carbon isotope values calculated following the equation by Kiljunen et al., 2006. All isotopic values are reported in per mil (‰).

3.2.1 Isotopic composition and segregation

The isotopic biplot ($\delta^{13}\text{C}$ vs. $\delta^{15}\text{N}$) revealed that species were distributed across a broad isotopic space, spanning approximately 4‰ in $\delta^{13}\text{C}$ and 7‰ in $\delta^{15}\text{N}$ (**Figure 3.2.1**).

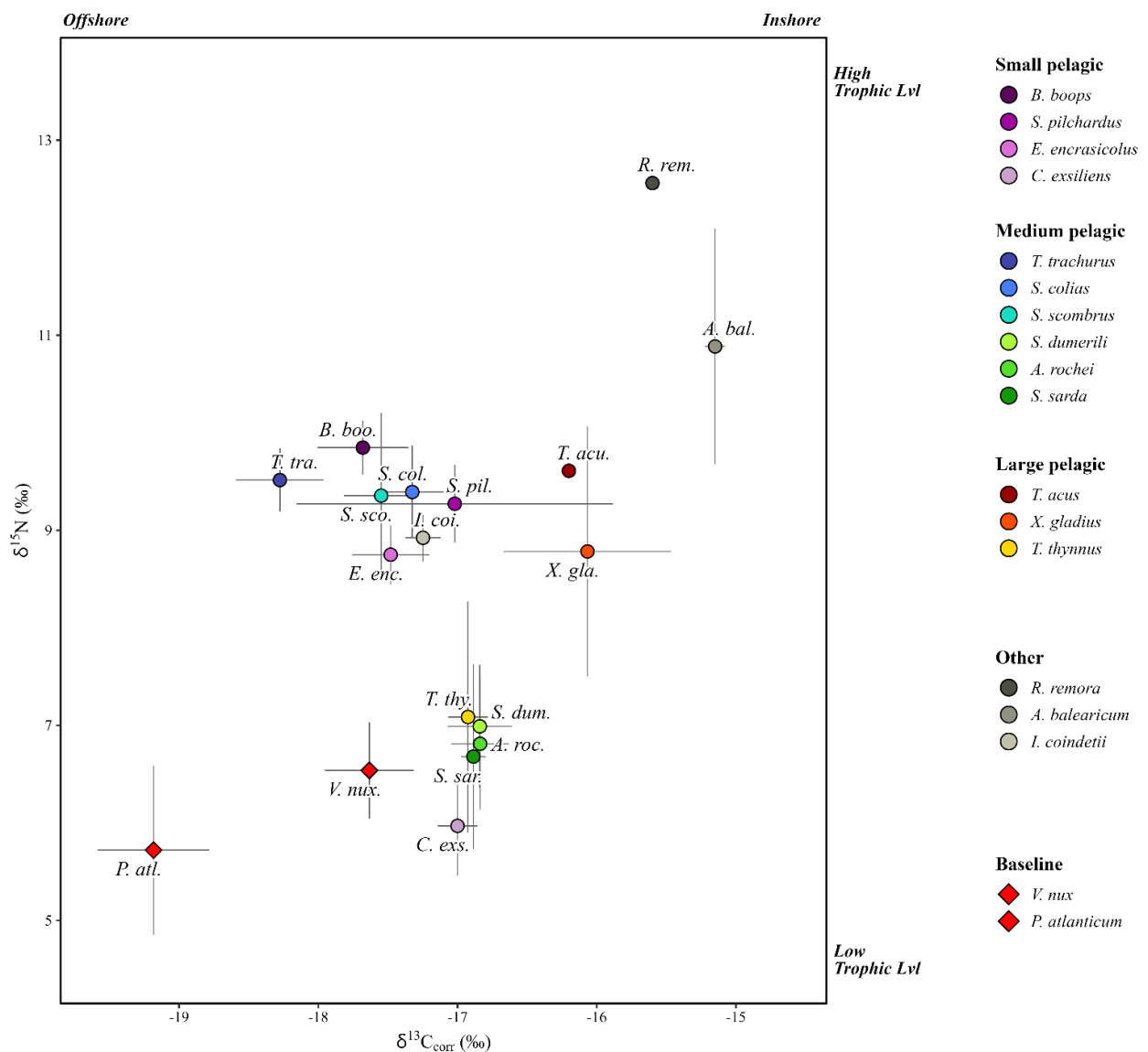
The pelagic tunicate *Pyrosoma atlanticum*, collected from deep offshore waters (339-575 m), displayed the most depleted C values ($\delta^{13}\text{C} \approx -19.2\text{‰}$), indicative of an offshore phytoplankton signal. In contrast, the bivalve *Venus nux*, sampled from the shallower continental shelf (36-77 m), exhibited enriched $\delta^{13}\text{C}$ values ($\delta^{13}\text{C} \approx -17.6\text{‰}$), reflecting a benthic coastal signature.

Significant inter-specific differences were detected for both carbon (**ANOVA**, $F_{17,143}=17.64$, $p<0.001$) and nitrogen signatures (**Kruskal-Wallis**, $\chi^2=135.5$, $p<0.001$) (**Appendix C**), confirming that the trophic segregation among the sampled species within the community. However, the structuring did not strictly follow size-based functional classifications:

- **Small pelagic fish** (*S. pilchardus*, *E. encrasicolus*, *B. boops*) formed a cohesive trophic group ($\delta^{15}\text{N}$ range: 8.8‰-9.9‰). Pairwise comparisons confirmed the high trophic overlap, with no significant differences in $\delta^{15}\text{N}$ detected among these species ($p>0.05$). Carbon signatures were also similar, although *Sardina pilchardus* was significantly enriched in $\delta^{13}\text{C}$ compared to *Boops boops* ($p=0.04$).
- **Medium pelagic fish** displayed wider dispersion, splitting into two distinct isotopic groups. *Trachurus trachurus* ($\delta^{15}\text{N} \approx 9.5\text{‰}$) and *Scomber* spp. ($\delta^{15}\text{N} \approx 9.4\text{‰}$) occupied trophic positions similar to those from the small pelagics ($p>0.05$). In contrast, *Auxis rochei* ($\delta^{15}\text{N} \approx 6.8\text{‰}$), *Sarda sarda* ($\delta^{15}\text{N} \approx 6.7\text{‰}$), and *Seriola dumerili* ($\delta^{15}\text{N} \approx 7.0\text{‰}$) exhibited significantly lower nitrogen values than *Scomber* spp. and *Trachurus trachurus* ($p<0.01$), overlapping with the lower range of the sampled food web.
- **Large pelagic fish** showed the most pronounced heterogeneity. *Remora remora* ($n = 1$) exhibited the highest recorded nitrogen signature ($\delta^{15}\text{N} \approx 12.6\text{‰}$). In contrast, *Thunnus thynnus* exhibited low nitrogen signatures ($\delta^{15}\text{N} \approx 7.1\text{‰}$) (due to the samples corresponded to juveniles), significantly lower

than medium predators like *Trachurus trachurus* ($p < 0.001$). *Xiphias gladius* occupied an intermediate trophic position ($\delta^{15}\text{N} \approx 8.8\text{‰}$), but was distinguished by its carbon signature, being significantly enriched in $\delta^{13}\text{C}$ compared to *Thunnus thynnus* ($p < 0.01$).

Figure 3.2.1 Case study 1 - Isotopic biplot of the Ebro Delta pelagic community. Scatterplot of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ mean values ($\pm$ SD) for small, medium, and large pelagic species (muscle tissue). Species are colour-coded according to their functional group shown as circles. Baseline organisms (*P. atlanticum* and *V. nux*) are shown as red diamonds.

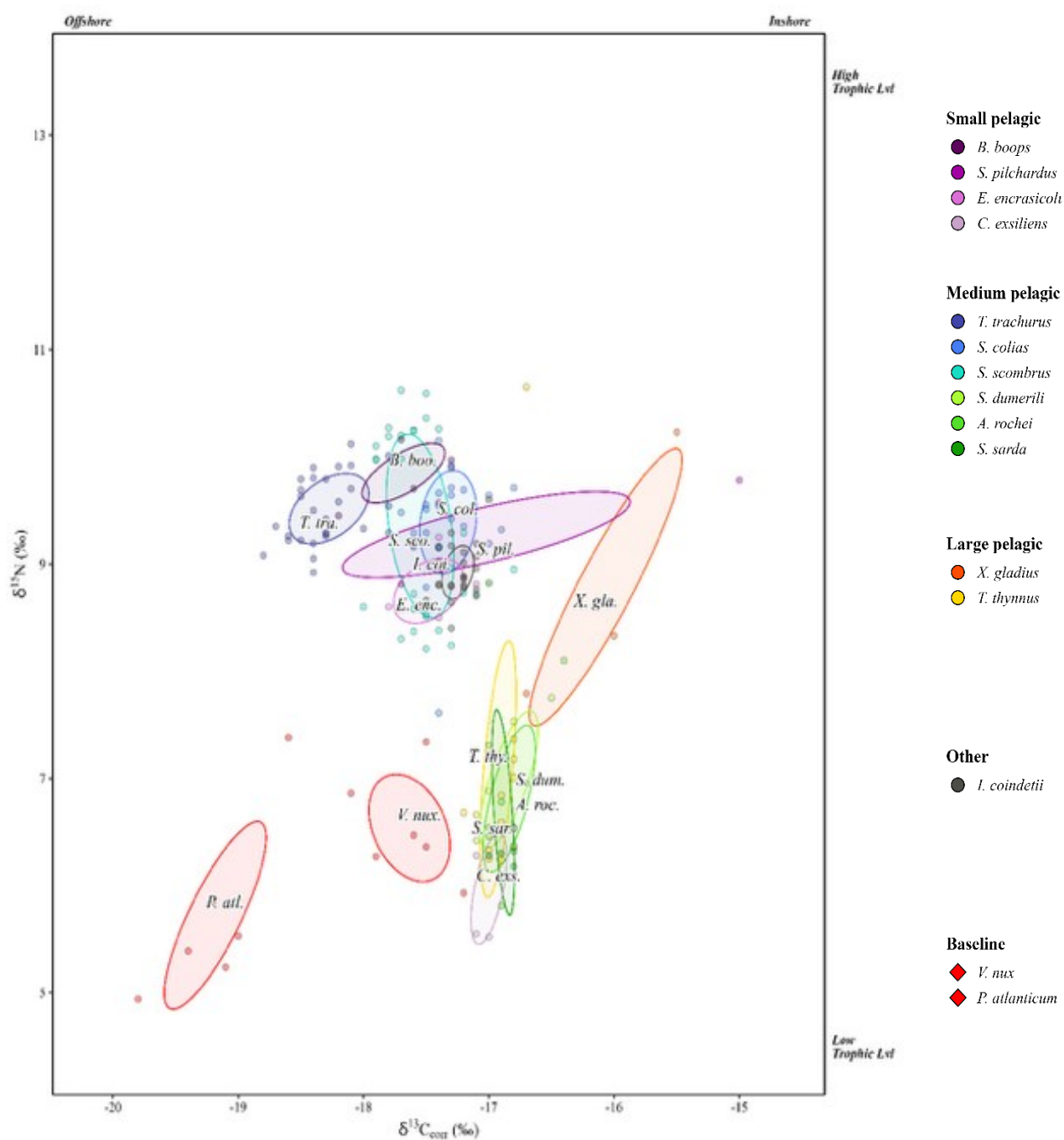


3.2.2 Isotopic niche width and community packing

To quantify the trophic diversity and degree of overlap among species, Standard ellipse areas corrected for small sample sizes (SEA_C) were calculated (**Figure 3.2.2; see Appendix Table C.2 for full metrics**). Species with insufficient sample sizes ($n < 3$; e.g., *Remora remora*, *Tylosurus acus*) were excluded from this analysis following Jackson's et al. (2011) rationale.

The SIBER analysis revealed substantial variation in isotopic niche breadth (SEA_C) among functional groups. *Xiphias gladius* exhibited the broadest isotopic niche ($SEA_C = 1.92 \text{ ‰}^2$), spanning the widest range of $\delta^{13}\text{C}$ values in the dataset. *Sardina pilchardus* displayed the second-largest niche ($SEA_C = 1.23 \text{ ‰}^2$), characterised by a wide dispersion along the $\delta^{13}\text{C}$ axis relative to other small pelagic fish. In contrast, the lowest niche width values were observed in *Illex coindetii* ($SEA_C = 0.10 \text{ ‰}^2$) and *Engraulis encrasicolus* ($SEA_C = 0.32 \text{ ‰}^2$), which occupied highly constrained areas within the biplot.

Figure 3.2.2. Case study 1 - Ebro Delta pelagic community. Isotopic biplot ($\delta^{13}\text{C}$ - $\delta^{15}\text{N}$) characterising the Ebro Delta pelagic community. Scatter points represent individual samples, coloured according to their functional group (small, medium, large pelagics and other). Standard ellipse area (SEA_C) representing the 40% core isotopic niche, are displayed only for species meeting the minimum sample size requirement ($n \geq 3$). *P. atlanticum* and *Venus nux* are plotted as isotopic baseline.



Niche segregation and overlap by functional group

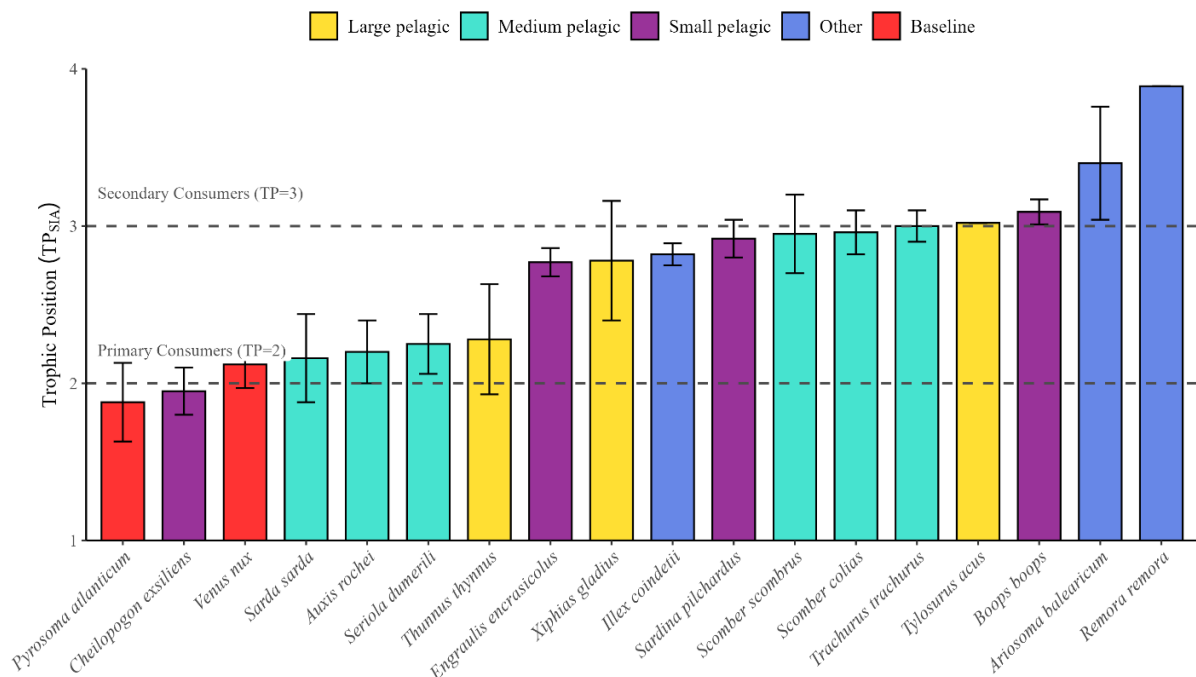
- **Small pelagic fish:** While this group occupied a broadly similar isotopic space, internal differences in ellipse geometry were evident. *Sardina pilchardus* ellipse was markedly larger than that of *Engraulis encrasicolus* ($SEAC$: 1.23 vs 0.32 ‰²), primarily due to greater variance in $\delta^{13}C$ values compared to the more clustered anchovy samples.
- **Medium pelagic fish:** *Scomber colias* ($SEAC = 0.35$ ‰²) and *S. scombrus* ($SEAC = 0.69$ ‰²) showed extensive overlap with the small pelagic cluster (*Sardina/Engraulis*). Conversely, *Seriola dumerili*, *Auxis rochei* and *Sarda sarda* occupied a distinct isotopic position with limited overlap, characterised by lower $\delta^{15}N$ values relative to the small pelagic fish and moderate niche widths ($SEAC$ range: 0.26 - 0.42 ‰²).
- **Large pelagic fish:** *Xiphias gladius* was segregated from other large predators by its enriched carbon signature and large ellipse area ($SEAC = 1.92$ ‰²). However, *Thunnus thynnus* occupied a smaller niche ($SEAC = 0.47$ ‰²) that overlapped extensively with the lower-trophic medium pelagics (*A. rochei*, *S. sarda*) along the $\delta^{15}N$ axis, reflecting the lower mean trophic position of the sampled individuals (juveniles) compared to *Xiphias gladius* or *Remora remora*.

3.2.3 Estimated trophic positions (TP_{SIA})

Estimates of trophic position (TP_{SIA}) were calculated relative to the isotopic baselines, which exhibited mean values of 1.88 ± 0.25 for *Pyrosoma atlanticum* and

2.12 ± 0.15 for *Venus nux*. The mean trophic positions for the pelagic community are summarised in **Figure 3.2.3** and detailed in **Table 3.2** (see above).

Figure 3.2.3. Mean estimated trophic positions ($TP_{SIA} \pm SD$) of the pelagic community in the Ebro Delta calculated with SIA. Species are arranged in ascending order of trophic position. Dashed horizontal lines represent theoretical trophic levels (TP=2.0: Primary consumers; TP=3.0: Secondary consumers). Colors indicate functional groups. Note that error bars are absent for *Remora remora* and *Tylosurus acus* due to single-specimen samples ($n=1$).



The sampled food web spanned approximately **two trophic levels**, ranging from a minimum of 1.83 in *Cheilopogon exsiliens* to a maximum of 3.89 in *Remora remora*. As illustrated in **Figure 3.2.3**, the community structure revealed that large pelagic species, represented by juvenile individuals (e.g. *T. thynnus*, mean BL = 28.5 ± 35.2 cm; *X. gladius*, mean BL = 37.0 ± 36.0 cm), occupied trophic positions (TP_{SIA} range: 2.16–2.78) that were lower than those recorded for adult small pelagic fish, such as *B. boops* (mean BL = 17.4 ± 3.3 cm, $TP_{SIA} = 3.09$) and *S. pilchardus* (mean BL = 13.88 ± 0.55 cm, $TP_{SIA} = 2.96$).

High trophic level species

The highest trophic position was recorded for the commensal species *Remora remora* ($TP_{SIA} = 3.89$, $n = 1$), although this value represents a single individual ($n = 1$). Among the schooling and demersal populations, the highest trophic roles were occupied by the benthic-pelagic feeder *Ariosoma balearicum* ($TP_{SIA} = 3.40 \pm 0.36$, $n = 2$) and the opportunistic small pelagic *Boops boops* ($TP_{SIA} = 3.09 \pm 0.08$). The medium pelagic *Trachurus trachurus* also occupied a comparable trophic level ($TP_{SIA} = 3.00 \pm 0.10$), exhibiting a narrow trophic niche similar to the high-trophic small pelagics.

Juvenile large predators and medium pelagics

Thunnus thynnus ($TP_{SIA} = 2.28 \pm 0.35$), *Auxis rochei* ($TP_{SIA} = 2.20 \pm 0.20$) and *Sarda sarda* ($TP_{SIA} = 2.16 \pm 0.28$) occupied positions only slightly above the baseline level ($TP_{SIA} \approx 2.0$). Even *Xiphias gladius*, typically considered a top predator, occupied an intermediate position ($TP_{SIA} = 2.78 \pm 0.38$, $n = 3$), lower than that of the small pelagic *Sardina pilchardus* ($TP_{SIA} = 2.96 \pm 0.12$).

Small pelagic fish and planktivores

The small pelagic group generally occupied robust secondary consumer roles ($TP_{SIA} \approx 2.8-3.0$), often exceeding the trophic positions of the large predators. *Sardina pilchardus* ($TP_{SIA} = 2.96 \pm 0.12$) and *Engraulis encrasicolus* ($TP_{SIA} = 2.70 \pm 0.09$) occupied intermediate trophic positions within the food web. Notably, *Cheilopogon exsiliens* presented the lowest value in the entire ecosystem ($TP_{SIA} = 1.83 \pm 0.15$), exhibiting values comparable to the primary consumer baselines.

3.2.4 Diet composition (SCA)

Stomach content analysis (SCA) was performed on all non-empty stomachs ($n_{\text{total}}=78$). The specific composition of prey items, including percentage by number (%N), weight (%W), frequency of occurrence (%F), and the index of relative importance (%IRI), is detailed in **Table 3.2.4**. To visualise broad trophic patterns, the relative frequency of occurrence (%F) of major prey taxa is presented ordering organisms by functional group in **Figure 3.2.4**.

Figure 3.2.4. Relative frequency of occurrence (%F) of major prey taxa in marine species from the Ebro Delta (Case study 1). Predator species are grouped by functional category (small, medium, large pelagic and other), with sample sizes of full stomachs shown in parentheses (n). All values represent pooled stomach contents at the species level. Detailed composition is provided in **Table 3.2.4**.

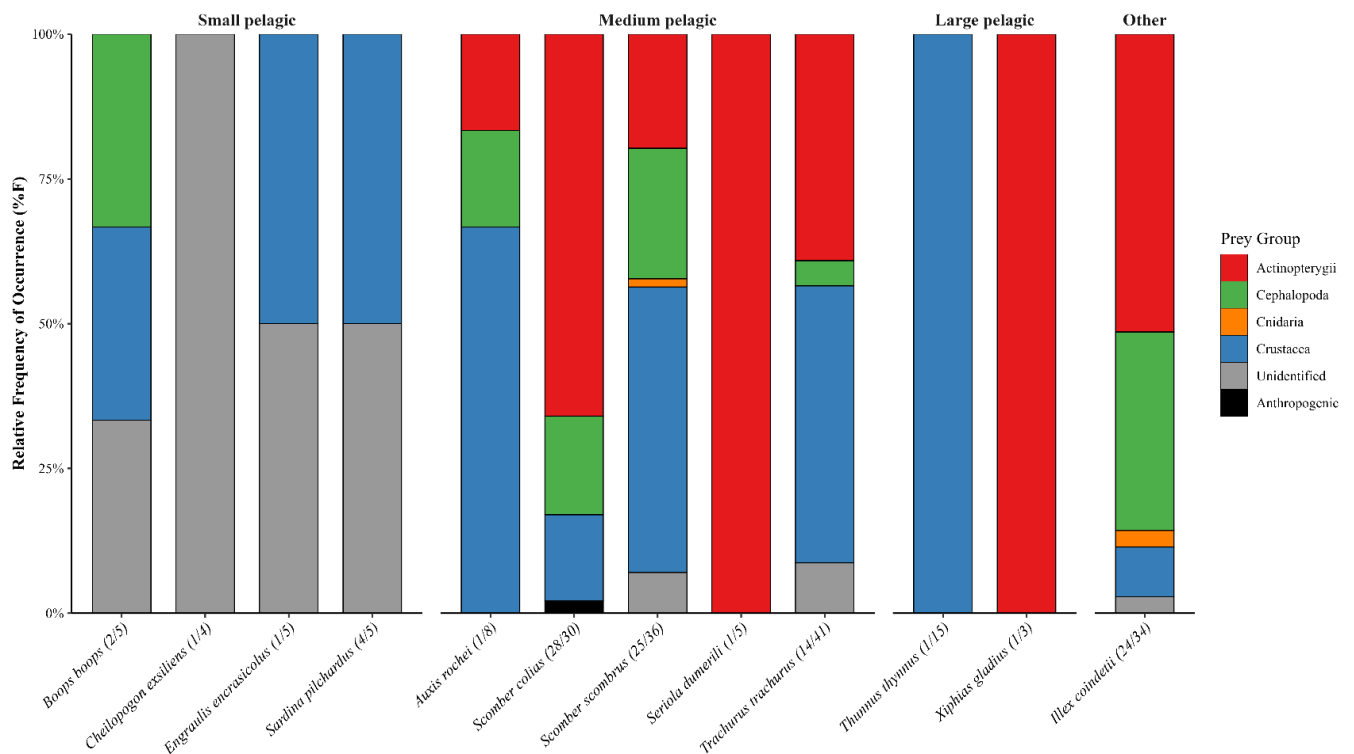


Table 3.2.4 Diet composition and stomach content analysis of the sampled species collected in the Delta del Ebro (Case study 1). Predators are grouped by functional category; parentheses (n/tot) represent the number of non-empty stomachs relative to the total number of individuals examined for each species. Diet metrics include percentage by number (%N), weight (%W), frequency of occurrence (%F), and the index of relative importance (%IRI).

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
Small pelagic									
<i>Boops boops</i> (2/5)	Crustacea					33.33	67.67	20	50.5
	MOLLUSCA	Cephalopoda				55.56	21.55	20	38.55
	Other invertebrate					11.11	10.78	20	10.94
<i>Cheilopogon exsiliens</i> (1/4)	Unidentified					100	100	33.33	100
<i>Engraulis encrasicolus</i> (1/5)	Crustacea					28.57	40	20	34.29
	Other Invertebrate					71.43	60	20	65.71
<i>Sardina pilchardus</i> (4/5)	Crustacea	Malacostraca	Decapoda			14.29	14.52	20	9.82
	Crustacea					42.86	21.77	20	22.03
	TOTAL Crustacea					57.14	36.29	40	63.68
	Other invertebrate					14.29	18.55	20	11.19
	Unidentified					28.57	45.16	20	25.13
Medium pelagic									
<i>Auxis rochei</i> (1/8)	CHORDATA	Actinopterygii				51.43	71.64	12.5	61.53
	Crustacea	Malacostraca	Amphipoda	Hyperiididae		2.86	0.21	12.5	1.53
	Crustacea	Malacostraca	Brachyura (Infraorder)			2.86	0.21	12.5	1.53

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
	Crustacea	Malacostraca	Euphausiacea			5.71	0.84	12.5	3.28
	Crustacea					8.57	7.56	12.5	8.07
	TOTAL Crustacea					20	8.82	12.5	14.41
	MOLLUSCA	Cephalopoda				28.57	19.54	12.5	24.05
<i>Scomber colias</i> (28/30)	CHORDATA	Actinopterygii	Clupeiformes	Engraulidae	<i>Engraulis encrasicolus</i>	12.3	70.53	36.67	23.08
	CHORDATA	Actinopterygii				36.07	22.93	66.67	29.89
	TOTAL CHORDATA					48.36	93.46	83.33	89.82
	Crustacea	Malacostraca	Amphipoda	Hyperiididae	<i>Phronima sedentaria</i>	3.28	0.28	6.67	0.18
	Crustacea	Malacostraca	Amphipoda	Hyperiididae		9.84	0.1	6.67	0.5
	Crustacea	Malacostraca	Brachyura (Infraorder)			1.64	0	3.33	0.04
	Crustacea	Malacostraca	Decapoda			2.46	0.17	3.33	0.07
	Crustacea	Malacostraca	Isopoda			2.46	0.96	3.33	0.09
	TOTAL Crustacea					19.67	1.51	16.67	2.68
	MOLLUSCA	Cephalopoda	Teuthida	Loliginidae	Alloteuthis	4.1	1.06	10	0.39
	MOLLUSCA	Cephalopoda	Teuthida	Ommastrephidae	<i>Illex coindetii</i>	4.1	3.64	6.67	0.39
	MOLLUSCA	Cephalopoda				23.77	0.34	10	1.83
	TOTAL MOLLUSCA					31.97	5.04	26.67	7.5
	Plastic					0	0	3.33	0
	CHORDATA	Actinopterygii	Clupeiformes	Engraulidae	<i>Engraulis encrasicolu</i>	0.6	51.06	2.78	2.02

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
<i>Scomber scombrus</i> (25/36)	CHORDATA	Actinopterygii	Gadiformes	Gadidae	<i>Micromesistius poutassou</i>	0.6	3.91	2.78	0.18
	CHORDATA	Actinopterygii				24.1	9.76	33.33	15.89
	TOTAL CHORDATA					25.3	64.73	38.89	49.31
	CNIDARIA	Scyphozoa				0.6	0.12	2.78	0.03
	Crustacea	Malacostraca	Amphipoda	Hyperiididae	<i>Phronima sedentaria</i>	4.22	0.09	5.56	0.34
	Crustacea	Malacostraca	Amphipoda	Caprellidae		1.2	0.1	5.56	0.1
	Crustacea	Malacostraca	Amphipoda	Hyperiididae		4.22	1.14	13.89	1.05
	Crustacea	Malacostraca	Amphipoda			1.81	0.08	5.56	0.15
	Crustacea	Malacostraca	Brachyura (Infraorder)			6.63	0.75	13.89	1.44
	Crustacea	Malacostraca	Decapoda	Galatheididae		2.41	0.22	2.78	0.1
	Crustacea	Malacostraca	Decapoda	Sergestidae		1.2	0.15	2.78	0.05
	Crustacea	Malacostraca	Decapoda			7.83	1.33	13.89	1.79
	Crustacea	Malacostraca	Stomatopoda	Squillidae		0.6	0.05	2.78	0.03
	Crustacea	Malacostraca	Euphausiacea			3.61	0.14	11.11	0.59
	Crustacea	Malacostraca	Isopoda			0.6	0.2	2.78	0.03
	Crustacea					12.05	4.07	16.67	3.78
	TOTAL Crustacea					46.39	8.34	30.56	23.55
	MOLLUSCA	Cephalopoda	Teuthida	Loliginidae	Alloteuthis.	1.81	18.63	8.33	2.4
	MOLLUSCA	Cephalopoda	Thecosomata	Cymbuliidae	<i>Cymbulia peronii</i>	0.6	0.04	2.78	0.03
	MOLLUSCA	Cephalopoda				19.88	5.74	33.33	12.03

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
	TOTAL MOLLUSCA					22.29	24.41	38.89	25.58
	Unidentified					5.42	2.4	13.89	1.53
<i>Seriola dumerili</i> (1/5)	CHORDATA	Actinopterygii				100	100	20	100
<i>Trachurus trachurus</i> (14/41)	CHORDATA	Actinopterygii				56.14	14.06	21.95	58.29
	Crustacea	Malacostraca	Amphipoda	Hyperiididae		1.75	0.09	2.44	0.17
	Crustacea	Malacostraca	Anomura (Infraorder)			1.75	0.57	2.44	0.21
	Crustacea	Malacostraca	Brachyura (Infraorder)			3.51	1.14	2.44	0.43
	Crustacea	Malacostraca	Decapoda	Portunidae	Liocarcinus	1.75	0.76	2.44	0.23
	Crustacea	Malacostraca	Decapoda	Caridea		1.75	0.57	2.44	0.21
	Crustacea	Malacostraca	Decapoda			7.02	5.79	4.88	2.36
	Crustacea					21.05	29.82	9.76	18.77
	TOTAL Crustacea					38.6	38.74	12.2	35.68
	MOLLUSCA	Cephalopoda	Teuthida	Loliginidae	Alloteuthis	1.75	37.79	2.44	3.65
	Unidentified					3.51	9.4	4.88	2.38
Large pelagic									
<i>Thunnus thynnus</i> (1)	Crustacea	Malacostraca	Amphipoda	Hyperiididae		100	100	6.67	100
<i>Xiphias gladius</i> (1/3)	CHORDATA	Actinopterygii	Clupeiformes	Engraulidae	<i>Engraulis encrasicolus</i>	100	100	33.33	100

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
Other									
<i>Illex coindetii</i> (24/34)	CHORDATA	Actinopterygii				38.89	60.12	52.94	64.08
	CNIDARIA	Scyphozoa				1.85	1.35	2.94	0.12
	Crustacea					9.26	1.53	8.82	1.16
	MOLLUSCA	Cephalopoda				48.15	31.54	35.29	34.38
	Unidentified					1.85	5.47	2.94	0.26

In the following sections, the specific dietary strategies observed for the species within each functional group are detailed, following the order presented in **Figure 3.2.4**.

Small pelagics

The diet of small pelagics was primarily invertebrate-based. *Sardina pilchardus* ($n = 4$) relied heavily on crustaceans (%IRI = 63.7%), particularly copepods and decapods. In the two *Boops boops* analysed, the diet was mixed, composed of crustaceans (%IRI = 50.5%) and cephalopods (%IRI = 38.6%). The single *Engraulis encrasicolus* stomach contained unspecified invertebrates (%IRI = 65.7%) and crustaceans, *Cheilopogon exsiliens* ($n = 1$) contained exclusively unidentified digested material.

Medium pelagics

This group displayed a clear piscivorous trend, though degrees of specialisation varied among species. *Scomber colias* ($n=28$) exhibited a highly specialised diet dominated by bony fishes (osteichthyes) (%IRI = 89.8%), specifically *Engraulis encrasicolus*. In contrast, *Scomber scombrus* ($n=25$) showed a more generalised feeding strategy, consuming fish (%IRI = 49.3%), molluscs (%IRI = 25.6%), and crustaceans (%IRI = 23.6%) in comparable proportions. *Trachurus trachurus* ($n=14$) preyed primarily on fish (%IRI = 58.3%) and crustaceans (%IRI = 35.7%), including decapods and amphipods. The limited samples for *Auxis rochei* ($n=1$) and *Seriola dumerili* ($n=1$) revealed a dominance of piscivory; *S. dumerili* was exclusively piscivorous (%IRI = 100%), while *Auxis rochei* was primarily piscivorous (%IRI = 61.5%) with a secondary consumption of cephalopods (%IRI = 24.1%).

Large pelagics

Sample sizes for this group were small ($n=1$ for all predator species), limiting broad population-level generalisations. The single *Xiphias gladius* stomach contained *Engraulis encrasicolus* (%IRI = 100%). Notably, the stomach of the juvenile *Thunnus thynnus* contained only hyperiid amphipods (%IRI = 100%), reflecting a specific predation event on zooplankton swarms rather than piscivory.

Other

The cephalopod *Illex coindetii* ($n=24$) functioned as an active predator. Stomach contents dominated by fish (%IRI = 64.1%) and other cephalopods (%IRI = 34.4%).

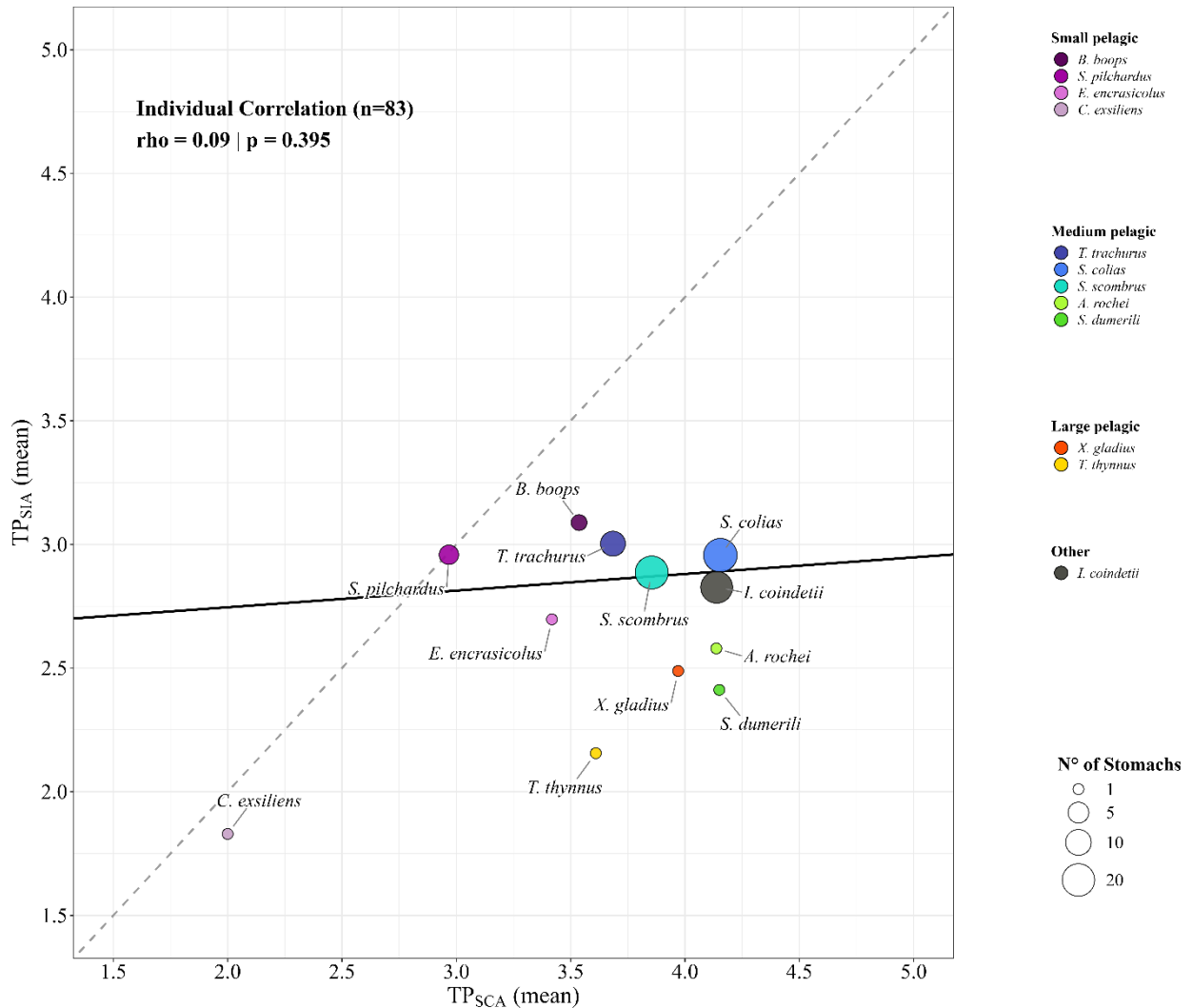
3.2.5 Cross-validation: Isotopes vs. Stomach contents (Testing H1)

To cross-validate trophic estimates, TP_{SIA} values were compared against TP_{SCA} for the Ebro Delta pelagic community. Spearman rank correlation, performed on all paired individuals ($n=83$), revealed a weak but positive relationship between the two methods ($\rho=0.09$, $p=0.395$) (**Figure 3.2.5**).

Visually, most species mean fell below the 1:1 identity line, indicating that TP_{SCA} estimates generally exceeded TP_{SIA} (**Figure 3.2.5**). The highest level of consistency was observed in the small pelagic guild, particularly for *Sardina pilchardus*, which showed nearly identical estimates ($TP_{SIA} = 2.96$ vs. $TP_{SCA} = 2.97$). In contrast, the greatest discrepancies occurred in medium and large pelagic predators (**Appendix C. Table C.3**). For instance, *Seriola dumerili* and *Auxis rochei* presented TP_{SCA} values exceeding 4.1, while their isotopic positions remained significantly lower, ranging between 2.4 and 2.6.

Five species were excluded from the comparative analysis due to high vacuity indices or the exclusive presence of unidentifiable digested matter, which precluded the calculation of TP_{SCA} . These taxa were *Cheilopogon exsiliens*, *Sarda sarda*, *Tylosurus acus*, *Ariosoma balearicum*, and *Remora remora*.

Figure 3.2.5. Cross-validation of trophic position estimates (TP_{SCA} vs. TP_{SIA}) for the Ebro Delta pelagic community. Data points represent species means ($\pm$ SD where applicable), while the underlying Spearman correlation (ρ) and p-value were calculated using all paired individual observations ($n=83$) to account for intra-specific variability. The dashed line indicates the 1:1 identity (perfect agreement), while the solid black line represents the linear regression trend. Point size is proportional to the number of stomachs analysed (N_{SCA}). Species are ranked in the legend by their mean TP_{SIA} within each functional guild.



3.3 CASE STUDY 2: THE LA AZOHÍA PELAGIC COMMUNITY

To reconstruct the trophic structure and isotopic dynamics of the Azohía pelagic community, stable isotope analysis was performed on six different tissues across 9 target species, comprising small, medium, and large pelagic fish. A total of 363 samples were analysed. A summary of the biometric data (body length, weight), sample sizes (N), and mean isotopic values ($\delta^{13}\text{C}$, $\delta^{13}\text{C}_{\text{corr}}$, $\delta^{15}\text{N}$) for each species and tissue is provided in **Table 3.3**.

Table 3.3. Summary of biological and isotopic metrics for the pelagic fish community of La Azohía (Case study 2). Data are grouped by functional group and include sample size (n) per tissue, biometric measurements (body length (BL) and weight) and mean ($\pm$ SD) isotopic signatures ($\delta^{13}\text{C}$, $\delta^{13}\text{C}_{\text{corr}}$, $\delta^{15}\text{N}$) and trophic position (TP_{SIA}) estimated for each analysed tissue (liver, gonads, muscle, dorsal fin, vertebrae, eyes).

Species	n	BL $\pm$ SD ¹	Weight $\pm$ SD	Maturity	Tissue ²	$\delta^{13}\text{C} \pm$ SD	$\delta^{13}\text{C}_{\text{cor}}$ $r \pm$ SD	$\delta^{15}\text{N} \pm$ SD	TP_{SIA} $\pm$ SD
Small pelagic fish									
<i>Engraulis encrasicolus</i>	6	13.80 $\pm$ 0.40	17.27 $\pm$ 1.52	Adults	Liver	-23.29 $\pm$ 1.10	-18.58 $\pm$ 0.19	7.42 $\pm$ 1.31	2.36 $\pm$ 0.38
					Gonads	-20.85 $\pm$ 0.59	-18.33 $\pm$ 0.10	7.40 $\pm$ 0.21	2.36 $\pm$ 0.06
					Muscle	-19.77 $\pm$ 0.59	-18.05 $\pm$ 0.10	8.83 $\pm$ 0.25	2.78 $\pm$ 0.07
					Dorsal fin	-22.39 $\pm$ 1.57	-17.90 $\pm$ 0.22	7.49 $\pm$ 0.41	2.38 $\pm$ 0.12
					Vertebra	-19.13 $\pm$ 1.64	-19.15 $\pm$ 1.65	7.00 $\pm$ 0.62	2.24 $\pm$ 0.18
					Eyes	-22.23 $\pm$ 1.35	-18.22 $\pm$ 0.28	6.41 $\pm$ 0.48	2.07 $\pm$ 0.14
<i>Sardina pilchardus</i>	6	14.00 $\pm$ 0.51	22.23 $\pm$ 2.79	Juveniles	Liver	-23.17 $\pm$ 0.52	-18.80 $\pm$ 0.19	8.28 $\pm$ 0.27	2.62 $\pm$ 0.08
					Gonads	-22.75 $\pm$ 1.00	-18.53 $\pm$ 0.22	8.31 $\pm$ 0.20	2.63 $\pm$ 0.06

Species	n	BL ± SD ¹	Weight ± SD	Maturity	Tissue ²	δ ¹³ C ± SD	δ ¹³ C _{cor} r ± SD	δ ¹⁵ N ± SD	TP _{SIA} ± SD
					Muscle	-20.09 ± 0.43	-18.08 ± 0.19	9.29 ± 0.24	2.91 ± 0.07
					Dorsal fin	-22.06 ± 0.84	-17.63 ± 0.32	7.35 ± 0.44	2.34 ± 0.13
					Vertebra	-19.57 ± 1.78	-19.18 ± 2.17	7.32 ± 0.53	2.33 ± 0.16
					Eyes	-21.28 ± 1.42	-17.80 ± 0.57	8.68 ± 0.93	2.74 ± 0.27
Medium pelagic fish									
					Liver	-20.34 ± 0.15	-17.60 ± 0.11	8.88 ± 0.44	2.80 ± 0.13
					Gonads	-19.18 ± 0.57	-17.60 ± 0.30	8.76 ± 0.25	2.76 ± 0.07
					Muscle	-18.69 ± 0.58	-17.15 ± 0.19	10.60 ± 0.34	3.30 ± 0.10
					Dorsal fin	-18.56 ± 1.85	-15.70 ± 0.85	7.44 ± 0.45	2.37 ± 0.13
					Vertebra	-23.21 ± 0.59	-22.82 ± 0.77	7.48 ± 0.49	2.38 ± 0.14
					Eyes	-18.99 ± 0.21	-17.48 ± 0.17	8.75 ± 0.93	2.76 ± 0.27
<i>Auxis thazard</i>	6	43.67 ± 1.41	1579.17 ± 121.10	Adults					
					Liver	-20.66 ± 0.52	-17.80 ± 0.28	9.13 ± 0.31	2.87 ± 0.09
					Gonads	-18.75 ± 0.88	-17.12 ± 0.41	9.37 ± 0.50	2.94 ± 0.15
					Muscle	-18.28 ± 0.62	-16.73 ± 0.35	10.23 ± 0.95	3.19 ± 0.28
					Dorsal fin	-17.55 ± 1.48	-15.17 ± 0.79	7.98 ± 0.87	2.53 ± 0.26
					Vertebra	-22.61 ± 0.87	-22.18 ± 1.30	8.00 ± 0.87	2.53 ± 0.25
<i>Euthynnus alletteratus</i>	6	51.63 ± 10.84	2583.33 ± 1550.62	Adults					

Species	n	BL ± SD ¹	Weight ± SD	Maturity	Tissue ²	δ ¹³ C ± SD	δ ¹³ C _{cor} r ± SD	δ ¹⁵ N ± SD	TP _{SIA} ± SD
					Eyes	-18.79 ± 0.59	-17.28 ± 0.48	9.40 ± 0.89	2.95 ± 0.26
<i>Sarda sarda</i>	6	54.62 ± 1.66	2471.67 ± 202.82	Adults	Liver	-21.03 ± 1.12	-17.53 ± 0.45	9.77 ± 0.66	3.06 ± 0.20
					Gonads	-19.24 ± 0.43	-17.23 ± 0.23	9.96 ± 0.51	3.11 ± 0.15
					Muscle	-17.48 ± 0.29	-16.10 ± 0.17	10.98 ± 0.53	3.41 ± 0.16
					Dorsal fin	-20.68 ± 0.53	-16.30 ± 0.33	10.42 ± 0.56	3.25 ± 0.17
					Vertebra	-22.41 ± 0.59	-21.95 ± 1.01	9.93 ± 0.49	3.10 ± 0.15
					Eyes	-18.43 ± 0.11	-16.98 ± 0.16	11.37 ± 0.64	3.53 ± 0.19
<i>Scomber colias</i>	6	25.60 ± 2.85	160.97 ± 62.38	Adults	Liver	-21.40 ± 0.56	-18.10 ± 0.15	8.50 ± 0.45	2.68 ± 0.13
					Gonads	-20.27 ± 1.13	-18.28 ± 0.40	9.08 ± 0.91	2.85 ± 0.27
					Muscle	-18.83 ± 0.25	-17.53 ± 0.27	9.21 ± 0.36	2.89 ± 0.11
					Dorsal fin	-20.54 ± 1.90	-16.88 ± 0.78	7.92 ± 0.46	2.51 ± 0.14
					Vertebra	-21.96 ± 1.12	-21.77 ± 0.91	8.05 ± 0.57	2.55 ± 0.17
					Eyes	-19.60 ± 0.40	-17.77 ± 0.25	8.71 ± 0.51	2.74 ± 0.15
Large pelagic fish									
<i>Mola mola</i>	11	37.63 ± 2.29	2813.09 ± 477.61	Indetermined (11)	Liver (11)	-24.24 ± 1.00	-18.89 ± 0.28	9.68 ± 0.40	3.03 ± 0.12
					Gonads (4)	-19.76 ± 0.31	-17.60 ± 0.24	9.99 ± 0.48	3.12 ± 0.14

Species	n	BL ± SD ¹	Weight ± SD	Maturity	Tissue ²	δ ¹³ C ± SD	δ ¹³ C _{cor} r ± SD	δ ¹⁵ N ± SD	TP _{SIA} ± SD
					Muscle (11)	-20.23 ± 0.41	-18.63 ± 0.39	9.39 ± 0.89	2.94 ± 0.26
					Dorsal fin (11)	-18.37 ± 1.68	-16.95 ± 0.53	11.22 ± 0.55	3.48 ± 0.16
					Vertebra (4)	-17.37 ± 0.53	-17.38 ± 0.53	9.74 ± 0.92	3.05 ± 0.27
					Eyes (2)	-18.78 ± 0.40	-17.25 ± 0.07	11.63 ± 0.26	3.60 ± 0.08
<i>Prionace glauca</i> ³	58	122.61 ± 38.36	8400.00 ± 1697.06	Adults (2), Indetermined (56)	Liver (2)	-21.40 ± 0.58	-17.85 ± 0.35	10.41 ± 0.57	3.24 ± 0.17
					Gonads (2)	-19.08 ± 0.79	-17.60 ± 0.85	10.22 ± 1.01	3.19 ± 0.30
					Muscle (58)	-17.63 ± 0.90	-16.11 ± 0.95	11.50 ± 1.38	3.56 ± 0.41
					Dorsal fin (2)	-18.52 ± 1.33	-17.15 ± 0.64	10.13 ± 1.08	3.16 ± 0.32
					Vertebra (2)	-15.54 ± 0.13	-15.50 ± 0.14	8.94 ± 1.11	2.81 ± 0.33
					Eyes (2)	-17.78 ± 0.18	-17.15 ± 0.64	9.09 ± 0.71	2.86 ± 0.21
<i>Tetrapturus belone</i>	6	141.47 ± 6.83	10883.33 ± 3111.54	Juveniles (1), Adults (5)	Liver	-20.33 ± 0.57	-17.47 ± 0.22	9.05 ± 0.37	2.84 ± 0.11
					Gonads	-18.54 ± 0.56	-16.65 ± 0.34	9.06 ± 0.46	2.85 ± 0.14
					Muscle	-17.73 ± 0.36	-16.77 ± 0.45	9.49 ± 0.38	2.97 ± 0.11
					Dorsal fin	-17.47 ± 1.26	-15.28 ± 0.54	7.30 ± 0.21	2.33 ± 0.06
					Vertebra	-16.67 ± 2.60	-16.52 ± 2.82	6.77 ± 0.58	2.17 ± 0.17
					Eyes	-18.24 ± 0.16	-16.77 ± 0.15	8.56 ± 0.54	2.70 ± 0.16

Species	n	BL $\pm$ SD ¹	Weight $\pm$ SD	Maturity	Tissue ²	$\delta^{13}\text{C} \pm$ SD	$\delta^{13}\text{C}_{\text{cor}} \pm$ r $\pm$ SD	$\delta^{15}\text{N} \pm$ SD	TP _{SIA} $\pm$ SD
Baseline species									
<i>Venus nux</i>	6	-	-	-	Whole body	-19.32 $\pm$ 0.44	-18.30 $\pm$ 0.97	6.18 $\pm$ 0.86	2.00 $\pm$ 0.25

Note: ¹**Body length (BL)** represents the specific morphometric standard for each species (TL, FL, LJFL, or ML) as detailed in **Section 2.2.1**. ²**Tissue** sample sizes correspond to the total number of individuals (n) listed for the species, unless a specific sample size is noted in parentheses. ³*P. glauca* muscle data are derived from samples sourced from Fernández et al., (2021).

3.3.1 Inter-tissue variability (Testing H2a)

To assess potential isotopic routing and tissue-specific turnover rates, $\delta^{13}\text{C}_{\text{cor}}$ and $\delta^{15}\text{N}$ signatures were compared across six tissues (muscle, liver, gonads, eyes, dorsal fin, and vertebrae) for the nine targeted species. The distribution of isotopic values is visualised in **Figure 3.3.1**, **(a)** $\delta^{13}\text{C}_{\text{cor}}$ and **(b)** $\delta^{15}\text{N}$.

Figure 3.3.1.a Distribution of (a) lipid-corrected carbon ($\delta^{13}\text{C}_{\text{corr}}$) stable isotope signatures across six tissue types for pelagic species in the ecosystem. Boxplots display the median (bold line), interquartile range (box), and outliers (points). The figure indicates the values distribution along an inshore to offshore gradient.

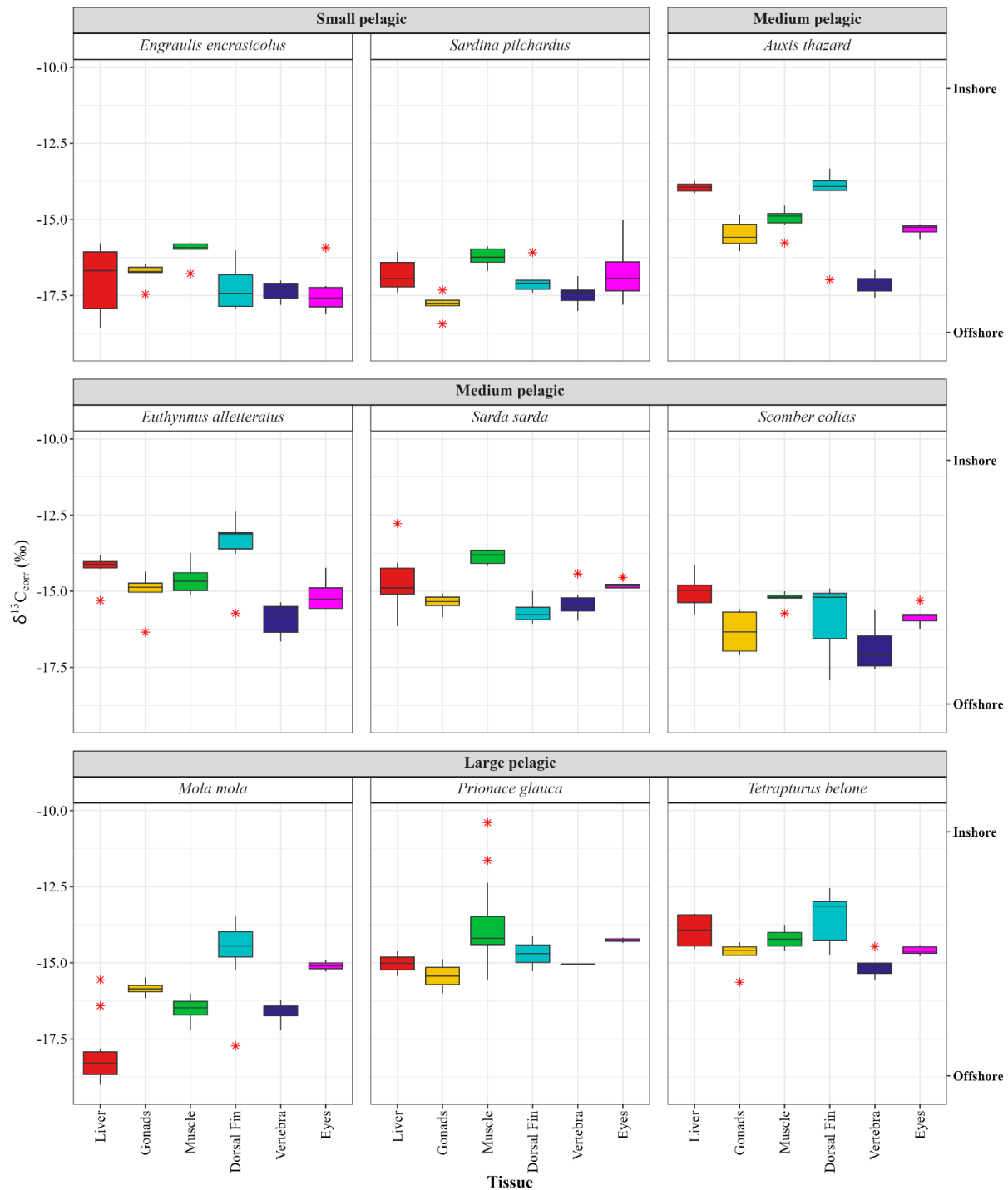
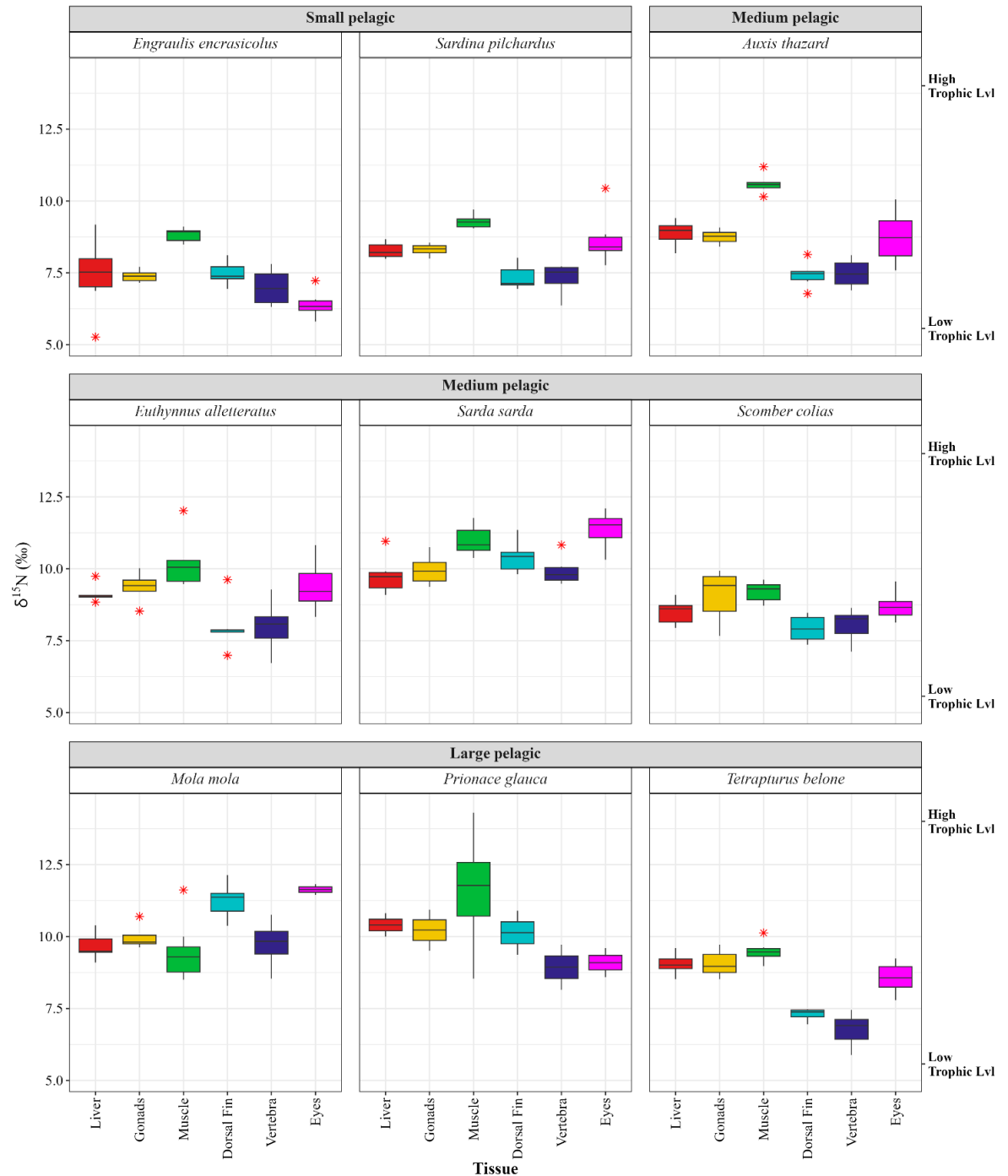


Figure 3.3.1.b. Distribution of nitrogen ($\delta^{15}\text{N}$) stable isotope signatures across six tissue types for pelagic species in the ecosystem. Boxplots display the median (bold line), interquartile range (box), and outliers (points). The figure indicates the values distribution along high to low trophic level (Trophic Lvl) gradient.



Carbon patterns across functional groups

Carbon signatures displayed significant inter-tissue variability across the Azohía community, with distinct patterns of enrichment and isotopic clustering observed within each functional group. Statistical pairwise comparisons across tissues are detailed in **Appendix D.1a**.

Small pelagic fish

In small pelagics, muscle tissue consistently emerged as the most enriched tissue. In *Sardina pilchardus*, muscle was significantly $\delta^{13}\text{C}$ enriched relative to gonads ($Z=-3.92$, $p<0.01$) and vertebrae ($Z=3.07$, $p=0.03$). Similarly, in *Engraulis encrasicolus*, muscle showed significant enrichment relative to eyes ($Z=-2.99$, $p=0.04$). Vertebrae in this group tended to be more depleted, though they remained statistically comparable to other non-muscle tissues. Intraspecific $\delta^{13}\text{C}$ were widest in the liver for both species, whereas vertebrae and eyes exhibited tighter isotopic clustering.

Medium pelagic fish

This group was characterised by high enrichment in both muscle and liver tissues compared to structural components, alongside varying degrees of intraspecific variability. In *Scomber colias*, both muscle ($Z=3.01$, $p=0.04$) and liver ($Z=3.45$, $p<0.01$) were significantly enriched relative to vertebrae. In *Auxis thazard*, liver ($Z=4.34$, $p<0.01$) and dorsal fin ($Z=4.43$, $p<0.01$) showed significant enrichment over vertebrae. Muscle tissue in *Sarda sarda* was significantly enriched compared to both vertebrae ($Z=3.23$, $p=0.02$) and gonads ($Z=-3.23$, $p=0.02$). High variability was observed in the dorsal fin of *Scomber colias* and the liver of *Sarda sarda*, while tighter isotopic clustering was noted in the structural tissues (vertebrae and eyes) of *Auxis thazard* and *Sarda sarda*.

Large pelagic fish

Large predators and the sunfish showed the most pronounced tissue-specific partitioning and structural-tissue variability. In *Tetrapturus belone*, significant $\delta^{13}\text{C}$ enrichment was observed in the dorsal fin ($Z=3.29$, $p=0.02$) and liver ($Z=3.04$, $p=0.03$)

relative to vertebrae. *Mola mola* exhibited a unique pattern where the dorsal fin was significantly $\delta^{13}\text{C}$ enriched relative to both muscle ($Z=3.16$, $p=0.02$) and liver ($Z=5.04$, $p<0.01$), marking the most distinct structural-tissue enrichment in the dataset. Regarding *Prionace glauca*, pairwise statistics could not be calculated as the species did not meet the minimum sample size criteria ($n \geq 3$ per tissue). However, visual inspection suggests that muscle tissue $\delta^{13}\text{C}$ values are the highest relative to other tissues, whereas gonads appear the most depleted. Wide isotopic ranges were found in the liver and dorsal fin of *Tetrapturus belone* and *Mola mola*; conversely, *Prionace glauca* and *Mola mola* showed more stable clustering in the vertebrae and eyes.

Nitrogen patterns across functional groups

Nitrogen signatures exhibited significant tissue-specific heterogeneity across the community, typically defined by a gradient where muscle tissue represents the most enriched tissue and vertebrae the most depleted, except for *Mola mola*. Statistical results for Nitrogen pairwise comparisons are detailed in **Appendix D.1b**.

Small pelagic fish

In small pelagics, muscle tissue was significantly $\delta^{15}\text{N}$ enriched relative to all other analysed tissues. In *Sardina pilchardus*, muscle enrichment was significant compared to liver (Mean Diff = +1.01‰, $p=0.02$), gonads (Mean Diff = +0.98‰, $p=0.02$), and vertebrae (Mean Diff = +1.98‰, $p<0.01$). A similar significant pattern was found in *Engraulis encrasicolus* for muscle relative to liver (Mean Diff = +1.41‰, $p<0.01$) and vertebrae (Mean Diff = -1.83‰, $p<0.01$). In *Engraulis encrasicolus*, nitrogen variability was highest in the liver, with values comparable to the vertebrae. For *Sardina pilchardus*, results were more clustered across all tissues, with slightly wider ranges observed only in the structural components (vertebrae and dorsal fin).

Medium pelagic fish

Medium pelagics displayed the most pronounced nitrogen partitioning, with muscle often showing the highest mean values in the entire dataset. Significant depletion in vertebrae relative to muscle was consistent across the group, ranging from a mean difference of -1.05‰ in *Sarda sarda* ($p=0.04$) to -3.11‰ in *Auxis thazard*

($p < 0.01$). In *Auxis thazard*, muscle was also significantly enriched relative to eyes (Mean Diff = +1.85‰, $p < 0.01$) and gonads (Mean Diff = +1.84‰, $p < 0.01$). In *Scomber colias*, muscle was significantly enriched relative to dorsal fin (Mean Diff = +1.28‰, $p < 0.01$), lacked a uniform pattern within this group: eyes emerged as one of the tissues with the widest nitrogen ranges in three of the four species, particularly in *Auxis thazard*. Additionally, the gonads in *Scomber colias* exhibited a wider isotopic range compared to other tissues.

Large pelagic fish

Large predators and the sunfish showed unique nitrogen distributions, including significant structural-tissue depletion. *Tetrapturus belone* displayed significant muscle enrichment relative to eyes (Mean Diff = +0.93‰, $p = 0.01$) and vertebrae (Mean Diff = -2.72‰, $p < 0.01$). Notably, vertebrae were significantly depleted relative to all metabolically active tissues (liver and gonads, $p < 0.01$). While pairwise statistics were not calculated for *Prionace glauca*, visual inspection identifies muscle as the most enriched tissue, exhibiting the widest nitrogen range in the entire dataset. *Mola mola* displayed a reversal of the general community trend; muscle tissue exhibited significantly lower nitrogen values than eyes (Mean Diff = -2.25‰, $p < 0.01$) and dorsal fin (Mean Diff = -1.84‰, $p < 0.01$). Both *Mola mola* and *Tetrapturus belone* showed high variability in the dorsal fin and liver, whereas isotopic clustering was tightest in the liver of *Prionace glauca*, and in the eyes and gonads of *Mola mola*.

3.3.2 Tissue-specific niche widths

Sample-size corrected standard ellipse areas (SEAc) were calculated for each tissue to quantify niche breadth. Community-level isotopic distributions are visualised in **Figure 3.3.2 (a and b)**, illustrating the relative positioning and overlap of tissue ellipses across the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ axes for each species and for all species and tissue. Detailed SIBER metrics and full statistical results are provided in **Appendix D (Table D.2 and D.3.)**

Figure 3.3.2.a Case study 2 – Azohía pelagic community. Isotopic niches of pelagic species across tissues in the $\delta^{13}\text{C}_{\text{corr}}$ – $\delta^{15}\text{N}$ space. Each point represents an individual sample, while standard ellipse areas (40% confidence interval) represent the core niche of each tissue type. Panels are grouped by functional categories (small, medium, and large pelagics). If samples had an $n < 3$, the ellipses were not calculated.

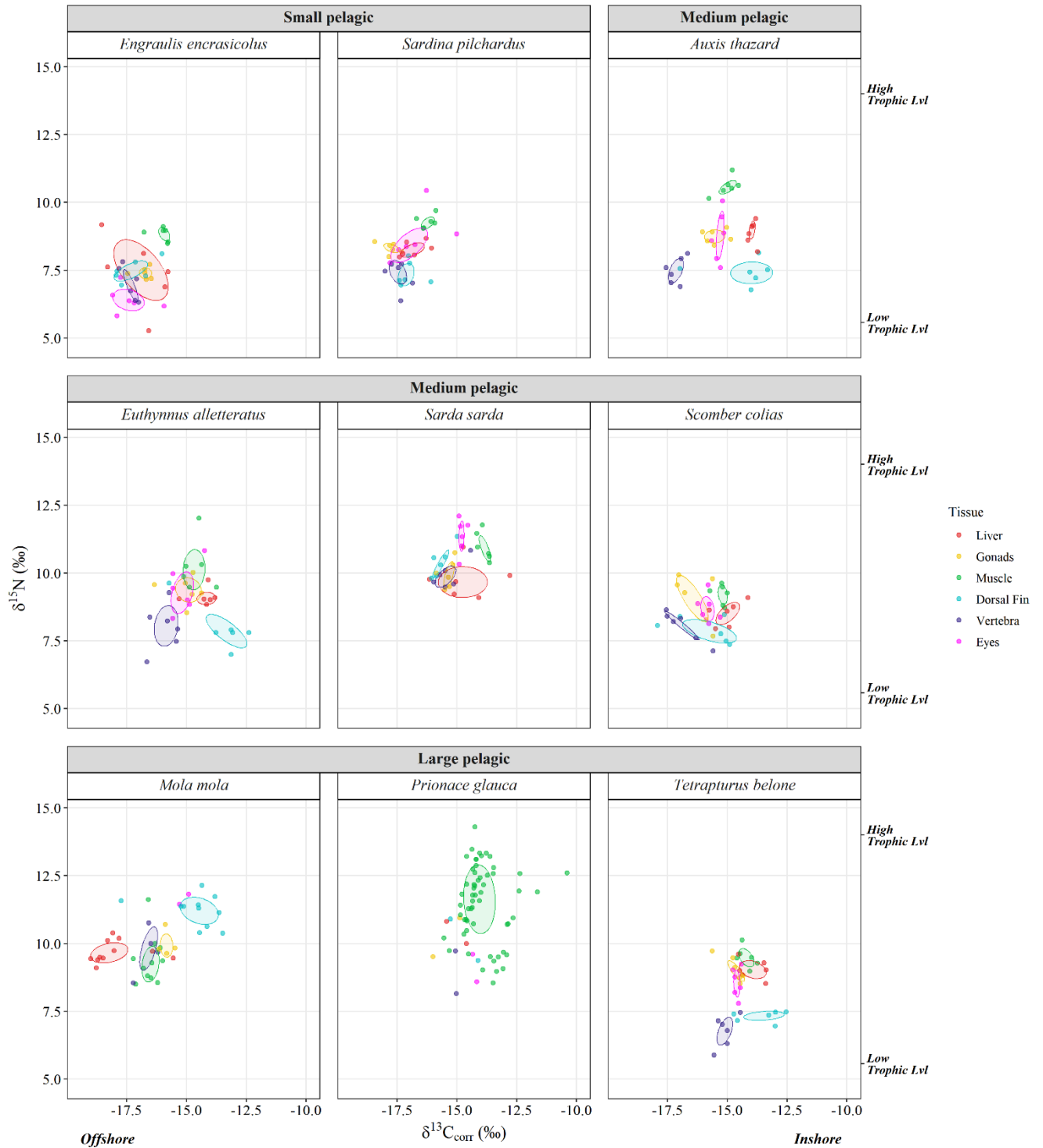
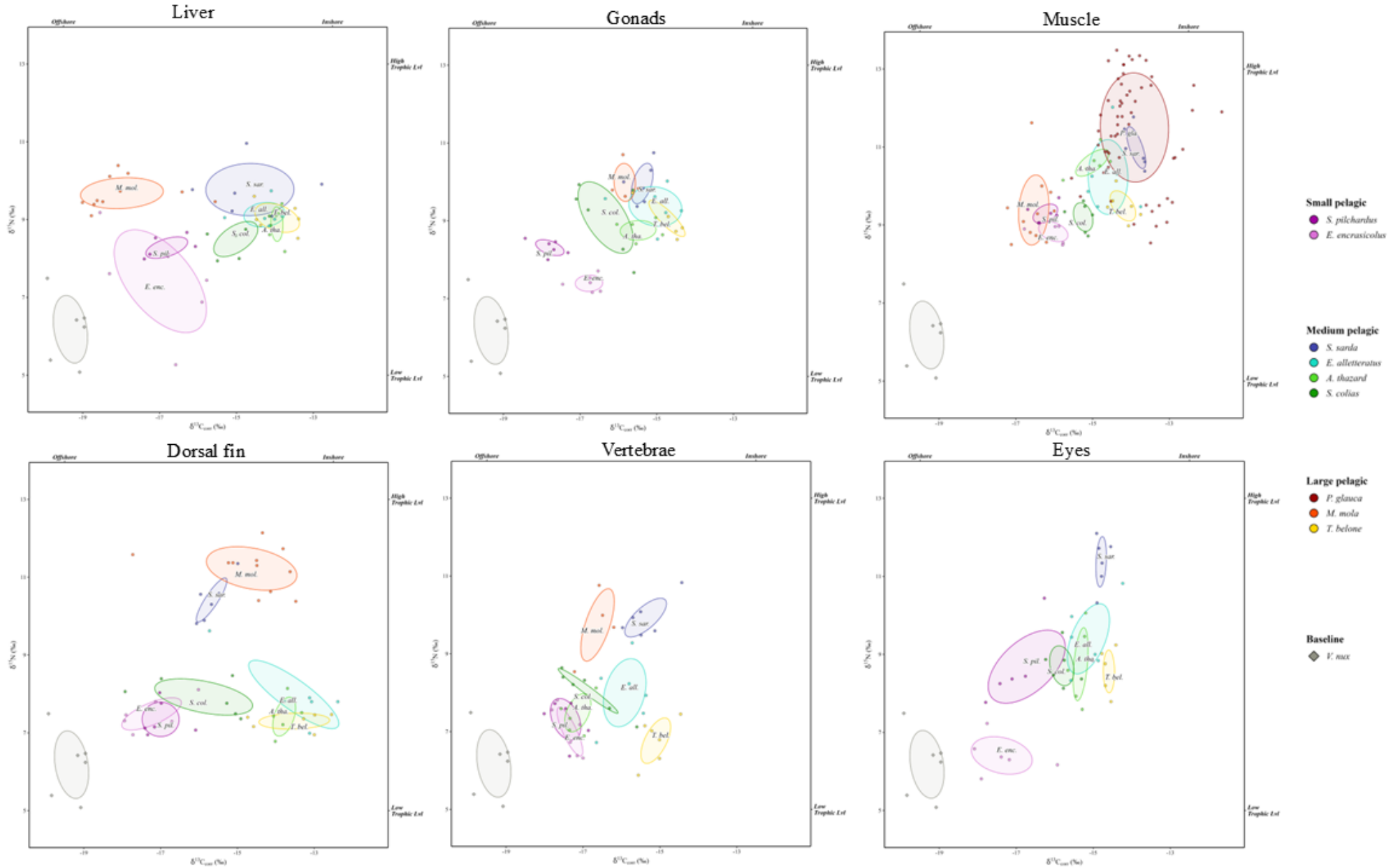


Figure 3.3.2.b. Case study 2 – Azohía pelagic community. Composite isotopic biplots showing inter-specific niche overlap across all tissue types analysed. Each panel represents the $\delta^{13}\text{C}_{\text{corr}}$ – $\delta^{15}\text{N}$ space for a specific tissue. Core niches are represented by standard ellipse areas (40% confidence interval). Species are color-coded by functional group, and the bivalve *Venus nux* is included as the benthic isotopic baseline (grey diamonds). If samples had an $n < 3$, the ellipses were not calculated.



Highly significant differences in isotopic niche width among tissues were observed for all analysed species ($p < 0.001$), revealing a high degree of intra-individual variability and tissue-specific integration times.

Small pelagic fish

Small pelagics displayed significant disparities in ellipse width across tissues. In *Engraulis encrasicolus*, the liver recorded the widest isotopic niche in the community ($SEA_C = 5.07 \text{ ‰}^2$), significantly exceeding those of muscle (0.35 ‰^2) or vertebrae (0.47 ‰^2 , $p < 0.001$). This liver niche was characterised by the widest expansion along the $\delta^{15}\text{N}$ axis for this tissue type ($\approx 6\text{--}8.7\text{‰}$). On the carbon axis, this expansion encompassed the entire range of *Sardina pilchardus* (≈ -18.8 to -16.5‰), which exhibited a tighter liver niche ($SEA_C = 0.49 \text{ ‰}^2$) positioned around 8‰ on the $\delta^{15}\text{N}$ axis. (**Figure 3.3.2.b**).

The most pronounced niche expansion for *Sardina pilchardus* occurred in the eye lenses (3.03 ‰^2), significantly exceeding all other tissues in the species ($p < 0.001$). This niche was defined by a range of nearly 2‰ along the $\delta^{15}\text{N}$ axis, showing the highest vertical dispersion for this tissue type. In contrast, the eye lenses of *E. encrasicolus* (1.41 ‰^2) were clustered at a lower trophic position. Additional tissue-specific observations for this functional group include:

- **Gonads:** The small pelagics exhibited similar isotopic niche widths (SEA_C range: $0.25\text{--}0.29 \text{ ‰}^2$). For both species, the isotopic niche distribution was driven primarily by dispersion along the $\delta^{13}\text{C}$ axis, while remaining highly constrained within a narrow $\delta^{15}\text{N}$ range. In terms of positioning, *S. pilchardus* exhibited a shift towards higher $\delta^{15}\text{N}$ values ($\approx 8.0\text{‰}$) and more positive $\delta^{13}\text{C}$ values ($\approx -17.5\text{‰}$), in comparison with *E. encrasicolus* ($\approx 7.5\text{‰}$ $\delta^{15}\text{N}$, $\approx -16.8\text{‰}$ $\delta^{13}\text{C}$).
- **Muscle:** Both taxa occupied the lowest vertical positions in the community ($\approx 9\text{‰}$ $\delta^{15}\text{N}$), with limited isotopic niche overlap. Despite exhibiting similarly narrow niche widths (*E. encrasicolus* $SEA_C = 0.35 \text{ ‰}^2$; *S. pilchardus* $SEA_C = 0.27 \text{ ‰}^2$), *S. pilchardus* occupied a slightly distinct, nitrogen-enriched position ($\approx 9.2\text{‰}$) relative to *E. encrasicolus* ($\approx 8.8\text{‰}$). For both species, the

residual intra-specific variability was driven exclusively by dispersion along the $\delta^{13}\text{C}$ axis (spanning $\approx -16.5\text{‰}$ to -15.5‰), while nitrogen values remained cohesive.

- **Dorsal fin:** Both species occupied a highly overlapping trophic space centered at $\approx 7.0\text{‰}$ $\delta^{15}\text{N}$. However, *Engraulis encrasicolus* exhibited a slightly broader isotopic niche ($\text{SEAc}=0.90\text{‰}^2$) compared to *Sardina pilchardus* ($\text{SEAc}=0.82\text{‰}^2$). This difference was driven by a wider dispersion of *E. encrasicolus* along the $\delta^{13}\text{C}$ axis (spanning from $\approx -18.0\text{‰}$ to -16.0‰), whereas *S. pilchardus* remained more spatially constrained ($\approx -17.5\text{‰}$ to -16.0‰).
- **Vertebrae:** Both taxa exhibited isotopic niches driven primarily by expansion along the $\delta^{15}\text{N}$ axis, with minimal variation in carbon (ellipses centered at $\approx -17.4\text{‰}$ $\delta^{13}\text{C}$). *Sardina pilchardus* displayed a wider isotopic niche ($\text{SEAc}=0.75\text{‰}^2$) relative to *E. encrasicolus* ($\text{SEAc}=0.47\text{‰}^2$), extending toward enriched nitrogen values relative to the anchovy.

Medium pelagic fish

Medium pelagics followed a consistent trend where metabolic tissues exhibited broader niches than structural components. In the liver biplot (Figure 3.3.2.b), the medium pelagic guild was characterised by a general pattern of habitat heterogeneity, with most ellipses primarily expanded along the $\delta^{13}\text{C}$ axis. A notable exception was *Auxis thazard*, which displayed a niche expansion along the $\delta^{15}\text{N}$ axis. In *Sarda sarda* the liver niche ($\text{SEAc} = 2.96 \text{‰}^2$) exhibited the greatest isotopic heterogeneity spanning a wide carbon gradient ($\delta^{13}\text{C} \approx -16.0$ to -13.5‰ , $\delta^{15}\text{N} (\approx 9\text{--}10.5\text{‰})$). This niche formed a trophic cluster with *A. thazard* ($\text{SEAc} = 0.27 \text{‰}^2$) and *E. alletteratus* ($\text{SEAc} = 0.64 \text{‰}^2$) within the shared nitrogen band of $9.0\text{--}9.5\text{‰}$. In contrast, *Scomber colias* ($\text{SEAc} = 0.83 \text{‰}^2$) was spatially and trophically segregated, occupying a lower trophic position ($\approx 8.5\text{‰}$) (**Figure 3.3.2.b**).

In the gonad biplot, *S. colias* has the largest niche width (2.03‰^2), significantly distinct from its muscle (0.34‰^2 , $p < 0.001$). This niche was shifted toward less negative $\delta^{13}\text{C}$ values, overlapping exclusively with *A. thazard* (0.46‰^2). *Euthynnus*

alletteratus exhibited the second-largest gonad niche (1.34 ‰^2), situated between *S. sarda* ($\text{SEA}_C = 0.50 \text{ ‰}^2$; $\delta^{15}\text{N} \approx 10.5\text{‰}$) and *A. thazard* ($\text{SEA}_C = 0.46 \text{ ‰}^2$; $\delta^{15}\text{N} \approx 9.0\text{‰}$).

- Muscle:** This tissue was characterised by high isotopic overlap among the majority of the guild. *E. alletteratus* recorded the widest niche (1.91 ‰^2) overlapping with *P. glauca* and *T. belone*. *A. thazard* (0.42 ‰^2) exhibited nearly complete isotopic overlap with *E. alletteratus* at a mid-trophic level. *S. sarda* (0.37 ‰^2) was distinguished as the most $\delta^{15}\text{N}$ -enriched member ($\approx 10\text{‰}$), while *S. colias* (0.34 ‰^2) was segregated by 1‰ decrease in $\delta^{15}\text{N}$ values ($\approx 9.0\text{‰}$), showing no overlap with other medium pelagics (**Figure 3.3.2.b**).
- Dorsal fin:** This guild exhibited the highest horizontal dispersion across the $\delta^{13}\text{C}$ axis in this tissue. *E. alletteratus* (2.09 ‰^2) and *S. colias* (2.01 ‰^2) recorded the widest niches, spanning the mid-section of the $\delta^{13}\text{C}$ axis. *S. colias* showed light overlap with *E. encrasicolus*. While the guild was generally defined by habitat heterogeneity, *S. sarda* and *A. thazard* were notable exceptions, exhibiting niches primarily expanded along the $\delta^{15}\text{N}$ axis. *S. sarda* recorded the smallest niche in this tissue (0.42 ‰^2), remaining vertically segregated at a higher trophic position ($\approx 10\text{‰}$; spanning $\approx 1\text{‰}$ along the $\delta^{15}\text{N}$ axis). Meanwhile, *E. alletteratus* and *A. thazard* (0.55 ‰^2) clustered toward higher $\delta^{13}\text{C}$ values with considerable niche overlap.
- Vertebrae:** This tissue was characterised by relatively constrained niches, with *Euthynnus alletteratus* recording the widest area ($\text{SEA}_C = 1.79 \text{ ‰}^2$). Most medium pelagics exhibited niches expanded along the $\delta^{15}\text{N}$ axis, forming a dense cluster between $8\text{--}10\text{‰}$. Within this group, *Auxis thazard* exhibited vertical overlap with the small pelagics. *Sarda sarda* (0.72 ‰^2) was segregated at the highest trophic level ($\approx 10\text{‰}$), while *E. alletteratus* was positioned at a mid-trophic level, overlapping primarily with *S. colias* (0.39 ‰^2).
- Eyes:** The medium pelagic guild was characterised by a shared isotopic pattern of trophic diversification, with ellipses primarily expanded along the $\delta^{15}\text{N}$ axis.

E. alletteratus displayed the largest niche area ($SEAC = 1.61 \text{ ‰}^2$), overlapping with both *A. thazard* (0.63 ‰^2) and *S. colias* (0.60 ‰^2). *S. sarda* reached the highest trophic levels ($\approx 12\text{‰}$) despite having the smallest niche area (0.33 ‰^2). In this tissue, the overlap between *A. thazard* and *E. encrasicolus* observed in vertebrae was absent; instead, *Scomber colias* showed niche overlap with the small pelagic *S. pilchardus* (**Figure 3.3.2.b**).

Large pelagic fish

Large pelagics exhibited the most pronounced niche partitioning and tissue-specific expansions. In *Mola mola*, liver ($SEAC = 1.46 \text{ ‰}^2$) and dorsal fin ($SEAC = 2.13 \text{ ‰}^2$) tissues displayed significantly wider niches than the muscle (1.20 ‰^2 , $p < 0.001$). The liver ellipse was located in the offshore extreme of the community, while the dorsal fin occupied the highest $\delta^{15}\text{N}$ levels (**Figure 3.3.2.b**). *Prionace glauca* presented a notable exception, with muscle tissue recording a niche width of 3.89 ‰^2 ($n = 58$). This elevated value was derived from a substantially larger sample size than other tissues for this species (where $n=2$) and was characterised by a $\delta^{15}\text{N}$ range stretching from 10‰ to 13‰ (**Figure 3.3.2.b**). In contrast, *Tetrapturus belone* maintained constrained niches across all tissues ($SEAC < 0.80 \text{ ‰}^2$), though the liver (0.77 ‰^2) remained broader than muscle (0.43 ‰^2 , $p < 0.001$). Additional tissue-specific observations for this guild include:

- **Liver:** *T. belone* was positioned at the extreme inshore end of the $\delta^{13}\text{C}$, overlapping with the medium pelagic community. In contrast, *M. mola* recorded a wider niche (1.46 ‰^2) characterized by a significant expansion along the $\delta^{13}\text{C}$ axis (spanning $\approx 2\text{‰}$), while simultaneously reaching one of the highest trophic positions for this tissue type (10‰ $\delta^{15}\text{N}$) (**Figure 3.3.2.b**).
- **Gonads:** *T. belone* (0.44 ‰^2) was situated at the inshore extreme of the carbon gradient, overlapping exclusively with *E. alletteratus*. While *M. mola* occupied the highest $\delta^{15}\text{N}$ position in the community, separated along the $\delta^{13}\text{C}$ axis from *T. belone* by approximately $2\text{--}3\text{‰}$ and overlapping slightly with *S. sarda* and *E. alletteratus* (**Figure 3.3.2.b**).

- **Muscle:** The large pelagic guild showed the highest trophic diversification (dispersion along the $\delta^{15}\text{N}$ axis). This was driven primarily by *P. glauca*, which presented the widest and highest niche area for muscle tissues ($\text{SEA}_C = 3.89$; $13\text{‰} \delta^{15}\text{N}$; $n = 58$). *M. mola* (1.2‰^2) exhibited a niche expansion along the $\delta^{15}\text{N}$ axis, showing vertical overlap with small pelagics. Meanwhile *T. belone* (0.43‰^2) displayed a lower trophic position and less negative $\delta^{13}\text{C}$ values, overlapping only with *E. alletteratus* (**Figure 3.3.2.b**).
- **Dorsal fin:** *T. belone* (0.73‰^2) was characterised by an expansion along the $\delta^{13}\text{C}$ axis, showing horizontal overlap with inshore scombrids but remaining at lower $\delta^{15}\text{N}$ values compared to *M. mola*. *Mola mola* recorded the widest niche area for this tissue ($\text{SEA}_C = 2.13 \text{‰}^2$), also expanding along the $\delta^{13}\text{C}$ axis at higher trophic position ($\approx 11.5\text{‰} \delta^{15}\text{N}$) and exhibiting a marginal isotopic overlap with *Sarda sarda* (**Figure 3.3.2.b**).
- **Vertebrae:** *Tetrapturus belone* recorded a constrained isotopic niche (0.69‰^2) situated at the inshore extreme ($\approx -15\text{‰} \delta^{13}\text{C}$; $\approx 7\text{‰} \delta^{15}\text{N}$). In contrast, *Mola mola* was bidimensionally segregated from *T. belone*, recording a wider niche (1.42‰^2) shifted offshore ($\approx -16.8\text{‰} \delta^{13}\text{C}$) and occupying a significantly higher trophic position ($\approx 10\text{‰} \delta^{15}\text{N}$) with a vertical expansion of approximately 2‰.
- **Eyes:** Due to insufficient replication for other large predators, this tissue was represented only by *T. belone*, which exhibited the most restricted niche for this tissue type (0.31‰^2). It was positioned at an intermediate trophic level ($\approx 9\text{‰} \delta^{15}\text{N}$), segregated slightly from the medium pelagic cluster and anchored at the inshore extreme of the community ($\approx -14.5\text{‰}$ in $\delta^{13}\text{C}$).

3.3.3 Inter-specific niche overlap

To quantify inter-specific isotopic interactions across different temporal windows, the degree of isotopic niche overlap was calculated for species pairs across six tissues (**Table D.4**).

Small pelagic fish

Overlap between the small pelagics *Engraulis encrasicolus* and *Sardina pilchardus* varied by tissue type, with the highest intersection recorded in the liver (72.3%) and the lowest in muscle tissue ($\approx 6\%$). In the vertebrae, was 44.7%. Additionally, an inter-guild intersection was observed in muscle tissue, where the medium pelagic *Scomber colias* exhibited a 100% overlap of the *S. pilchardus* isotopic niche.

Medium pelagic fish

Within the medium pelagic guild, *Auxis thazard* and *Euthynnus alletteratus* exhibited a gradient of overlap. These species displayed 0% overlap in the vertebrae, while overlap reached 69.3% in the eye lenses, and ranged between 52% and 68% in muscle and liver tissues. Interactions between *Sarda sarda* and *E. alletteratus* showed 0% overlap in both muscle and eyes, though intersections were recorded in the liver (56.6%) and gonads (41.4% of *S. sarda* overlapped by *E. alletteratus*).

Large pelagic fish

Mola mola showed limited intersection with other species in metabolic and structural tissues; liver overlap was 0% across all species, and dorsal fin overlap with *Sarda sarda* was 7.2%. However, in muscle tissue, an intersection of 67.5% was observed between *Mola mola* and *Sardina pilchardus*. For *Prionace glauca* and *Tetrapturus belone*, inter-specific overlap values remained near 0% across most tissues (Table D.4).

3.3.4 Intra-individual variability across tissues

To assess the stability of individual isotopic offsets, $\delta^{13}\text{C}_{\text{corr}}$ and $\delta^{15}\text{N}$ values were tracked per individual across tissues. These trajectories are visualised in **Figure 3.3.4.a** ($\delta^{13}\text{C}_{\text{corr}}$) and **Figure 3.3.4.b** ($\delta^{15}\text{N}$).

Figure 3.3.4.a. Individual $\delta^{13}\text{C}_{\text{corr}}$ trajectories across the tissues. Lines connect tissues within the same individual (same colour), ordered by metabolic turnover rate (fast-to-slow integration).

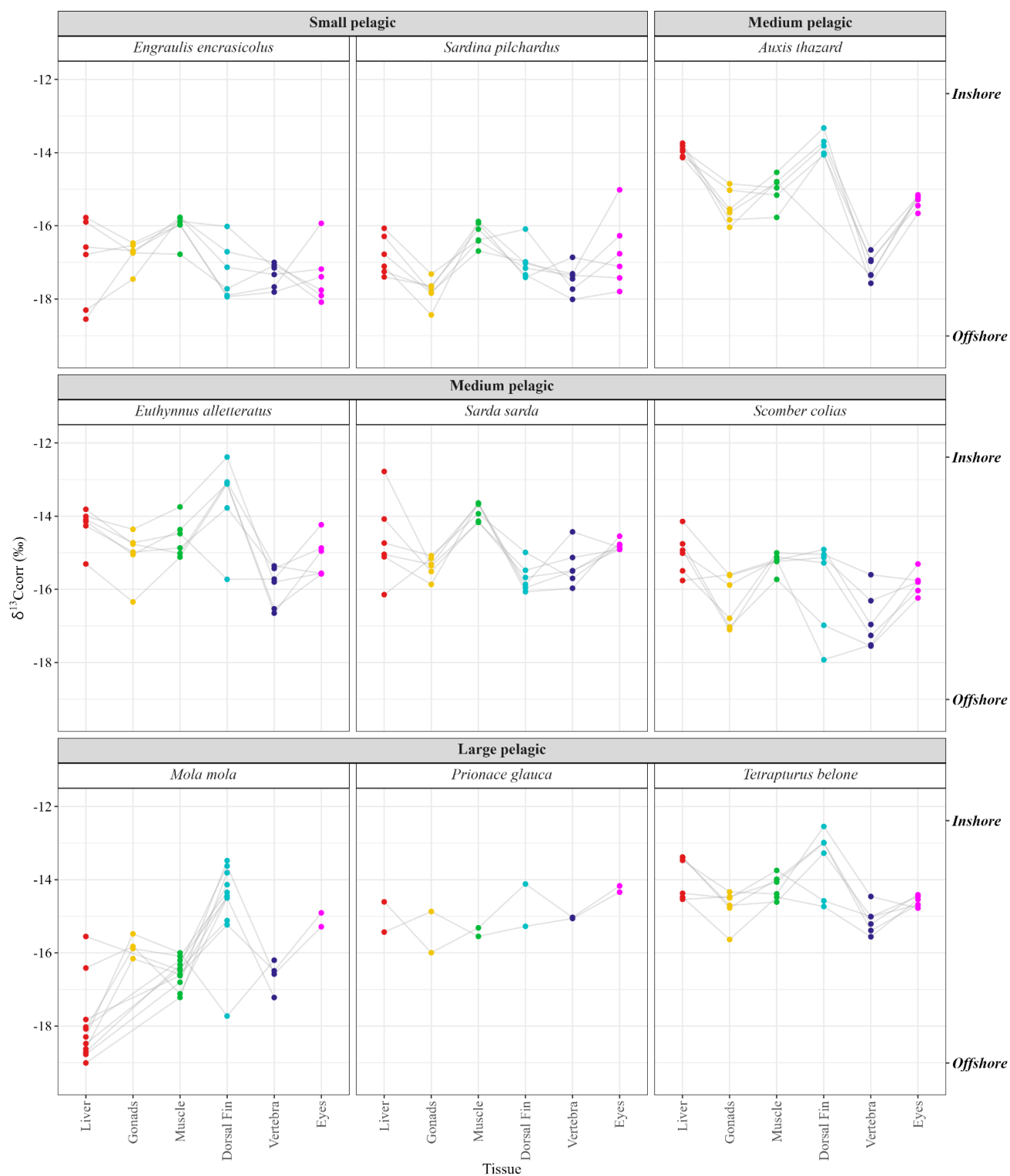
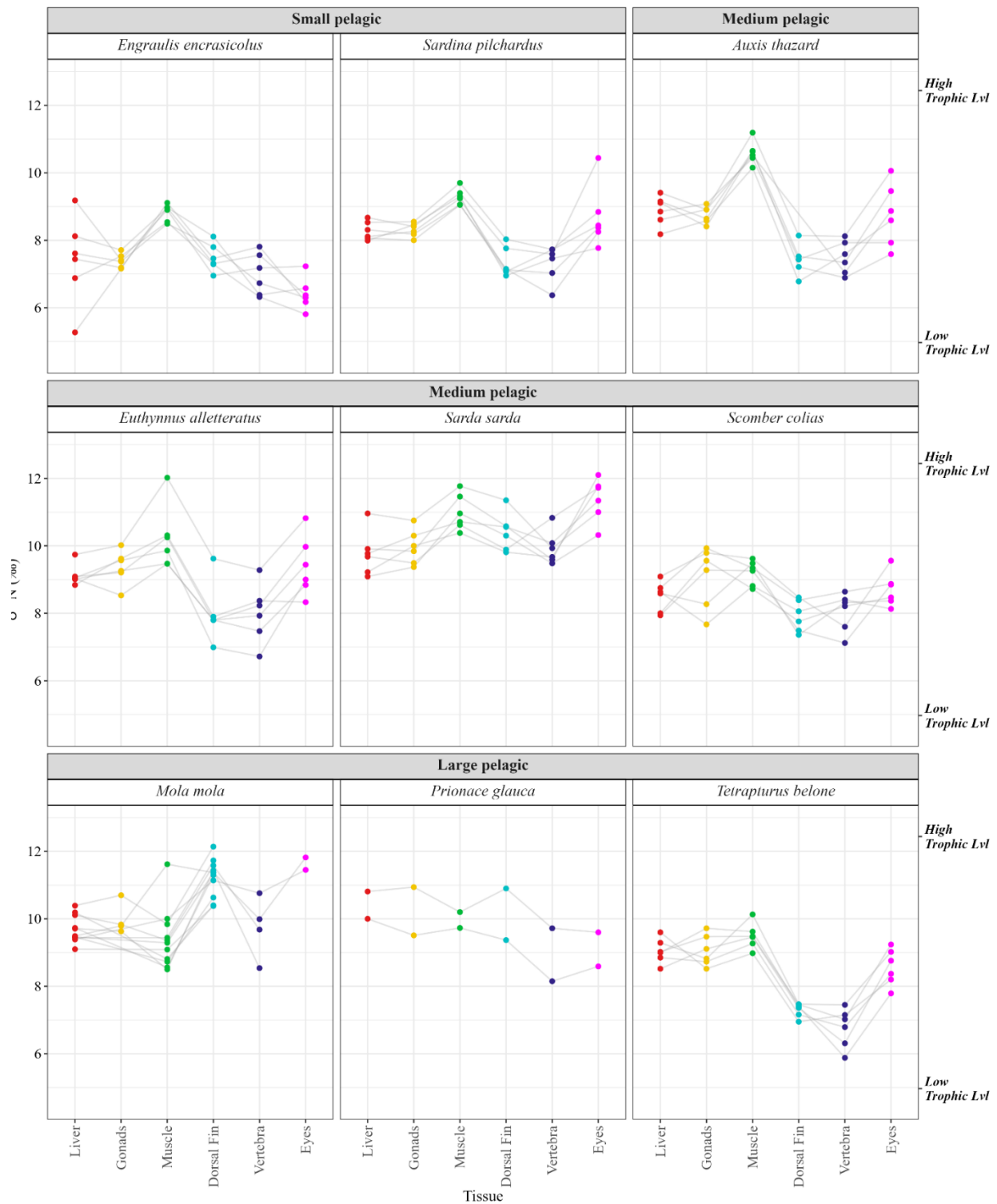


Figure 3.3.4.b. Individual $\delta^{15}\text{N}$ trajectories across the tissues. Paired-line plots represent the trophic history of individuals (same colour).



Carbon trajectories

The paired-line plots for $\delta^{13}\text{C}_{\text{corr}}$ revealed distinct levels of individual consistency across functional groups (**Figure 3.3.4.a**).

Small pelagic fish

Both species ($n=6$ each) displayed a shared pattern of isotopic dispersion, characterised by elevated variability in the liver, dorsal fin, and eyes, contrasted by strong cohesion in the muscle, gonads, and vertebrae.

In the **liver**, *E. encrasicolus* recorded the widest dispersion in the guild (2.7‰), while *S. pilchardus* exhibited moderate heterogeneity (1.4‰). This variability persisted in the **dorsal fin** (2.0‰ for anchovy; 1.4‰ for sardine) and in the **eyes**, where both species reached their maximum dispersion. *E. encrasicolus* eyes spanned a 2.1‰ gradient, while *S. pilchardus* recorded the widest range in the guild (3.0‰), driven by a single distinct outlier ($\approx -15.0\text{‰}$). In contrast, the muscle, vertebrae, and gonads demonstrated high individual consistency across both taxa. For *E. encrasicolus*, **gonads** and **muscle** converged into a tight cluster (between -16.0‰ and -16.5‰), while **vertebrae** restricted all individuals to a narrow 0.8‰ range. Similarly, *S. pilchardus* maintained high cohesion in these tissues, with individual ranges narrowing to $\leq 1.0\text{‰}$ (except for a single gonadal outlier).

Medium pelagic fish

The small tunas exhibited a shared pattern of high individual consistency in the liver and eyes, where variability driven primarily by specific outliers rather than general dispersion. In *A. thazard*, the **liver**, **dorsal fin**, **vertebrae**, and **eyes** formed highly synchronised clusters, restricting individual variability to ranges $\leq 0.8\text{‰}$. Slight dispersion was observed only in the **muscle** and **gonads** (1.1–1.3‰), driven in both cases by single individuals shifting toward depleted carbon values. *E. alletteratus* followed a similar trend of metabolic cohesion; the **liver** and **eyes** maintained tight clustering (main groups $\leq 0.5\text{‰}$), though single outliers expanded the total ranges. However, this species differed in the following tissues: the **dorsal fin** recorded a

massive expansion (3.3‰) driven by an extreme outlier, and the **vertebrae** and **gonads** displayed moderate, continuous dispersion (1.3–2.0‰) without distinct subgroupings.

Sarda sarda and *Scomber colias* displayed a contrasting pattern. While both species shared highest individual consistency in the **muscle** and **eyes** (ranges narrowing to $\leq 0.9\text{‰}$), they diverged significantly in the drivers of their variability. In *S. sarda*, the **liver** exhibited the widest dispersion for the species (3.4‰); while intermediate ranges were observed in the **vertebrae** (1.6‰) and **dorsal fin** (1.0‰). In the case of *S. colias* the variability was driven by structural tissues: the **dorsal fin** and **vertebrae** recorded the maximum intra-specific variability (3.0‰ and 2.0‰, respectively) characterised by distinct subgroups. By contrast, the **liver** and **gonads** displayed moderate, continuous dispersion ($\approx 1.5\text{‰}$).

Large pelagic fish

Mola mola revealed the most pronounced isotopic heterogeneity in the entire study, characterised by a marked contrast between highly variable and cohesive tissues. Intra-specific dispersion was maximised in the **dorsal fin** (4.2‰, $n = 11$) and **liver** (3.5‰, $n = 11$), where distributions were defined by dense clusters at one extreme and distinct outliers at the other. In contrast, the remaining tissues exhibited significantly reduced dispersion ($\leq 1.2\text{‰}$), following a gradient of cohesion from the **eyes** (0.5‰, $n=2$) and **gonads** (0.8‰, $n=4$) to the **vertebrae** (1.0‰, $n=4$) and **muscle** (1.2‰, $n=11$). In particular, the gonads formed a tightly grouped point of similarity ($\approx -16.0\text{‰ } \delta^{13}\text{C}$), while the muscle showed high cohesion with even spacing among individuals.

Prionace glauca ($n=2$) and *Tetrapturus belone* ($n=6$) followed a shared pattern where the **dorsal fin** drove the maximum intra-specific variability, while **muscle** and **eyes** acted as centers of cohesion. In both species, the dorsal fin recorded the widest dispersion (*P. glauca*: 1.2‰; *T. belone*: 2.3‰), characterized by segregation into distinct subgroups or outliers. Intermediate dispersion was observed in the **liver** and **gonads** (1.2–1.5‰), where individuals often split into pairs or clusters. Conversely, the **muscle** and **eyes** demonstrated high individual consistency: *T. belone* eyes formed a tightly grouped cluster (0.4‰), and *P. glauca* recorded its narrowest ranges in the

muscle and eyes ($\leq 0.3\text{‰}$). Notably, the **vertebrae** showed divergent behaviors between the two: they converged into a synchronized point for *P. glauca*, whereas they maintained moderate dispersion (1.1‰) for *T. belone*.

Nitrogen trajectories

The paired-line plots for $\delta^{15}\text{N}$ were characterised by a dynamic pattern, where specific tissues maintained high cohesion while others drove significant intra-specific variability (**Figure 3.3.4.b**).

Small pelagic fish

The nitrogen profiles of *Engraulis encrasicolus* and *Sardina pilchardus* ($n=6$ each) revealed that while most tissues remained cohesive, variability was concentrated in specific tissues. In *E. encrasicolus*, the **liver** recorded the maximum intra-specific variability for the species (3.9‰), driven by a single outlier at the minimum value ($5.3\text{‰ } \delta^{15}\text{N}$) while the remaining individuals clustered between 6.9‰ and $9.2\text{‰ } \delta^{15}\text{N}$. Moderate dispersion was observed in the **vertebrae** and **eyes** ($1.4\text{--}1.5\text{‰}$), where individuals were partitioned into subgroups or separated by outliers. In contrast, the gonads and muscle functioned as centers of cohesion. The **gonads** formed a highly cohesive 0.5‰ cluster centered at $\approx 7.4\text{‰ } \delta^{15}\text{N}$, representing the lowest variability for the species, while the **muscle** maintained high synchrony (0.6‰) with a slight split into two distinct groups.

Sardina pilchardus displayed a different pattern of metabolic stability. Unlike the anchovy, its **liver** and **gonads** exhibited high individual cohesion, with narrow ranges of $0.7\text{--}0.8\text{‰}$ and no distinct outliers. The **muscle** similarly maintained low variability (0.7‰), with most individuals forming a dense band at the upper end of the distribution. However, variability increased significantly in the structural tissues. The **vertebrae** expanded to a 1.7‰ range due to a minimum outlier, and the **eyes** recorded the maximum intra-specific variability for the entire guild (3.0‰). This extreme eye dispersion was driven by a single individual at $10.6\text{‰ } \delta^{15}\text{N}$, segregated from the main cluster ($7.6\text{--}8.8\text{‰ } \delta^{15}\text{N}$).

Medium pelagic fish

Auxis thazard, *Euthynnus alletteratus* and *Scomber colias* exhibited a shared pattern of metabolic stability, characterised by high individual cohesion in the **liver** (0.9–1.3‰). However, they diverged in their primary drivers of isotopic variation. In the tunas, the **eyes** recorded the maximum variability (≈ 2.5 –2.6‰), driven by distinct outlier pairs or similar spacing. In contrast, for *S. colias*, the **gonads** displayed the widest dispersion (2.3‰), driven by strong segregation between a single minimum outlier and the main cluster. Regarding structural tissues, the species were separated according to their stability. *E. alletteratus* was characterised by outlier-driven variability across the **dorsal fin**, **vertebrae**, and **muscle** (≈ 2.5 –2.6‰). Conversely, *A. thazard* and *S. colias* maintained high synchrony in these tissues. *S. colias* achieved its highest consistency in the **muscle** (0.9‰), while *A. thazard* was most cohesive in the **gonads** (0.7‰).

Sarda sarda displayed a distinctive pattern where the **liver** drove the maximum intra-specific dispersion (1.9‰), comparable only to the **eyes** (1.8‰). Notably, the eyes were characterised by even spacing across the nitrogen gradient rather than the distinct clustering seen in the other species. The remaining tissues exhibited moderate cohesion, following a gradient of stability from the **vertebrae** (1.3‰) to the **muscle** and **gonads** (1.4‰), and finally the **dorsal fin** (1.5‰). These distributions were generally defined by a primary cluster of individuals with single outliers at the range margins, rather than the massive segregation observed in *Euthynnus*.

Large pelagic fish

Mola mola exhibited a distinct pattern where isotopic dispersion was maximised in the muscle and vertebrae. The **muscle** ($n=11$) recorded the highest intra-specific variability for the species (3.1‰), driven by a single extreme outlier (11.6‰ $\delta^{15}\text{N}$) segregated from the primary cluster. Similarly, the **vertebrae** ($n=4$) displayed wide, uneven spacing across a 2.3‰ gradient. The **dorsal fin** followed this trend of expansion, converging toward higher values within a 1.7‰ range. In contrast, the **liver** ($n=11$) and **gonads** ($n=4$) showed moderate dispersion (1.1–1.3‰), characterised by even spacing or tight clustering near the mean. The highest individual cohesion was observed in the **eyes** ($n=2$), which maintained a narrow 0.4‰ range at the upper end of the trophic range.

Prionace glauca ($n=2$) and *Tetrapturus belone* ($n=6$) displayed a shared pattern where variability was maximised in the **vertebrae** (1.6‰ for both), contrasting with high consistency in the muscle. *P. glauca* recorded its lowest variability in the **muscle** (0.5‰), while *T. belone* formed a cohesive group (1.1‰) interrupted only by a single outlier. The species diverged notably in the **dorsal fin**, *P. glauca* exhibited wide dispersion (1.5‰), while *T. belone* recorded its maximum stability (0.6‰), forming a tight band. In the remaining tissues, *T. belone* maintained moderate dispersion in the **liver** and **gonads** (1.1‰–1.2‰), similar to its **eyes** (1.5‰). *P. glauca* displayed a split in the **gonads** (1.4‰) compared to higher cohesion in the **liver** (0.8‰) and **eyes** (1.0‰).

3.3.5 Ontogenetic trophic shifts (testing H2b)

Size-based isotopic trends

To test the hypothesis of size-dependent trophic shifts, linear regression analyses were performed between body length (BL) and $\delta^{15}\text{N}$ signatures for each species and tissue. The regression scatter plots are visualised in **Figure 3.3.5**, and the complete statistical parameters (Slope β , R^2 , p-value) are detailed in **Appendix D (Table D.5)**.

Statistical relationships

A significant positive relationship between body size and $\delta^{15}\text{N}$ was observed in the muscle tissue of *Prionace glauca* ($n=58$, $\beta=0.013$, $p=0.008$), indicating a consistent enrichment in nitrogen signatures as individuals increase in size. Conversely, *Scomber colias* exhibited a significant negative relationship in vertebrae ($n=6$, $\beta=-0.180$, $p=0.016$), with body size explaining 80% of the isotopic variance ($R^2=0.80$).

Marginal positive trends ($0.05 < p < 0.10$) were identified in the archival eye lenses of the medium pelagic guild, specifically in *Auxis thazard* ($\beta=0.530$, $p=0.053$) and *Scomber colias* ($\beta=0.132$, $p=0.092$), indicating an increase in nitrogen signatures recorded in this tissue as length increases.

While limited sample sizes ($n=6$) and narrow size ranges precluded statistical significance for other taxa, distinct directional trends were noted. *Engraulis*

engrasicolus displayed positive slopes across five of six tissues, particularly in muscle ($\beta=0.459$, $p=0.10$) and liver ($\beta=0.564$, $p=0.74$). In contrast, *Sardina pilchardus* exhibited negative slopes, including in muscle ($\beta=-0.328$, $p=0.12$) and dorsal fin ($\beta=-0.524$, $p=0.19$). For this species, eye lenses provided the only positive ontogenetic slope ($\beta=0.240$, $p=0.80$), consistent with the pattern observed in the medium pelagic eyes tissues.

Figure 3.3.5. Linear regression relationships between Body length (cm) and $\delta^{15}\text{N}$ signatures across functional groups and tissues. Solid lines indicate statistically significant ($p<0.05$) or marginal ($0.05<p<0.10$) relationships. **Note** the varying scales on x-axes to accommodate species-specific size ranges and trophic positions.

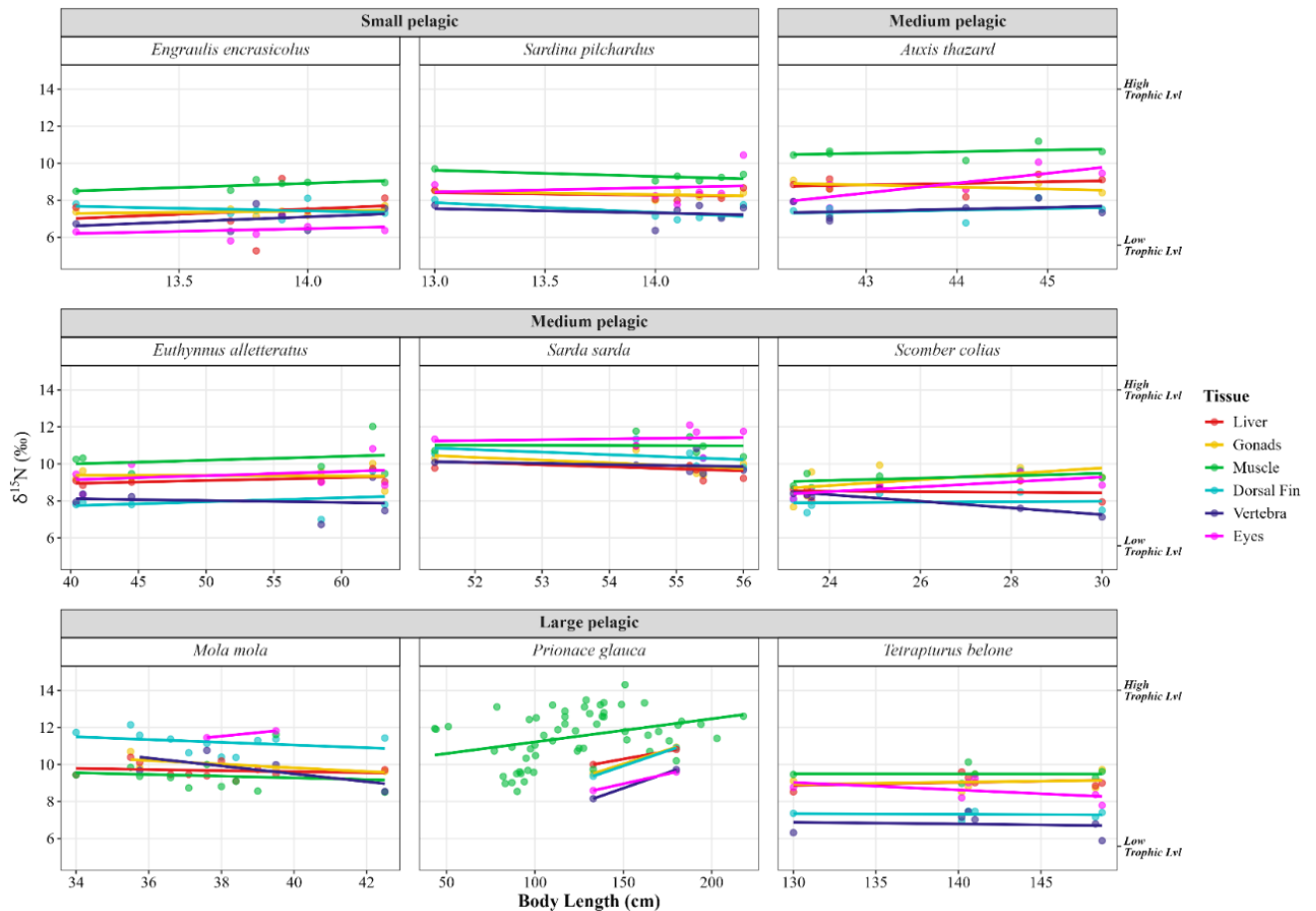
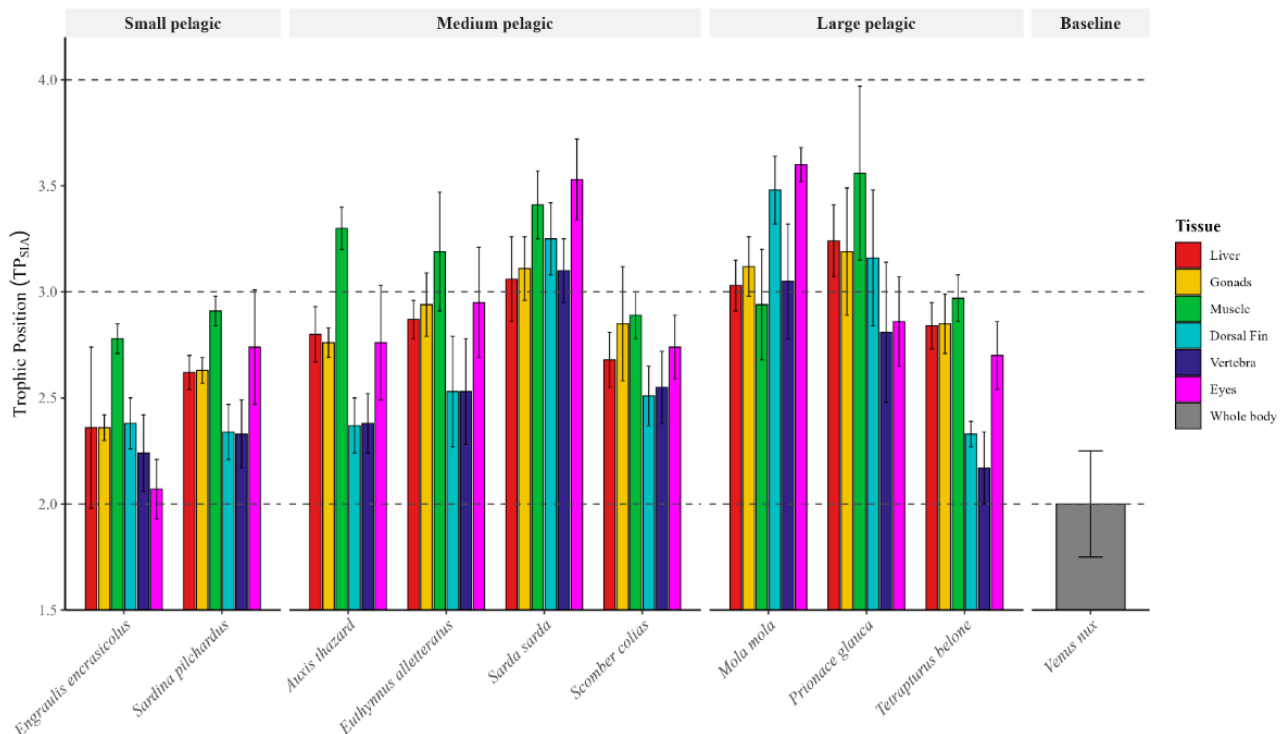


Figure 3.3.6. Trophic position estimates (TP_{SIA}) across distinct tissue types for the Azohía pelagic community. Estimates are calculated relative to the filter-feeding bivalve *Venus nux* ($TP_{baseline}=2.0$). Bars represent the mean $TP \pm SD$. Dashed lines indicate discrete trophic levels (Secondary consumer = 3.0; Tertiary consumer = 4.0).



3.3.6 Trophic positions estimates

Trophic positions (TP_{SIA}) were estimated for each species and tissue relative to the local isotopic baseline (*Venus nux*, $TP_{Baseline}=2.0$). The mean trophic positions across tissues are visualised in **Figure 3.3.6**, and the corresponding values are detailed in **Table 3.3**.

Small pelagic fish

Engraulis encrasicolus and *Sardina pilchardus* occupied the lowest trophic levels in the community. TP_{SIA} values for *E. encrasicolus* ranged from a minimum of 2.07 in eyes to a maximum of 2.78 in muscle. *S. pilchardus* exhibited a similar range, varying from 2.33 in vertebrae to 2.91 in muscle. In both species, muscle recorded the highest isotopic position relative to other tissues.

Medium pelagic fish

Scomber colias exhibited intermediate values ($TP_{SIA_Muscle} = 2.89$), while the tunas *Auxis thazard* and *Euthynnus alletteratus* displayed distinct tissue stratification. In these scombrids, structural tissues recorded moderate estimates ($TP_{SIA_Vert} \approx 2.3$ - 2.5), whereas muscle TP was significantly higher ($TP_{SIA_Muscle} \approx 3.2$ - 3.3). *Sarda sarda* occupied the highest trophic position within this functional group, with values consistently exceeding 3.0 across most tissues ($TP_{SIA_Muscle} = 3.41$; $TP_{SIA_Eyes} = 3.53$),

Large pelagic fish

Prionace glauca reached a peak TP_{SIA} of 3.56 in muscle tissue, the highest recorded value in the dataset, significantly exceeding its vertebral collagen average ($TP_{SIA_Vert} = 2.81$). *Mola mola* displayed consistently high values across tissues ($TP_{SIA_Muscle} = 2.94$), with a peak in eye lenses ($TP_{SIA_Eyes} = 3.60$). *Tetrapturus belone* showed a marked discrepancy between long-term and short-term tissues, where vertebral TP_{SIA} was relatively low (2.17) compared to muscle (2.97).

Tissue-specific patterns

A systematic pattern of higher trophic position in muscle tissue relative to vertebrae was observed across most species and functional groups. In species such as *Auxis thazard*, *Euthynnus alletteratus*, and *Tetrapturus belone*, muscle TP_{SIA} was approximately 0.5 to 0.9 trophic levels higher than vertebral TP_{SIA} . Muscle consistently represented the highest trophic position for every species sampled, while vertebrae and dorsal fin generally recorded the lowest TP estimates (with the exception of *Sarda sarda* and *Mola mola*).

3.3.7 Diet composition (SCA)

The relative frequency of occurrence (%F) of prey categories is visualised in **Figure 3.3.7**, and detailed metrics including percentage number (%N), percentage weight (%W), and the index of relative importance (%IRI) are provided in **Table 3.5**.

Figure 3.3.7. Relative frequency of occurrence (%F) of major prey taxa in marine species from La Azohía (Case study 2). Predators are grouped by functional category (Small, medium and large pelagic fish), with sample sizes of full stomachs shown in parentheses (n). Nematoda parasites are indicated by a crosshatch pattern. All values represent pooled stomach contents at the species level. Detailed composition is provided in **Table 3.3.7**.

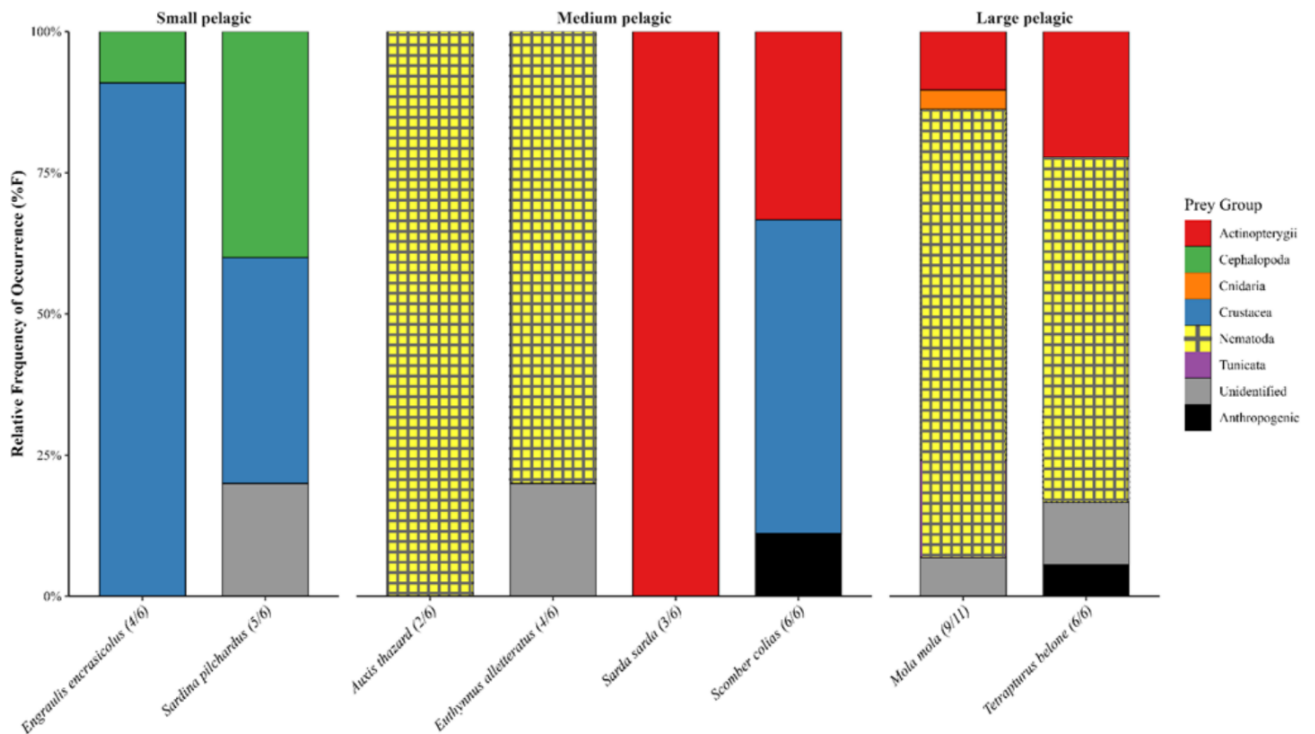


Table 3.3.7. Diet composition and stomach content analysis of pelagic species sampled in La Azohía (Case study 2). Predators are grouped by functional category; parentheses (n/tot) represent the number of non-empty stomachs relative to the total number of individuals examined for each species. Diet metrics include percentage by number (%N), weight (%W), frequency of occurrence (%F), and the index of relative importance (%IRI). Prey items are displayed hierarchically by Phylum and sorted by taxonomic resolution. For taxa lacking specific weight measurements, a nominal weight ($W \approx 0$) was assigned to facilitate %IRI calculations; for these entries, the index primarily reflects the contribution of %N and %F. Parasites and non-biological remains are reported for ecological completeness but distinguished from primary dietary items.

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
Small pelagic									
<i>Engraulis encrasicolus</i> (4/6)	Crustacea	Copepoda	Harpacticoida	Ectinosomatidae	Microsetella	8.33	0.00	16.67	2.15
	Crustacea	Copepoda	Calanoida	Calanidae		37.50	0.00	33.33	19.35
	Crustacea	Copepoda	Cyclopoida	Oncaeidae		16.67	0.00	16.67	4.30
	Crustacea	Malacostraca	Decapoda			4.17	0.00	16.67	1.08
	Crustacea	Malacostraca	Euphausiacea			12.50	0.00	33.33	6.45
	Crustacea	Malacostraca	Isopoda			8.33	0.00	16.67	2.15
	Crustacea					4.17	0.00	16.67	1.08
	TOTAL Crustacea					95.83	0.00	66.67	98.92
	MOLLUSCA	Cephalopoda				4.17	0.00	16.67	1.08
<i>Sardina pilchardus</i> (5/6)	Crustacea	Malacostraca	Euphausiacea			18.18	0.00	33.33	5.62
	Crustacea	Copepoda				9.09	0.00	16.67	1.41
	Crustacea					9.09	0.00	16.67	1.41
	TOTAL Crustacea					36.36	0.00	50.00	16.86

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
	MOLLUSCA	Cephalopoda				45.45	59.83	66.67	65.10
	Unidentified					18.18	40.17	33.33	18.04
Medium pelagic									
<i>Auxis thazard</i> (2/6)	NEMATODA					0.00	0.00	33.33	0.00
<i>Euthynnus alletteratus</i> (4/6)	NEMATODA					0.00	0.00	66.67	0.00
	Unidentified					100.00	0.00	16.67	100.00
<i>Sarda sarda</i> (3/6)	CHORDATA	Actinopterygii				100.00	100.00	50.00	100.00
<i>Scomber colias</i> (6/6)	CHORDATA	Actinopterygii	Clupeiformes	Engraulidae	<i>Engraulis encrasicolus</i>	1.45	85.54	16.67	7.89
	CHORDATA	Actinopterygii				4.35	11.77	83.33	7.31
	TOTAL CHORDATA					5.80	97.31	100.00	56.08
	Crustacea	Malacostraca	Amphipoda	Hyperiididae		4.35	0.10	33.33	0.81
	Crustacea	Malacostraca	Decapoda			30.43	0.87	66.67	11.35
	Crustacea					59.42	1.72	66.67	22.17
	TOTAL Crustacea					94.20	2.69	83.33	43.92

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
	Anthropogenic - Fish net					0.00	0.00	16.67	0.00
	Anthropogenic - Plastic					0.00	0.00	16.67	0.00
	TOTAL Anthropogenic					0.00	0.00	33.33	0.00
Large pelagic									
<i>Mola mola</i> (9/11)	CHORDATA	Actinopterygii	Pleuronectiformes			5.88	4.29	9.09	2.13
	CHORDATA	Actinopterygii				11.76	3.43	18.18	6.35
	TOTAL CHORDATA					17.65	7.73	18.18	10.60
	CHORDATA / Tunicata	Ascidiacea	Phlebobranchia	Ascidiidae	Ascidia	23.53	21.46	27.27	28.19
	CHORDATA / Tunicata	Thaliacea	Salpida	Salpidae		35.29	4.29	18.18	16.54
	TOTAL CHORDATA / Tunicata					58.82	25.75	27.27	53.00
	CNIDARIA	Scyphozoa				5.88	0.00	9.09	1.23

Predator species	Prey species								
	PHYLUM / Subphylum	Class	Order	Family	Genus / Species	%N	%W	%F	%IRI
	NEMATODA	Secernentea	Rhabditida	Anisakidae	Anisakis	0.00	0.00	81.82	0.00
	Unidentified					17.65	66.52	18.18	35.17
<i>Tetrapturus belone</i> (6/6)	CHORDATA	Actinopterygii				20.00	99.38	66.67	55.44
	Crustacea	Malacostraca	Amphipoda			4.00	0.00	16.67	0.46
	MOLLUSCA	Cephalopoda	Octopoda	Argonautidae	<i>Argonauta argo</i>	12.00	0.00	33.33	2.79
	MOLLUSCA	Cephalopoda				44.00	0.62	83.33	25.91
	TOTAL MOLLUSCA					56.00	0.62	100.00	39.45
	NEMATODA	Secernentea	Rhabditida	Anisakidae	Anisakis	0.00	0.00	33.33	0.00
	NEMATODA					0.00	0.00	16.67	0.00
	TOTAL NEMATODA					0.00	0.00	50.00	0.00
	Unidentified					20.00	0.00	33.33	4.64
	Anthropogenic - Plastic					0.00	0.00	16.67	0.00

Small pelagic fishes

Engraulis encrasicolus ($n = 4$) exhibited a diet dominated by crustaceans (%IRI = 98.92). Copepods were the most abundant prey item, particularly the order Calanoida (%N = 37.50%), followed by euphausiids, which occurred in 33.33% of stomachs. *Sardina pilchardus* ($n = 5$) presented a different composition characterised by a high proportion of cephalopods (%IRI = 65.10) and unidentified digested organic matter (%IRI = 18.04). While crustaceans such as euphausiids and copepods were present, cephalopods accounted for most of the stomach content biomass (%W = 59.83%).

Medium pelagic fish

Sarda sarda ($n = 3$) prey items were identified exclusively as Actinopterygii (osteichthyes) across all metrics (%N, %W, %F = 100%). *Scomber colias* *Scomber colias* ($n = 6$) displayed a discrepancy between numerical abundance and biomass; numerically, the diet was dominated by crustaceans (%N = 94.20%), specifically decapods and amphipods, whereas Actinopterygii (primarily *Engraulis encrasicolus*) constituted most of the diet by weight (%W = 97.31%). Anthropogenic debris, including plastics and fishing net fibers, was recorded in two of the six stomachs (33.33%) analysed of *Scomber colias*. Stomach contents for the tunas *Auxis thazard* ($n = 2$) and *Euthynnus alletteratus* ($n = 4$) were highly digested; *Euthynnus alletteratus* contents were classified as 100% unidentified material, and nematode parasites were frequently observed in both species (%F = 33–67%).

Large pelagic fish

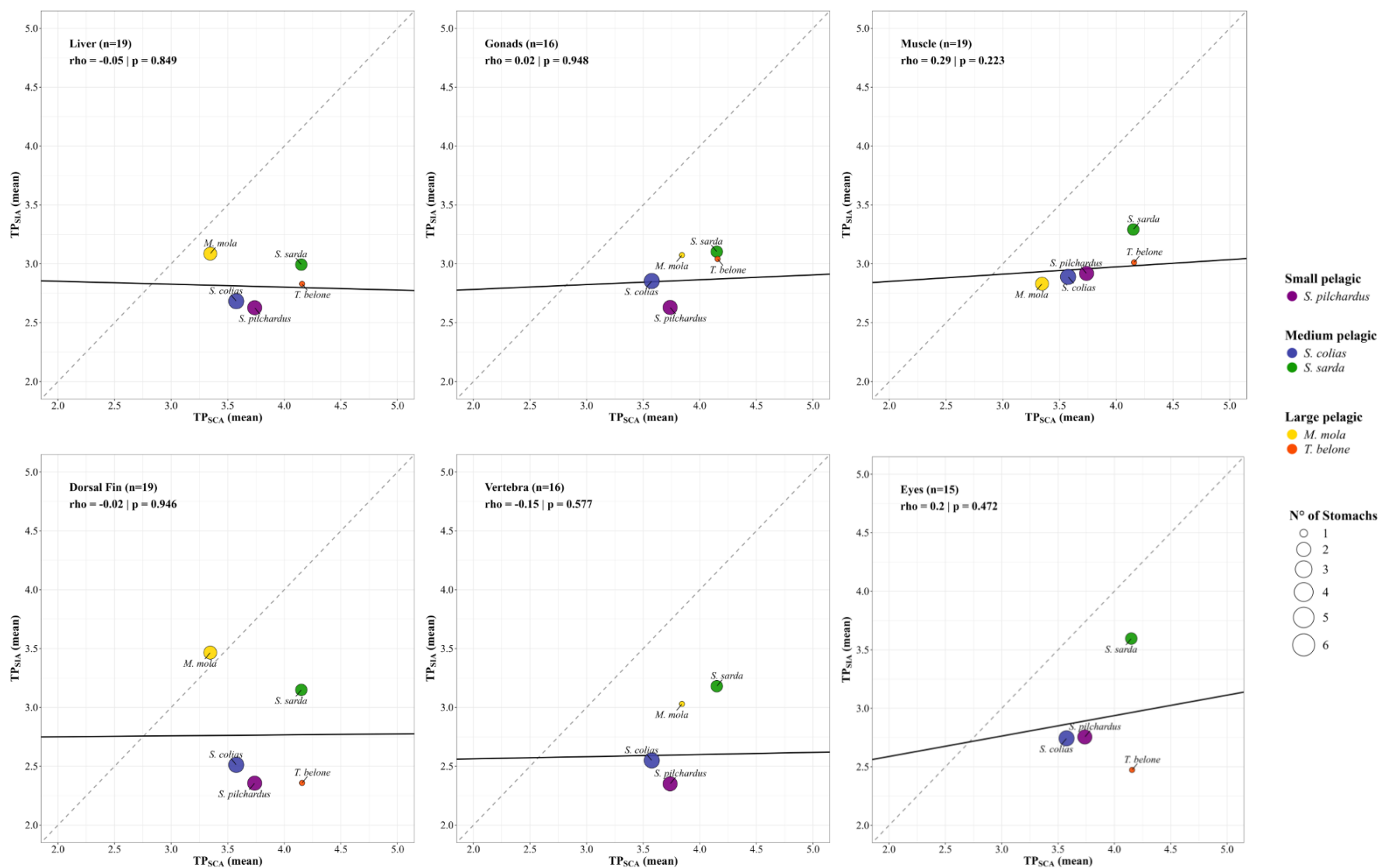
Mola mola ($n = 9$) consumed primarily tunicates, with Ascidiacea and Thaliacea (salps) constituting the most important prey group (%IRI = 53.00%), followed by unidentified digested material (%IRI = 35.17%) and Cnidaria (%IRI = 1.23%). *Tetrapturus belone* ($n = 6$) contents were dominated by Actinopterygii, which accounted for 99.38% of the prey weight (%IRI = 55.44%). Cephalopods, including *Argonauta argo*, were a secondary prey item (%IRI = 39.45%) and occurred in in all

six (100%) of the sampled stomachs. Anthropogenic plastic debris was identified in one of the six (16.67%) analysed *Tetrapturus belone* individuals.

3.3.8 Trophic position comparisons (TP_{SIA} vs TP_{SCA})

To validate the isotopic analysis against recent feeding activity, the mean isotopic trophic position (TP_{SIA}) calculated with isotopic values of the different tissues was compared with the trophic position calculated from stomach content analysis (TP_{SCA}) for the subset of species where both data were available. The tissue-specific relationship is visualised in **Figure 3.3.8**, and the complete comparative dataset is provided in **Appendix D (Table D.6)**.

Figure 3.3.8.a. Spearman rank correlation analyses comparing trophic positions estimated from stable isotope analysis (TP_{SIA}) and stomach content analysis (TP_{SCA}) across different tissues for the Azohía fish community. Data points represent species-level mean values, with point size proportional to the number of non-empty stomachs analysed (N_{SCA}). Spearman correlation coefficients (ρ) and the solid regression lines were calculated using individual-level data to incorporate intra-specific variability. The dashed line represents the 1:1 relationship. Plots are faceted by tissue type, in this order: liver, gonads, muscle, dorsal fin, vertebrae, and eyes.



Statistical comparison

Spearman rank correlation tests performed separately for each tissue revealed no significant linear relationship between TP_{SCA} and TP_{SIA} ($p > 0.05$ for all tissues). **Muscle** tissue exhibited the highest positive, though non-significant, correlation coefficient ($\rho = 0.29$, $p = 0.223$), followed by the **eye lenses** ($\rho = 0.20$, $p = 0.472$). Other tissues displayed weak negative or negligible trends, with coefficients ranging from $\rho = -0.02$ in the **dorsal fin** ($p = 0.946$) to $\rho = -0.15$ in the vertebrae ($p = 0.577$). Despite the lack of statistical correlation, a systematic discrepancy was observed across the community: TP_{SCA} estimates were consistently higher than TP_{SIA} estimates for nearly all analysed species.

Species-specific discrepancies

Discrepancies between methods varied by functional group. Within the small pelagic fish, *Sardina pilchardus* exhibited a consistent downward offset in isotopic estimates, with the **muscle** tissue showing a difference of 0.82 trophic levels ($TP_{SCA} = 3.74$ vs. $TP_{SIA} = 2.92$). In the medium pelagic guild, *Sarda sarda* and *Scomber colias* displayed similar patterns, with TP_{SIA} values in muscle being 0.74 and 0.69 levels lower than their respective stomach-based estimates. Among large pelagic fish, *Tetrapturus belone* recorded the most pronounced gap; the stomach-based estimate ($TP_{SCA} = 4.16$, $n = 1$) was more than one full trophic level higher than the isotopic signal recorded across all six tissues, which ranged from 2.17 in the vertebrae to 2.97 in the muscle. In contrast, *Mola mola* showed the highest convergence between the two methods with a difference of only 0.41 in the muscle ($TP_{SCA} = 3.35$ vs. $TP_{SIA} = 2.94$) and slight inverse trend in dorsal fin tissues ($Diff = 0.13$).

Chapter 4: Discussion

The present study aimed to reconstruct the trophic dynamics of pelagic fish communities in the western Mediterranean Sea. The integration of stomach content analysis (SCA) and multi-tissue stable isotope analysis (SIA) in the first case study 1 produced important information about the trophic structure of this pelagic community of the Ebro Delta shelf region and highlighted some divergences between trophic position estimates. These divergences could be explained by the information provided by these two methods, with distinct temporal resolution according to stomach digestion and assimilation by different tissues, distinguishing between a momentary snapshot of ingestion and the long-term assimilation of the diet. The second case study presented a novel multi-tissue analysis, which revealed that isotopic signals change across tissues and that can be related to species- and tissue-specific assimilation, to the turnover rate and to the physiology of the species.

4.1 CASE STUDY 1: TROPHIC STRUCTURE AND METHODOLOGICAL COMPARISONS

4.1.1 Trophic structure

In order to compare the trophic structure of both study areas, muscle tissue was selected as the standard comparison tissue.

In the Ebro Delta, the trophic structure was characterised by a narrow distribution along the $\delta^{13}\text{C}$ axis while there was a clear vertical segregation along the $\delta^{15}\text{N}$. The narrow carbon range indicates that, with the exception of specific outliers, most species relied on a shared habitat use (Barnes et al., 2009; Albo-Puigserver et al., 2016). On the contrary, the clear vertical segregation can suggest a different use of trophic resources across the pelagic community (Navarro et al., 2011). This stratification may be linked to the heterogeneity in maturity stages among the sampled species, which led to the presence of juveniles of medium and large pelagic fish in the

samples analysed, which placed them lower than expected in the food web (Albo-Puigserver et al., 2016; Ortega et al., 2025).

Small pelagic fish shared a similar isotopic niche, consistent with previous studies conducted in this area (Albo-Puigserver et al., 2016). However, *Sardina pilchardus* showed a wider isotopic niche (SEAc), primarily driven by high variability in $\delta^{13}\text{C}$. As the literature suggests, this is likely due to its greater feeding flexibility and the fact that sardines can switch from particulate (selective) to filter-feeding (non-selective) (James, 1986; Albo-Puigserver et al., 2016). This allows them to access to a broader and more diverse of prey, from phytoplankton to mesozooplankton (Costalago et al., 2012; Albo-Puigserver et al., 2016). The switch, from one feeding mode to the other, depends primarily on the concentration of food (Bulgakova, 1996) and can also shift in response to changes in the presence and abundance of particular prey items (Van der Lingen, 1994). These changes in feeding mode were also visible when comparing the $\delta^{15}\text{N}$ values with size, where for sardine a negative relationship was visible.

The medium and large pelagic fish were divided into two groups based on the isotopic niche: the adults (e.g. *Scomber* spp., *T. trachurus*) occupied higher trophic levels, while the juveniles (e.g. *A. rochei*, *S. dumerili*) showed the lowest trophic levels across the whole community, overlapping with primary consumers. These low trophic levels are likely due to the growth dilution process (Hesslein et al., 1993). In rapidly growing juveniles, isotopic turnover is mainly driven by the synthesis of new tissue rather than the metabolic replacement of existing tissue (Herzka & Holt, 2000). Consequently, although stomach content analysis confirms that they had recently consumed fish, the isotopic signal in muscle tissue is still affected by the biomass synthesised during their previous zooplanktivorous stage (Hesslein et al., 1993; Herzka & Holt, 2000). In this case, the higher trophic diet of piscivory has not been assimilated completely from the muscle tissue in order to dilute the older, lower isotopic signature (Herzka, 2005; Madigan et al., 2012). In contrast, for *Thunnus thynnus* juveniles, the low trophic level directly reflected a zooplanktivorous diet, as

confirmed by stomach content analysis conducted in other studies (Varela et al., 2018; Pardo et al., 2025).

Differences in $\delta^{13}\text{C}$ defined the habitat use of the the pelagic community, with *Trachurus trachurus* showing depleted values reflecting the use of offshore habitats, in line with what is reported for this species (Rumolo et al., 2017). Meanwhile, *Ariosoma balearicum* showed enriched $\delta^{13}\text{C}$ values consistent with benthic habitats, which are also used by conger species (Mytilineou et al., 2005).

If we compare the trophic structure of the Ebro Delta obtained with muscle samples with the one obtained from the second case study, from La Azohía, in the second one we see a narrower distribution along the $\delta^{15}\text{N}$ axis. The pelagic fish of the La Azohía community showed a continuous distribution of $\delta^{15}\text{N}$ values within a narrower $\delta^{13}\text{C}$ range. This indicates that, while all species relied on a shared habitat use, the community was highly structured vertically, with species segregated across distinct trophic levels (Valls et al., 2014).

In this second case study, the species community was structured by small pelagic species (*S. pilchardus*, *E. encrasicolus*) at the base of the food web. While medium pelagic tunas (*A. thazard*, *E. alletteratus*, *S. sarda*) occupied an intermediate trophic level. The higher $\delta^{15}\text{N}$ values placed them higher than the small pelagic fish, consistent with a piscivorous diet of secondary consumers (Navarro et al., 2017). However, *Scomber colias*, *Tetrapturus belone* and *Mola mola* showed low trophic levels overlapping with plankton feeders rather than other pelagic fish of their functional group; likely related to a higher consumption of zooplankton (Bachiller and Irigoien, 2015). As noted by Cardona et al. (2012) and Bachiller and Irigoien (2015), the assimilation of these low-trophic prey items effectively lowers the trophic level of these pelagic fish. In this context, the blue shark seemed to occupy the highest trophic level in the pelagic community, confirming its role as a top predator (Estrada et al., 2006). The stratified trophic structure of the La Azohía was similar to that described for another oligotrophic environment: the Balearic Sea (Valls et al., 2014).

In both study areas, important diet overlap was observed not only within functional groups but also across them. This overlap is consistent with the findings of Bachiller and Irigoien (2015), who observed that zooplankton remain a shared prey across the entire pelagic community. As Bachiller and Irigoien (2013) explained, this pattern can be linked to the development of the feeding apparatus: as fish grow, they gain the ability to consume larger prey without losing the efficiency to retain small zooplankton, that continue to be an important part of their diet. Consequently, larger pelagic fish do not replace their diet but rather expand it, maintaining a significant diet overlap with smaller pelagic fish (Bachiller and Irigoien, 2013).

4.1.2 Comparison between trophic positions estimates

The comparison of the trophic level positions, derived from stomach content analysis (TP_{SCA}) and stable isotope analysis (TP_{SLA}), revealed divergences between both methods across both study areas. For the Ebro Delta, the comparison was conducted using the TP calculated from muscle tissue; while in La Azohía, the comparison included all sampled tissues.

In general, trophic positions derived from stomach contents (TP_{SCA}) were consistently higher than those estimated from stable isotopes (TP_{SLA}). However, the degree of this variation differed among the functional groups which could be related to their different prey source range (Bode et al., 2004). These divergences could be also related to the fact that to calculate TP_{SCA} we used the trophic level values of a food web model that included both juvenile and adult individuals within each modelled species (Coll et al., 2024), which can overestimate the trophic level of the group and the TP_{SCA} .

In these regards, small pelagic fish showed the highest consistency between the trophic position estimate from the two methods. In the Ebro Delta, species like *Sardina pilchardus* showed similar trophic position estimates, suggesting that for planktivores on lower trophic levels, the dietary snapshot provided by stomach contents aligns well with the long-term assimilated diet in the tissues (Bode et al., 2004). In contrast, medium and large pelagic fish exhibited more pronounced mismatches, where stomach

contents analyse frequently overestimated the trophic position compared to the trophic level calculated through stable isotopes. This mismatch could also indicate that these pelagic species were feeding at higher trophic level than their assimilated diet actually reflected (Baker et al., 2014). For example, in the Ebro Delta, juvenile fish species had high content of fish species in their stomachs (TP_{SCA}), despite their trophic level calculated through isotopic signatures is low (TP_{SLA}). A similar pattern was observed in La Azohía, where large pelagic fish appeared to be top predators based on the stomach content analysis, while isotopic signatures placed them significantly lower.

Another driver of this discrepancy could be related to the difference in temporal resolution provided by the two methods used (Baker et al., 2014). Stomach content analysis provides a snapshot of the most recent prey ingested by the individuals, whereas stable isotopes integrate dietary information over weeks or months (Baker et al., 2014; Post, 2002). The consistently higher TP_{SCA} suggests that, while pelagic fish consume high-trophic prey (such as small fish), their long-term assimilated diet indicated lower-trophic prey (Fernández-Corredor et al., 2023). However, these lower-trophic preys are less visible and often underestimated in stomach samples because their soft tissues are digested rapidly (Fernández-Corredor et al., 2023). While prey items with resistant structures, such as fish bones and cephalopod beaks, persist longer in the stomach and are easily identified, leading to an increase in the trophic position (TP_{SCA}) (Hyslop, 1980). In contrast, preys with soft tissue (e.g. gelatinous zooplankton) are rapidly digested and frequently classified as unidentified material, causing it to be underestimated in SCA despite being fully assimilated into the tissues (Hyslop, 1980; Varela et al., 2019).

These combined mechanisms are particularly relevant for the species sampled in La Azohía, like *Tetrapturus belone* and *Scomber colias*, described by Cardona et al. (2012), as opportunistic consumers of gelatinous plankton in the Mediterranean. The rapid digestion of these low-trophic organisms likely resulted in their underestimation in stomach contents, leading to an unbalanced calculation of the TP_{SCA} .

In conclusion, the findings of this study did not support the initial hypothesis: the two methodologies did not produce broadly consistent trophic estimates as

expected. However, the divergence between the two methodologies highlighted their complementary information about the diet of species (Post, 2002; Varela et al. 2019). While SCA captures recent feeding events, SIA reveals more diet sources supporting the pelagic fish's population (Fernández-Corredor et al., 2023). This distinction is evident in the Ebro Data results: while stomach content confirm that juvenile pelagic fish are currently consuming fish, their isotopic signatures remained low reflecting the zooplankton -based diet during previous weeks/months (Varela et al., 2018). Thus, the combination and comparison of both methodologies was necessary in order to determine recent and long-term assimilated diet (Bachiller et al., 2020; Fernández-Corredor et al., 2023).

4.2 CASE STUDY 2: TISSUES COMPARISON

4.2.1 General patterns of isotopic variability across tissues

The second objective of this work aimed to understand how isotopic signals vary across different tissues. Following the principles of physiological turnover described by Navarro et al. (2020), this study evaluated the isotopic signals of metabolically active tissues (e.g., liver, muscle), expected to act as temporal windows of recent diet, and inert tissues (vertebrae, eyes), expected to serve as biological archives.

The results for the **metabolically active** tissue group revealed a pattern driven primarily by variability $\delta^{13}\text{C}$. Since these tissues renew their cells over short to medium timeframes, they reflect the resources consumed during recent weeks, months or year of the species life (Madigan et al., 2012). The high variation observed along the $\delta^{13}\text{C}$ axis indicates that, for the pelagic community sampled there was a high variability in the habitat use during this time frame (Barnes et al., 2009). This suggests a dynamic movement of fish across different habitats or feeding at different depths in the water column (Trueman et al., 2012). Consequently, these tissues acted as indicators of the spatial dynamics of the community sampled.

In contrast, the **inert tissues** (vertebrae, eye lenses) expected to function as long-term archives, revealed patterns driven primarily by variability in $\delta^{15}\text{N}$. Because these

tissues do not undergo metabolic replacement once formed, the analysis of whole homogenized tissue provides an average isotopic signal accumulated throughout the fish's life (Wallace et al., 2014; Young et al., 2021). The significant variation observed along the $\delta^{15}\text{N}$ axis reflects the average trophic history of the individuals sampled (Tzadik et al., 2017). Thus, rather than spatial movement, these tissues provided insight into the ontogenetic shifts (changes in diet with growth), preserving the lower nitrogen values typical of early life stages (Vander Zanden et al., 1998).

4.2.2 Specific tissue variability

To evaluate the inter-individual isotopic variability between tissues, a comparison of niche widths (SEA_C) was performed. Across the pelagic community, each tissue revealed general patterns according to their distinct physiological properties. The **liver** exhibited the highest intra-individual variability and widest isotopic niches (SEA_C) across most species. This high variability is driven by its rapid protein turnover rate, which integrates dietary signals over a scale of days to weeks, allowing it to detect recent shifts in diet or habitats that are not yet visible in slower turnover rates (Madigan et al., 2012; Buchheister & Latour, 2010).

Regarding reproductive tissues, **gonads** generally showed low isotopic variability across the community. This isotopic consistency suggests that energy for gamete production is derived from a similar trophic diet across different individuals (Minagawa and Wada, 1984; Post et al., 2002; Yoneda et al., 2025). Indeed, the isotopic variability in gonads tissue may be affected from the different strategies of the species' reproductive strategy (Yoneda et al., 2025). In case of capital breeding species, utilise stored energy over extended periods for reproductive purpose (McBride et al., 2015). This mobilisation of energy reserves likely stabilises the short-term isotopic signals, probably resulting in lower isotopic variability across individuals (McBride et al., 2015; Yoneda et al., 2025).

In contrast, the higher $\delta^{13}\text{C}$ variability observed in medium pelagic fish may suggest an income breeding strategy, where reproduction is supported by current energy intake (McBride et al., 2015). Since $\delta^{13}\text{C}$ serves as a proxy for spatial habitat

use (Trueman et al., 2012), this variability indicates that individuals were synthesizing gametes while exploiting spatially distinct habitats, mirroring the diet consumed before spawning (McBride et al., 2015; Yoneda et al., 2025).

Muscle consistently recorded the highest trophic position (TP_{SIA}) estimates. This can be due to its slower turnover rate than metabolically active tissues described above. As established by MacNeil et al. (2006), isotopic turnover in muscle is driven largely by the addition of new tissue from the growth of the species; rather than the rapid metabolic replacement observed in organs like the liver (Madigan et al., 2012). Consequently, it acts as a stable reference for the assimilated diet over months, suggesting that the main diet supporting growth is higher in the trophic position compared to what resulted from the other tissues (Pinnegar & Polunin, 1999; Madigan et al., 2012).

Dorsal fin was characterised mainly by horizontal dispersion along the $\delta^{13}\text{C}$ axis, exhibiting significantly higher variability than muscle. Although fin tissue is widely proposed as a non-lethal surrogate for muscle due to assumed similarities in isotopic turnover (Willis et al., 2013), our results revealed a distinct divergence in isotopic variability between these two tissues. This discrepancy is likely due to the sampling methodology, which involved homogenising the entire fin tissue (Willis et al., 2013). The fin tissue comprises two distinct physiological components, the soft inter-ray membrane, which turns over at a rate comparable to muscle, and the mineralised rays, which grow by accretion (Willis et al., 2013). Thus, the resulting isotopic signal can represent a mix of recent metabolic activity and long-term assimilation from the two different fin tissues (Tzadik et al., 2017; Willis et al., 2013). Consequently, instead of mirroring the stable muscle signal, this mixed tissue may have captured a wider range of medium-term habitat heterogeneity (Barnes et al., 2009).

Finally, although both **vertebrae** and **eyes** acted as archival tissues, they showed distinct isotopic values and different levels of intra-individual variability. This difference can be driven by their specific growth mechanisms. Teleost vertebrae undergo metabolic remodelling to maintain mineral homeostasis (Tzadik et al., 2017).

Consequently, vertebral bone provides an average signal of the fish's lifetime diet, which can be partially overwritten by later metabolic activity (Andrews et al., 2023). In contrast, eye lens grows through the sequential addition of concentric protein layers (crystallins) which remain chemically stable after deposition (Wallace et al., 2014). Therefore, the homogenised eye lens may have preserved the complete lifetime trophic history, including distinct signals from early life stages (Curtis et al., 2020; Andrews et al., 2023). The retention of these isotopically distinct early-life layers can explain the higher inter-individual isotopic variability observed in the eye lens compared to the vertebrae results (Wallace et al., 2014; Tzadik et al., 2017).

Beyond these tissues general differences, across the majority of the species a consistent pattern was observed: **muscle** tissue showed significantly higher levels of $\delta^{15}\text{N}$ in comparison with results from the **vertebrae**. The observed differences can be linked to the differences in the biochemical composition between these two tissues. As Pinnegar and Polunin (1999) observed, muscle tissue requires for its main functions and maintenance high levels of specific amino acids (e.g., glutamic acid), typically enriched in $\delta^{15}\text{N}$. In contrast, bone collagen of the vertebrae is rich in glycine, which typically carries lower $\delta^{15}\text{N}$ values (Tzadik et al., 2017). The *Mola mola* was the only exception to this general pattern. The results showed an inverted isotopic pattern, where structural tissues were enriched in $\delta^{15}\text{N}$ relative to the muscle. This distinct pattern is probably due to the unique anatomical characteristics of the sunfish (Davenport et al., 2018). Unlike the dense muscle of other fish, sunfish muscle is exceptionally watery (>80% water), which may result in a lower retention of heavy isotopes ($\delta^{15}\text{N}$) compared to the keratin structural tissues (Davenport et al., 2018; Cardona et al., 2012).

4.2.3 Species-specific patterns of isotopic variability

Consistent with the general physiological patterns described above, the liver exhibited significantly broader isotopic niches (SEAC) and higher intra-individual variability compared to muscle and archive tissues. This pattern was most evident in *Sarda sarda*, *Engraulis encrasicolus*, and *Mola mola*.

For *Sarda sarda*, the SEAc value of the liver tissue was driven primarily by $\delta^{13}\text{C}$ variability. This result contrasts with the findings of Navarro et al. (2020), who typically observed wider niches in the fin due to seasonal integration. However, in this study the results in $\delta^{13}\text{C}$ of the liver demonstrates a high level of recent, short-term variability in habitat use (Barnes et al., 2009; Madigan et al., 2012). Since coastal waters are typically enriched in $\delta^{13}\text{C}$ compared to offshore waters, this variability suggests that individuals captured in the same location had recently arrived from different inshore and offshore habitats in the days/weeks prior to capture (Barnes et al., 2009). For *E. encrasicolus*, the liver showed a high significant variation in both isotopes: this could indicate a recent broader diet and habitat use (Madigan et al., 2012; Bachiller et al., 2020). This isotopic variability aligns with the stomach content analysis for this species, which revealed a diverse diet of prey items (Bachiller et al., 2020). While the liver captured the recent diet, the muscle (with a slower turnover tissue) showed less isotopic variability. This finding is probably related to the fact that the muscle integrated these recent feeding activities into a stable, long-term isotopic signal (Boecklen et al., 2011; Madigan et al., 2012).

In the case of *Mola mola*, the wider isotopic liver niche and the high isotopic intra-individual variability in habitat use indicate individuals were exploiting different habitats (Barnes et al., 2009; Madigan et al., 2012). This can be related to the findings of Syväranta et al. (2012), who suggested that the sunfish are not restricted to offshore pelagic waters but also actively forage in coastal or benthic habitats. Consequently, for these three species (*S. sarda*, *E. encrasicolus*, *M. mola*), the high variability in active tissues allows us to suggest that these species were feeding a diverse range of prey resources in the days or weeks before capture (Barnes et al., 2009; Madigan et al., 2012).

On the contrary, *Auxis thazard*, *Euthynnus alletteratus* and *Sardina pilchardus* unexpectedly showed a higher isotopic variability in archive tissues compared to metabolically active ones. For example, in *S. pilchardus* the high variability in the eyes was driven significantly by $\delta^{13}\text{C}$. This spatial heterogeneity likely reflects the

variability of habitats use over the average lifetime of the analysed individuals (Barnes et al., 2009; Wallace et al., 2014).

As a result, it can be concluded that the expectation of higher inter-individual variability in metabolically active tissues is not generally applicable but instead may be dependent on the interaction between **tissue physiology** and **individuals' ecological dynamics**. For species like *E. encrasicolus* and *M. mola* the high variability in active tissues likely indicated a recent heterogeneity in diet and habitat; acting as a high-resolution snapshot of recent species-environmental interactions (Buchheister & Latour, 2010; Syväranta et al., 2012). However, for *A. thazard*, *E. alletteratus*, and *S. pilchardus*, the low isotopic variability across the individuals likely indicated that individuals were sharing the same resources in the short term (Madigan et al., 2012). In contrast, the wider isotopic niches observed in the archive tissues of these species may reflect the integration of both dietary and spatial information over longer temporal scales (Wallace et al., 2014). This aligns with the findings of Navarro et al. (2020), who observed that tissues with slower turnover rates (such as the dorsal fin) may also exhibit wider isotopic niches because they capture long-term seasonal variability or shifts in habitats that are homogenised by the fast turnover of active tissues.

4.2.4 Ontogenetic shifts

This study also shows that archival tissues can be used to reconstruct ontogenetic shifts exhibiting $\delta^{15}\text{N}$ enrichment correlated with body size. A significant positive correlation was observed in the muscle tissue of the top predator, *Prionace glauca*. This indicates that as these sharks grow, they progressively shift to eating larger, higher-trophic prey (Estrada et al., 2006). Regarding the specific capacity of archival tissues, a similar positive correlation was observed in the eye lenses of *Auxis thazard*. Although this pattern was significant only for this species, it suggests that eye lenses can integrate the signal of trophic enrichment associated with ontogenetic growth (Hussey et al., 2014; Wallace et al., 2014).

However, for species such as *Scomber colias*, the expected pattern was significantly reversed. Instead of increasing, nitrogen levels in the vertebrae

significantly decreased in larger individuals. This pattern may indicate the presence of a baseline trophic shift through the species life (Vander Zanden & Rasmussen, 1999). Nitrogen levels at the base of the food web can vary significantly between regions and years (Valls et al., 2014). The depletion of $\delta^{15}\text{N}$ in vertebrae tissue of larger mackerel may suggest that adults have moved into waters with a naturally lower nitrogen baseline compared to their previous habitats (Herzka, 2005). Consequently, this may result in a negative correlation between $\delta^{15}\text{N}$ and body size.

Finally, it is necessary to highlight that isotopic variability is not exclusively a product of trophic ecology but also depends on tissue-specific physiological mechanisms, such as turnover rates and diet-tissue discrimination factors (DTDF) (Navarro et al., 2020; Madigan et al., 2012). Therefore, a deeper exploration of the multi-tissue approach is essential for distinguishing between recent feeding activity and habitat use, and the long-term species dynamics (Nielsen et al., 2017; Pardo et al., 2025).

4.3 LIMITATIONS OF THE STUDY

The findings of this thesis must be interpreted within the context of specific methodological constraints. A primary limitation was the restricted sample size of the individuals that were sampled, and the short temporal resolution of the sampling events. For several functional groups, particularly large pelagic predators, low number of samples limited the statistical power to detect specific isotopic patterns, while some of the predators were only sampled as juveniles. Furthermore, because sampling was conducted during a single season, the results represent a momentary dietary event, rather than a year window.

The study applied standard fish equations for lipid correction and tissue turnover to species with different metabolic rates. Finally, the outcome of life-history reconstruction was constrained by the bulk analysis of vertebrae and eyes. Further analysis that looks at bands and regions of vertebra and eyes could provide further insights into the ontogenetic shifts of isotopic values (Wallace et al., 2014; Andrews et al., 2023).

4.4 FUTURE DIRECTIONS

To address the outlined limitations and refine the ecological resolution of future studies, several methodological advances should be implemented.

First, to overcome the digestive bias of SCA, future research could employ DNA metabarcoding. As demonstrated by Berry et al. (2015) and Albaina et al. (2016), this genetic approach has the capacity to identify prey that are digested rapidly and are not visually identifiable, making it useful for more accurate prey identification.

Second, there is an insufficient number of studies on species-specific tissue turnover rates and lipid content, and thus we relied on a general assumption. Future experimental work should aim to calibrate species-specific lipid correction equations (Logan et al., 2008) and turnover rates (Madigan et al., 2012).

Furthermore, to better resolve the timing of ontogenetic shifts, analysis should transition from bulk tissues to the laminar analysis of eye lenses and vertebral rings. By physically peeling the concentric protein layers of the lens and growth rings from vertebra it is possible to reconstruct a high-resolution chronological record from the juvenile to the adult life stage (Wallace et al., 2014; Quaeck-Davies et al., 2018; Young et al., 2021).

Finally, to better interpret the differences in the habitat use observed, future studies could benefit from applying Compound-Specific Isotope Analysis of Amino Acids (CSIA-AA). As reviewed by McMahon & McCarthy (2016), this method distinguishes between "source" amino acids (reflecting baseline geography) and "trophic" amino acids (reflecting diet), allowing researchers to effectively filter out the noise of spatial baseline shifts. There are already some studies applied to the study area that show promising results (Gimenez et al., 2023).

Chapter 5: Conclusions

Overall, this work generated interesting new insights about the structure of pelagic communities of the western Mediterranean Sea and contributed a few key insights to the ecology of these species.

According to the results, the **divergence between trophic position estimates** was mostly driven by the distinct temporal resolutions of the methods. This mismatch highlights the distinction between a momentary snapshot of ingestion and the long-term assimilation of the diet. While stomach contents often overestimated trophic positions due to digestive bias and predation on larger prey, stable isotopes revealed that generally, larger fish species growth is largely sustained by lower-trophic resources. This confirms that relying on a single metric could fail to capture trophic dynamics.

Secondly, the multi-tissue approach revealed that **isotopic variability** across tissues is likely shaped by the differences between species- and tissue-specific assimilation and turnover rate. The hypothesis that metabolically active tissues should be more variable was not generally true. For species exhibiting recent heterogeneity in diet or habitat, the rapid turnover of the liver successfully captured these short-term dynamics. However, for other species this pattern was reversed, highlighting the importance of examining multiple temporal windows.

Furthermore, this study highlighted that the reconstructed trophic structure is not static but varies according to the tissue analysed. Consequently, the **multi-tissue approach** can improve our understanding of complex trophic structures dynamics, which would be missed by a single-tissue analysis. By integrating tissue-specific turnover rates, future assessments could move beyond static trophic webs and reconstruct life-history dynamics, revealing how species interact with their environment at different timescales.

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Appendices

Appendix A

(1) Taxonomic keys for SCA findings

Table A.1 Taxonomic key mapping identified prey items from stomach content analysis (SCA) of Ebro Delta pelagic predators, with corresponding functional groups within the western Mediterranean Ecopath model (GSA 06 and GSA 07). Trophic position (TP) values are derived from the estimated trophic levels (TL) of the balanced ecosystem model presented in Coll et al. (2024). **Note:** Numbers in parentheses within the "Ecopath Functional Group" column refer to the specific functional group indices as established in Table 1 of the Western Mediterranean food-web model.

Major group	Identified prey item	Ecopath Functional Group (ID)	TP
Chordata	Actinopterygii	Small demersal fishes (continental shelf) (31)	3.15
	<i>Engraulis encrasicolus</i>	Juvenile European anchovy (39)	2.97
	<i>Micromesistius poutassou</i>	Blue whiting (24)	3.23
Cnidaria	Scyphozoa	Gelatinous plankton (7)	2.75
Crustacean	Squillidae	Spot tail mantis shrimp (10)	2.99
	Caridea	Other shrimps (13)	2.88
	Brachyura	Crabs (14)	2.82
	Galatheididae	Suprabenthos (9)	2.13
	Anomura	Suprabenthos (9)	2.13
	Decapoda	Suprabenthos (9)	2.13
	Crustacea	Suprabenthos (9)	2.13
	Liocarcinus	Suprabenthos (9)	2.13
	Hyperiididae	Macrozooplankton (6)	2.61
	Euphausiacea	Macrozooplankton (6)	2.61
	<i>Phronima sedentaria</i>	Macrozooplankton (6)	2.61
	Sergestidae	Macrozooplankton (6)	2.61
	Caprellidae	Macrozooplankton (6)	2.61
	Amphipoda	Macrozooplankton (6)	2.61
	Isopoda	Macrozooplankton (6)	2.61
Mollusca	Alloteuthis	Benthopelagic cephalopods (continental shelf) (18)	3.51
	<i>Illex coindietti</i>	Benthopelagic cephalopods (continental shelf) (18)	3.51
	<i>Cymbulia peronii</i>	Gelatinous plankton (7)	2.75
Other invertebrates	Other invertebrates	Macrozooplankton (6)	2.61
Unidentified	Unidentified	Detritus (57)	1.00
Anthropogenic	Plastic	Detritus (57)	1.00

Table A.2 Taxonomic key for the Azohía dataset mapping identified prey taxa to functional groups within the Western Mediterranean Ecopath model (GSA 06 and GSA 07). Trophic Position (TP) values correspond to the estimated Trophic Levels (TL) provided in the balanced food-web model by Coll et al. (2024). Parasites and certain non-prey items are excluded from trophic calculations but reported for ecological completeness. **Note:** Numbers in parentheses within the "Ecopath Functional Group" column refer to the specific functional group indices as established in Table 1 of the Western Mediterranean food-web model.

Major Group	Identified Prey Item	Ecopath Functional Group (ID)	TP
Chordata	Osteichthyes	Small demersal fishes (31)	3.15
	Flat fish larvae	Small demersal fishes (31)	3.15
	<i>Engraulis encrasicolus</i>	Juvenile European anchovy (39)	2.97
Cnidaria and Tunicata	Scyphozoa	Gelatinous plankton (7)	2.75
	Salpidae	Gelatinous plankton (7)	2.75
	Ascidia	Gelatinous plankton (7)	2.75
Crustacean	Crustacea	Suprabenthos (9)	2.13
	Decapoda	Suprabenthos (9)	2.13
	Amphipoda	Macrozooplankton (6)	2.61
	Isopoda	Macrozooplankton (6)	2.61
	Euphausiacea	Macrozooplankton (6)	2.61
	Hyperiididae	Macrozooplankton (6)	2.61
	Copepoda	Micro and mesozooplankton (5)	2.02
	Calanidae	Micro and mesozooplankton (5)	2.02
	Oncaeidae	Micro and mesozooplankton (5)	2.02
	<i>Microsetella</i>	Micro and mesozooplankton (5)	2.02
Mollusca	Cephalopoda	Benthopelagic cephalopods (18)	3.51
	Cephalopod (eye lens)	Benthopelagic cephalopods (18)	3.51
	<i>Argonauta argo</i>	Benthopelagic cephalopods (18)	3.51
Nematoda	Nematoda	Excluded from trophic analysis	NA
	Parasites	Excluded from trophic analysis	NA
	<i>Anisakis</i>	Excluded from trophic analysis	NA
Unidentified	Unidentified	Detritus (57)	1.00
Anthropogenic	Plastic and Fish net	Detritus (57)	1.00

Appendix B

(1) Statistical data quality checks

Table B.1. Results of statistical assumption tests for Study 1 (Ebro Delta). (a) Normality results (Shapiro-Wilk) tested across species. (b) Homogeneity of variance results (Levene's test) tested across species. (-): non-calculated. (*): non-normal / non-homogeneous.

(a) Normality Check

Species	n	Shapiro N p-value	Shapiro C p-value
<i>Ariosoma balearicum</i>	2	-	-
<i>Auxis rochei</i>	8	0.66	0.04*
<i>Boops boops</i>	5	0.70	0.44
<i>Cheilopogon exsiliens</i>	4	0.21	0.16
<i>Engraulis encrasicolus</i>	5	0.15	0.66
<i>Illex coindetii</i>	25	0.41	0.15
<i>Pyrosoma atlanticum</i>	6	0.08	0.98
<i>Remora remora</i>	1	-	-
<i>Sarda sarda</i>	7	< 0.01*	0.06
<i>Sardina pilchardus</i>	5	0.78	< 0.01*
<i>Scomber colias</i>	24	< 0.01*	0.14
<i>Scomber scombrus</i>	29	< 0.01*	0.26
<i>Seriola dumerili</i>	5	0.32	0.69
<i>Thunnus thynnus</i>	12	< 0.01*	0.74
<i>Trachurus trachurus</i>	29	0.27	0.03*
<i>Tylosurus acus</i>	1	-	-
<i>Venus nux</i>	6	0.79	0.77
<i>Xiphias gladius</i>	3	0.41	0.82

(b) Homogeneity check

Isotope	Df	F_Value	p_Value
d ¹⁵ N*	17	2.4	0.002
d ¹³ C _{corr}	17	1.6	0.07

Table B.2. Results of statistical assumption tests for Study 2 (La Azohía). (a) Normality results (Shapiro-Wilk) tested across tissues for each species. (b) Homogeneity of variance results (Levene's test) tested across tissues. (-): non-calculated. (*): non-normal.

(a) Normality check

Species	Tissue	n	Shapiro_W_C	Shapiro C p-value	Shapiro_W_N	Shapiro N p-value
<i>Auxis thazard</i>	Dorsal fin	6	0.69	< 0.01*	0.96	0.84
	Eyes	6	0.88	0.25	0.98	0.92
	Gonads	6	0.94	0.62	0.95	0.71
	Liver	6	0.95	0.72	0.96	0.82
	Muscle	6	0.91	0.41	0.93	0.56
	Vertebra	6	0.95	0.70	0.95	0.76
<i>Engraulis encrasicolus</i>	Dorsal fin	6	0.89	0.33	0.96	0.84
	Eyes	6	0.86	0.19	0.94	0.70
	Gonads	6	0.78	0.04*	0.95	0.74
	Liver	6	0.87	0.24	0.97	0.89
	Muscle	6	0.70	0.01*	0.86	0.19
	Vertebra	6	0.90	0.36	0.92	0.50
<i>Euthynnus alletteratus</i>	Dorsal fin	6	0.82	0.08	0.78	0.04*

	Eyes	6	0.84	0.13	0.97	0.87
	Gonads	6	0.82	0.09	0.95	0.78
	Liver	6	0.77	0.03*	0.74	0.02*
	Muscle	6	0.92	0.51	0.81	0.07
	Vertebra	6	0.87	0.21	0.99	0.98
<i>Mola mola</i>	Dorsal fin	11	0.79	0.01*	0.93	0.41
	Eyes	2	-	-	-	-
	Gonads	4	0.97	0.86	0.79	0.08
	Liver	11	0.81	0.01*	0.91	0.27
	Muscle	11	0.95	0.69	0.85	0.04*
	Vertebra	4	0.92	0.56	0.98	0.88
<i>Prionace glauca</i>	Dorsal fin	2	-	-	-	-
	Eyes	2	-	-	-	-
	Gonads	2	-	-	-	-
	Liver	2	-	-	-	-
	Muscle	58	0.89	< 0.01*	0.97	0.10
	Vertebra	2	-	-	-	-
<i>Sarda sarda</i>	Dorsal fin	6	0.92	0.47	0.92	0.53
	Eyes	6	0.85	0.15	0.95	0.70
	Gonads	6	0.93	0.59	0.96	0.83
	Liver	6	0.96	0.79	0.89	0.33
	Muscle	6	0.82	0.09	0.94	0.62
	Vertebra	6	0.92	0.50	0.87	0.22

<i>Sardina pilchardus</i>	Dorsal fin	6	0.81	0.07	0.83	0.10
	Eyes	6	0.93	0.61	0.83	0.11
	Gonads	6	0.91	0.46	0.97	0.86
	Liver	6	0.92	0.49	0.91	0.46
	Muscle	6	0.93	0.56	0.92	0.49
	Vertebra	6	0.97	0.92	0.83	0.10
<i>Scomber colias</i>	Dorsal fin	6	0.78	0.04*	0.92	0.53
	Eyes	6	0.94	0.66	0.94	0.63
	Gonads	6	0.82	0.09	0.88	0.25
	Liver	6	0.97	0.90	0.91	0.44
	Muscle	6	0.80	0.06	0.91	0.42
	Vertebra	6	0.88	0.28	0.90	0.36
<i>Tetrapturus belone</i>	Dorsal fin	6	0.85	0.15	0.85	0.15
	Eyes	6	0.93	0.56	0.97	0.91
	Gonads	6	0.80	0.06	0.95	0.73
	Liver	6	0.77	0.03*	0.98	0.96
	Muscle	6	0.95	0.73	0.96	0.80
	Vertebra	6	0.94	0.68	0.96	0.80

(b) Homogeneity check

Species	Levene C p-value	Levene N p-value
<i>Auxis thazard</i>	0.43	0.07
<i>Engraulis encrasicolus</i>	0.08	0.05
<i>Euthynnus alletteratus</i>	0.90	0.56
<i>Mola mola</i>	0.47	0.55
<i>Prionace glauca</i>	0.72	0.59
<i>Sarda sarda</i>	0.06	0.99
<i>Sardina pilchardus</i>	0.22	0.54
<i>Scomber colias</i>	0.23	0.54
<i>Tetrapturus belone</i>	0.09	0.29

Appendix C

(1) Statistical results for Case study 1 (Ebro Delta)

Table C.1. Pairwise statistical comparisons of **(a)** $\delta^{13}\text{C}_{\text{corr}}$ and **(b)** $\delta^{15}\text{N}$ among pelagic fish species in the Ebro Delta. Carbon differences (a) were assessed via **ANOVA + Tukey HSD** (Mean Diff in ‰). Nitrogen differences (b) were assessed via **Kruskal-Wallis + Dunn-Holm** (Z-Score). Adjusted p-values (P_{adj}) are shown; (*) denotes significance ($p < 0.05$).

a) Carbon ($\delta^{13}\text{C}$) comparisons - ANOVA + Tukey

Species1	Species2	Mean Diff (‰)	P_Value_Adj
<i>Boops boops</i>	<i>Auxis rochei</i>	-0.84	< 0.01 *
<i>Cheilopogon exsiliens</i>	<i>Auxis rochei</i>	-0.16	1.00
<i>Engraulis encrasicolus</i>	<i>Auxis rochei</i>	-0.64	0.02 *

<i>Illex coindetii</i>	<i>Auxis rochei</i>	-0.41	0.06
<i>Sarda sarda</i>	<i>Auxis rochei</i>	-0.05	1.00
<i>Sardina pilchardus</i>	<i>Auxis rochei</i>	-0.18	1.00
<i>Scomber colias</i>	<i>Auxis rochei</i>	-0.49	< 0.01 *
<i>Scomber scombrus</i>	<i>Auxis rochei</i>	-0.71	< 0.01 *
<i>Seriola dumerili</i>	<i>Auxis rochei</i>	-0.00	1.00
<i>Thunnus thynnus</i>	<i>Auxis rochei</i>	-0.09	1.00
<i>Trachurus trachurus</i>	<i>Auxis rochei</i>	-1.44	< 0.01 *
<i>Xiphias gladius</i>	<i>Auxis rochei</i>	0.77	0.02 *
<i>Cheilopogon exsiliens</i>	<i>Boops boops</i>	0.68	0.06
<i>Engraulis encrasicolus</i>	<i>Boops boops</i>	0.20	1.00
<i>Illex coindetii</i>	<i>Boops boops</i>	0.43	0.17
<i>Sarda sarda</i>	<i>Boops boops</i>	0.79	< 0.01 *
<i>Sardina pilchardus</i>	<i>Boops boops</i>	0.66	0.04 *
<i>Scomber colias</i>	<i>Boops boops</i>	0.36	0.47
<i>Scomber scombrus</i>	<i>Boops boops</i>	0.13	1.00
<i>Seriola dumerili</i>	<i>Boops boops</i>	0.84	< 0.01 *
<i>Thunnus thynnus</i>	<i>Boops boops</i>	0.75	< 0.01 *
<i>Trachurus trachurus</i>	<i>Boops boops</i>	-0.60	< 0.01 *
<i>Xiphias gladius</i>	<i>Boops boops</i>	1.61	< 0.01 *
<i>Engraulis encrasicolus</i>	<i>Cheilopogon exsiliens</i>	-0.48	0.49
<i>Illex coindetii</i>	<i>Cheilopogon exsiliens</i>	-0.25	0.95
<i>Sarda sarda</i>	<i>Cheilopogon exsiliens</i>	0.11	1.00
<i>Sardina pilchardus</i>	<i>Cheilopogon exsiliens</i>	-0.02	1.00
<i>Scomber colias</i>	<i>Cheilopogon exsiliens</i>	-0.32	0.75

<i>Scomber scombrus</i>	<i>Cheilopogon exsiliens</i>	-0.55	0.05 *
<i>Seriola dumerili</i>	<i>Cheilopogon exsiliens</i>	0.16	1.00
<i>Thunnus thynnus</i>	<i>Cheilopogon exsiliens</i>	0.07	1.00
<i>Trachurus trachurus</i>	<i>Cheilopogon exsiliens</i>	-1.28	< 0.01 *
<i>Xiphias gladius</i>	<i>Cheilopogon exsiliens</i>	0.93	< 0.01 *
<i>Illex coindetii</i>	<i>Engraulis encrasicolus</i>	0.23	0.94
<i>Sarda sarda</i>	<i>Engraulis encrasicolus</i>	0.59	0.06
<i>Sardina pilchardus</i>	<i>Engraulis encrasicolus</i>	0.46	0.46
<i>Scomber colias</i>	<i>Engraulis encrasicolus</i>	0.16	1.00
<i>Scomber scombrus</i>	<i>Engraulis encrasicolus</i>	-0.07	1.00
<i>Seriola dumerili</i>	<i>Engraulis encrasicolus</i>	0.64	0.06
<i>Thunnus thynnus</i>	<i>Engraulis encrasicolus</i>	0.55	0.04 *
<i>Trachurus trachurus</i>	<i>Engraulis encrasicolus</i>	-0.80	< 0.01 *
<i>Xiphias gladius</i>	<i>Engraulis encrasicolus</i>	1.41	< 0.01 *
<i>Sarda sarda</i>	<i>Illex coindetii</i>	0.36	0.22
<i>Sardina pilchardus</i>	<i>Illex coindetii</i>	0.23	0.95
<i>Scomber colias</i>	<i>Illex coindetii</i>	-0.08	1.00
<i>Scomber scombrus</i>	<i>Illex coindetii</i>	-0.30	0.02 *
<i>Seriola dumerili</i>	<i>Illex coindetii</i>	0.41	0.25
<i>Thunnus thynnus</i>	<i>Illex coindetii</i>	0.32	0.13
<i>Trachurus trachurus</i>	<i>Illex coindetii</i>	-1.03	< 0.01 *
<i>Xiphias gladius</i>	<i>Illex coindetii</i>	1.18	< 0.01 *
<i>Sardina pilchardus</i>	<i>Sarda sarda</i>	-0.13	1.00
<i>Scomber colias</i>	<i>Sarda sarda</i>	-0.44	0.05
<i>Scomber scombrus</i>	<i>Sarda sarda</i>	-0.66	< 0.01 *

<i>Seriola dumerili</i>	<i>Sarda sarda</i>	0.05	1.00
<i>Thunnus thynnus</i>	<i>Sarda sarda</i>	-0.04	1.00
<i>Trachurus trachurus</i>	<i>Sarda sarda</i>	-1.39	< 0.01 *
<i>Xiphias gladius</i>	<i>Sarda sarda</i>	0.82	< 0.01 *
<i>Scomber colias</i>	<i>Sardina pilchardus</i>	-0.30	0.71
<i>Scomber scombrus</i>	<i>Sardina pilchardus</i>	-0.53	0.03 *
<i>Seriola dumerili</i>	<i>Sardina pilchardus</i>	0.18	1.00
<i>Thunnus thynnus</i>	<i>Sardina pilchardus</i>	0.09	1.00
<i>Trachurus trachurus</i>	<i>Sardina pilchardus</i>	-1.26	< 0.01 *
<i>Xiphias gladius</i>	<i>Sardina pilchardus</i>	0.95	< 0.01 *
<i>Scomber scombrus</i>	<i>Scomber colias</i>	-0.22	0.29
<i>Seriola dumerili</i>	<i>Scomber colias</i>	0.48	0.07
<i>Thunnus thynnus</i>	<i>Scomber colias</i>	0.40	0.02 *
<i>Trachurus trachurus</i>	<i>Scomber colias</i>	-0.95	< 0.01 *
<i>Xiphias gladius</i>	<i>Scomber colias</i>	1.26	< 0.01 *
<i>Seriola dumerili</i>	<i>Scomber scombrus</i>	0.71	< 0.01 *
<i>Thunnus thynnus</i>	<i>Scomber scombrus</i>	0.62	< 0.01 *
<i>Trachurus trachurus</i>	<i>Scomber scombrus</i>	-0.73	< 0.01 *
<i>Xiphias gladius</i>	<i>Scomber scombrus</i>	1.48	< 0.01 *
<i>Thunnus thynnus</i>	<i>Seriola dumerili</i>	-0.09	1.00
<i>Trachurus trachurus</i>	<i>Seriola dumerili</i>	-1.44	< 0.01 *
<i>Xiphias gladius</i>	<i>Seriola dumerili</i>	0.77	0.04 *
<i>Trachurus trachurus</i>	<i>Thunnus thynnus</i>	-1.35	< 0.01 *
<i>Xiphias gladius</i>	<i>Thunnus thynnus</i>	0.86	< 0.01 *
<i>Xiphias gladius</i>	<i>Trachurus trachurus</i>	2.21	< 0.01 *

b) Nitrogen ($\delta^{15}\text{N}$) comparison - Kruskal-Wallis + Dunn - Holm

Species1	Species2	Z-Score	P _{adj}
<i>Auxis rochei</i>	<i>Boops boops</i>	-4.27	< 0.01 *
<i>Auxis rochei</i>	<i>Cheilopogon exsiliens</i>	0.50	1.00
<i>Boops boops</i>	<i>Cheilopogon exsiliens</i>	4.08	< 0.01 *
<i>Auxis rochei</i>	<i>Engraulis encrasicolus</i>	-1.47	1.00
<i>Boops boops</i>	<i>Engraulis encrasicolus</i>	2.52	0.56
<i>Cheilopogon exsiliens</i>	<i>Engraulis encrasicolus</i>	-1.71	1.00
<i>Auxis rochei</i>	<i>Illex coindetii</i>	-2.64	0.43
<i>Boops boops</i>	<i>Illex coindetii</i>	2.78	0.30
<i>Cheilopogon exsiliens</i>	<i>Illex coindetii</i>	-2.56	0.51
<i>Engraulis encrasicolus</i>	<i>Illex coindetii</i>	-0.48	1.00
<i>Auxis rochei</i>	<i>Sarda sarda</i>	0.14	1.00
<i>Boops boops</i>	<i>Sarda sarda</i>	4.27	< 0.01 *
<i>Cheilopogon exsiliens</i>	<i>Sarda sarda</i>	-0.38	1.00
<i>Engraulis encrasicolus</i>	<i>Sarda sarda</i>	1.55	1.00
<i>Illex coindetii</i>	<i>Sarda sarda</i>	2.67	0.39
<i>Auxis rochei</i>	<i>Sardina pilchardus</i>	-2.83	0.26
<i>Boops boops</i>	<i>Sardina pilchardus</i>	1.30	1.00
<i>Cheilopogon exsiliens</i>	<i>Sardina pilchardus</i>	-2.86	0.24
<i>Engraulis encrasicolus</i>	<i>Sardina pilchardus</i>	-1.22	1.00
<i>Illex coindetii</i>	<i>Sardina pilchardus</i>	-1.11	1.00
<i>Sarda sarda</i>	<i>Sardina pilchardus</i>	-2.87	0.23
<i>Auxis rochei</i>	<i>Scomber colias</i>	-4.54	< 0.01 *
<i>Boops boops</i>	<i>Scomber colias</i>	1.18	1.00

<i>Cheilopogon exsiliens</i>	<i>Scomber colias</i>	-4.00	< 0.01 *
<i>Engraulis encrasicolus</i>	<i>Scomber colias</i>	-2.06	1.00
<i>Illex coindetii</i>	<i>Scomber colias</i>	-2.74	0.33
<i>Sarda sarda</i>	<i>Scomber colias</i>	-4.48	< 0.01 *
<i>Sardina pilchardus</i>	<i>Scomber colias</i>	-0.49	1.00
<i>Auxis rochei</i>	<i>Scomber scombrus</i>	-4.18	< 0.01 *
<i>Boops boops</i>	<i>Scomber scombrus</i>	1.58	1.00
<i>Cheilopogon exsiliens</i>	<i>Scomber scombrus</i>	-3.70	0.01 *
<i>Engraulis encrasicolus</i>	<i>Scomber scombrus</i>	-1.71	1.00
<i>Illex coindetii</i>	<i>Scomber scombrus</i>	-2.19	1.00
<i>Sarda sarda</i>	<i>Scomber scombrus</i>	-4.13	< 0.01 *
<i>Sardina pilchardus</i>	<i>Scomber scombrus</i>	-0.11	1.00
<i>Scomber colias</i>	<i>Scomber scombrus</i>	0.67	1.00
<i>Auxis rochei</i>	<i>Seriola dumerili</i>	-0.09	0.93
<i>Boops boops</i>	<i>Seriola dumerili</i>	3.76	0.01 *
<i>Cheilopogon exsiliens</i>	<i>Seriola dumerili</i>	-0.53	1.00
<i>Engraulis encrasicolus</i>	<i>Seriola dumerili</i>	1.24	1.00
<i>Illex coindetii</i>	<i>Seriola dumerili</i>	2.08	1.00
<i>Sarda sarda</i>	<i>Seriola dumerili</i>	-0.21	1.00
<i>Sardina pilchardus</i>	<i>Seriola dumerili</i>	2.47	0.64
<i>Scomber colias</i>	<i>Seriola dumerili</i>	3.67	0.01 *
<i>Scomber scombrus</i>	<i>Seriola dumerili</i>	3.34	0.05 *
<i>Auxis rochei</i>	<i>Thunnus thynnus</i>	-0.51	1.00
<i>Boops boops</i>	<i>Thunnus thynnus</i>	4.13	< 0.01 *
<i>Cheilopogon exsiliens</i>	<i>Thunnus thynnus</i>	-0.94	1.00

<i>Engraulis encrasicolus</i>	<i>Thunnus thynnus</i>	1.14	1.00
<i>Illex coindetii</i>	<i>Thunnus thynnus</i>	2.39	0.78
<i>Sarda sarda</i>	<i>Thunnus thynnus</i>	-0.64	1.00
<i>Sardina pilchardus</i>	<i>Thunnus thynnus</i>	2.59	0.48
<i>Scomber colias</i>	<i>Thunnus thynnus</i>	4.58	< 0.01 *
<i>Scomber scombrus</i>	<i>Thunnus thynnus</i>	4.18	< 0.01 *
<i>Seriola dumerili</i>	<i>Thunnus thynnus</i>	-0.34	1.00
<i>Auxis rochei</i>	<i>Trachurus trachurus</i>	-4.97	< 0.01 *
<i>Boops boops</i>	<i>Trachurus trachurus</i>	0.92	1.00
<i>Cheilopogon exsiliens</i>	<i>Trachurus trachurus</i>	-4.30	< 0.01 *
<i>Engraulis encrasicolus</i>	<i>Trachurus trachurus</i>	-2.37	0.81
<i>Illex coindetii</i>	<i>Trachurus trachurus</i>	-3.35	0.05 *
<i>Sarda sarda</i>	<i>Trachurus trachurus</i>	-4.88	< 0.01 *
<i>Sardina pilchardus</i>	<i>Trachurus trachurus</i>	-0.77	1.00
<i>Scomber colias</i>	<i>Trachurus trachurus</i>	-0.48	1.00
<i>Scomber scombrus</i>	<i>Trachurus trachurus</i>	-1.21	1.00
<i>Seriola dumerili</i>	<i>Trachurus trachurus</i>	-3.99	< 0.01 *
<i>Thunnus thynnus</i>	<i>Trachurus trachurus</i>	-5.10	< 0.01 *
<i>Auxis rochei</i>	<i>Xiphias gladius</i>	-1.73	1.00
<i>Boops boops</i>	<i>Xiphias gladius</i>	1.72	1.00
<i>Cheilopogon exsiliens</i>	<i>Xiphias gladius</i>	-1.94	1.00
<i>Engraulis encrasicolus</i>	<i>Xiphias gladius</i>	-0.46	1.00
<i>Illex coindetii</i>	<i>Xiphias gladius</i>	-0.17	1.00
<i>Sarda sarda</i>	<i>Xiphias gladius</i>	-1.81	1.00
<i>Sardina pilchardus</i>	<i>Xiphias gladius</i>	0.60	1.00

<i>Scomber colias</i>	<i>Xiphias gladius</i>	1.11	1.00
<i>Scomber scombrus</i>	<i>Xiphias gladius</i>	0.81	1.00
<i>Seriola dumerili</i>	<i>Xiphias gladius</i>	-1.54	1.00
<i>Thunnus thynnus</i>	<i>Xiphias gladius</i>	-1.46	1.00
<i>Trachurus trachurus</i>	<i>Xiphias gladius</i>	1.33	1.00

Table C.2. Isotopic niche metrics for the Ebro Delta pelagic community. **Abbreviations:** **n** = sample size; **TA** = Total Area (Convex Hull); **SEA** = Standard ellipse area; **SEAc** = Standard ellipse area corrected for small sample sizes; **SEAb (95% CI)** = Bayesian Standard ellipse area, reported as the posterior mean followed by the 95% Credible Interval in brackets.

Functional Group	Species	n	TA (‰ ²)	SEA (‰ ²)	SEAc (‰ ²)	SEAb (95% CI)
Small Pelagic	<i>Boops boops</i>	5	0.23	0.21	0.28	0.57 [0.24 - 1.31]
	<i>Engraulis encrasicolus</i>	5	0.26	0.24	0.32	0.46 [0.26 - 0.84]
	<i>Sardina pilchardus</i>	5	0.90	0.92	1.23	1.80 [0.51 - 5.38]
Medium Pelagic	<i>Auxis rochei</i>	8	0.53	0.33	0.38	0.69 [0.48 - 0.99]
	<i>Illex coindetii</i>	25	0.32	0.09	0.10	0.77 [0.33 - 1.73]
	<i>Sarda sarda</i>	7	0.27	0.22	0.26	0.35 [0.23 - 0.53]
	<i>Scomber colias</i>	24	1.15	0.33	0.35	0.40 [0.15 - 1.00]
	<i>Scomber scombrus</i>	29	1.84	0.67	0.69	0.30 [0.21 - 0.43]
	<i>Seriola dumerili</i>	5	0.36	0.31	0.42	0.10 [0.06 - 0.14]
	<i>Trachurus trachurus</i>	29	0.97	0.29	0.30	0.24 [0.08 - 0.67]

Large Pelagic	<i>Cheilopogon exsiliens</i>	4	0.16	0.19	0.28	0.25 [0.10 - 0.63]
	<i>Thunnus thynnus</i>	12	0.87	0.43	0.47	1.18 [0.45 - 2.88]
	<i>Xiphias gladius</i>	3	0.53	0.96	1.92	0.28 [0.13 - 0.61]
Baseline	<i>Pyrosoma atlanticum</i>	6	0.71	0.61	0.77	0.28 [0.11 - 0.68]
	<i>Venus nux</i>	6	0.67	0.47	0.59	0.36 [0.18 - 0.71]

Table C.3. Quantitative comparison of trophic position estimates derived from stable isotope analysis (TP_{SIA}) and stomach content analysis (TP_{SCA}) for the Ebro Delta pelagic community. **Note:** N_{SIA} represents the total isotopic sample size, while N_{SCA} denotes the subset of individuals with non-empty stomachs used for dietary calculations.

Category	Species	N_{SIA}	Mean TP_{SIA}	N_{SCA}	Mean TP_{SCA}
Small pelagic	<i>Boops boops</i>	5	3,09	2	3,54
	<i>Sardina pilchardus</i>	5	2,96	4	2,97
	<i>Engraulis encrasicolus</i>	5	2,70	1	3,42
	<i>Cheilopogon exsiliens</i>	4	1,83	1	2
Medium pelagic	<i>Trachurus trachurus</i>	29	3,00	9	3,68
	<i>Scomber colias</i>	24	2,96	22	4,15
	<i>Scomber scombrus</i>	29	2,89	21	3,85
	<i>Auxis rochei</i>	8	2,58	1	4,14
	<i>Seriola dumerili</i>	5	2,41	1	4,15
Large pelagic	<i>Xiphias gladius</i>	3	2,49	1	3,97
	<i>Thunnus thynnus</i>	12	2,16	1	3,61
Other	<i>Illex coindetii</i>	25	2,83	19	4,14

Appendix D

Statistical results for Case study 2 (Azohía)

Table D.1: Pairwise statistical comparisons of (a) lipid-corrected $\delta^{13}\text{C}$ and (b) $\delta^{15}\text{N}$ among tissue types for pelagic species in the Azohía ecosystem. Carbon differences (a) were assessed via **Kruskal-Wallis + Dunn-Holm** (Z-Score). Nitrogen differences (b) were assessed via **ANOVA + Tukey HSD** (Mean Diff in ‰). Adjusted p-values (P_{adj}) are shown; (*) denotes significance ($p < 0.05$).

a) Carbon ($\delta^{13}\text{C}$) comparisons - Kruskal-Wallis + Dunn - Holm

Species	Tissue1	Tissue2	Z-Score	P_{adj}
<i>Auxis thazard</i>	Dorsal fin	Eyes	2.69	0.08
	Dorsal fin	Gonads	2.90	0.05 *
	Eyes	Gonads	0.23	0.82
	Dorsal fin	Liver	0.30	1.00
	Eyes	Liver	-2.51	0.11
	Gonads	Liver	-2.73	0.08
	Dorsal fin	Muscle	1.91	0.45
	Eyes	Muscle	-0.82	1.00
	Gonads	Muscle	-1.04	1.00
	Liver	Muscle	1.69	0.55
	Dorsal fin	Vertebra	4.43	< 0.01 *
	Eyes	Vertebra	1.83	0.47
	Gonads	Vertebra	1.61	0.54
	Liver	Vertebra	4.34	< 0.01 *
	Muscle	Vertebra	2.65	0.08
<i>Engraulis encrasicolus</i>	Dorsal fin	Eyes	0.36	1.00
	Dorsal fin	Gonads	-1.23	1.00
	Eyes	Gonads	-1.59	1.00
	Dorsal fin	Liver	-0.77	1.00

	Eyes	Liver	-1.12	1.00
	Gonads	Liver	0.47	1.00
	Dorsal fin	Muscle	-2.63	0.11
	Eyes	Muscle	-2.99	0.04 *
	Gonads	Muscle	-1.40	1.00
	Liver	Muscle	-1.86	0.75
	Dorsal fin	Vertebra	0.16	0.87
	Eyes	Vertebra	-0.19	1.00
	Gonads	Vertebra	1.40	1.00
	Liver	Vertebra	0.93	1.00
	Muscle	Vertebra	2.79	0.07
<i>Euthynnus alletteratus</i>	Dorsal fin	Eyes	2.44	0.18
	Dorsal fin	Gonads	2.08	0.41
	Eyes	Gonads	-0.36	0.72
	Dorsal fin	Liver	0.63	1.00
	Eyes	Liver	-1.81	0.71
	Gonads	Liver	-1.45	1.00
	Dorsal fin	Muscle	1.37	1.00
	Eyes	Muscle	-1.07	1.00
	Gonads	Muscle	-0.71	1.00
	Liver	Muscle	0.74	1.00
	Dorsal fin	Vertebra	3.84	< 0.01 *
	Eyes	Vertebra	1.40	1.00
	Gonads	Vertebra	1.75	0.72
	Liver	Vertebra	3.21	0.02 *
	Muscle	Vertebra	2.47	0.18
<i>Mola mola</i>	Dorsal fin	Eyes	0.23	1.00

	Dorsal fin	Gonads	1.02	1.00
	Eyes	Gonads	0.48	1.00
	Dorsal fin	Liver	5.04	< 0.01 *
	Eyes	Liver	2.57	0.12
	Gonads	Liver	2.66	0.10
	Dorsal fin	Muscle	3.16	0.02 *
	Eyes	Muscle	1.52	1.00
	Gonads	Muscle	1.29	1.00
	Liver	Muscle	-1.88	0.59
	Dorsal fin	Vertebra	2.49	0.14
	Eyes	Vertebra	1.47	1.00
	Gonads	Vertebra	1.21	1.00
	Liver	Vertebra	-1.20	1.00
	Muscle	Vertebra	0.18	0.86
<i>Sarda sarda</i>	Dorsal fin	Eyes	-2.49	0.15
	Dorsal fin	Gonads	-0.66	1.00
	Eyes	Gonads	1.84	0.66
	Dorsal fin	Liver	-2.16	0.33
	Eyes	Liver	0.33	1.00
	Gonads	Liver	-1.51	0.92
	Dorsal fin	Muscle	-3.89	< 0.01 *
	Eyes	Muscle	-1.40	0.81
	Gonads	Muscle	-3.23	0.02 *
	Liver	Muscle	-1.73	0.67
	Dorsal fin	Vertebra	-0.66	1.00
	Eyes	Vertebra	1.84	0.60
	Gonads	Vertebra	0.00	1.00

	Liver	Vertebra	1.51	0.79
	Muscle	Vertebra	3.23	0.02 *
<i>Sardina pilchardus</i>	Dorsal fin	Eyes	-0.19	0.85
	Dorsal fin	Gonads	2.19	0.31
	Eyes	Gonads	2.38	0.21
	Dorsal fin	Liver	-0.47	1.00
	Eyes	Liver	-0.27	1.00
	Gonads	Liver	-2.66	0.10
	Dorsal fin	Muscle	-1.73	0.76
	Eyes	Muscle	-1.53	1.00
	Gonads	Muscle	-3.92	< 0.01 *
	Liver	Muscle	-1.26	1.00
	Dorsal fin	Vertebra	1.34	1.00
	Eyes	Vertebra	1.53	0.87
	Gonads	Vertebra	-0.85	1.00
	Liver	Vertebra	1.81	0.71
	Muscle	Vertebra	3.07	0.03 *
<i>Scomber colias</i>	Dorsal fin	Eyes	0.88	1.00
	Dorsal fin	Gonads	1.40	1.00
	Eyes	Gonads	0.52	1.00
	Dorsal fin	Liver	-1.34	1.00
	Eyes	Liver	-2.22	0.29
	Gonads	Liver	-2.74	0.08
	Dorsal fin	Muscle	-0.90	1.00
	Eyes	Muscle	-1.78	0.67
	Gonads	Muscle	-2.30	0.26
	Liver	Muscle	0.44	0.66

	Dorsal fin	Vertebra	2.11	0.35
	Eyes	Vertebra	1.23	1.00
	Gonads	Vertebra	0.71	1.00
	Liver	Vertebra	3.45	< 0.01 *
	Muscle	Vertebra	3.01	0.04 *
<i>Tetrapturus belone</i>	Dorsal fin	Eyes	1.95	0.57
	Dorsal fin	Gonads	2.16	0.36
	Eyes	Gonads	0.22	0.83
	Dorsal fin	Liver	0.25	1.00
	Eyes	Liver	-1.70	0.80
	Gonads	Liver	-1.92	0.55
	Dorsal fin	Muscle	0.58	1.00
	Eyes	Muscle	-1.37	1.00
	Gonads	Muscle	-1.59	0.90
	Liver	Muscle	0.33	1.00
	Dorsal fin	Vertebra	3.29	0.02 *
	Eyes	Vertebra	1.34	1.00
	Gonads	Vertebra	1.12	1.00
	Liver	Vertebra	3.04	0.03 *
	Muscle	Vertebra	2.71	0.09

b) Nitrogen ($\delta^{15}\text{N}$) comparison - ANOVA + Tukey

Species	Tissue1	Tissue2	Mean Diff (‰)	P_Value_Adj
<i>Auxis thazard</i>	Eyes	Dorsal fin	1.33	< 0.01 *
	Gonads	Dorsal fin	1.34	< 0.01 *
	Liver	Dorsal fin	1.47	< 0.01 *
	Muscle	Dorsal fin	3.18	< 0.01 *
	Vertebra	Dorsal fin	0.07	1.00
	Gonads	Eyes	0.00	1.00
	Liver	Eyes	0.13	1.00
	Muscle	Eyes	1.85	< 0.01 *
	Vertebra	Eyes	-1.26	< 0.01 *
	Liver	Gonads	0.13	1.00
	Muscle	Gonads	1.84	< 0.01 *
	Vertebra	Gonads	-1.27	< 0.01 *
	Muscle	Liver	1.71	< 0.01 *
	Vertebra	Liver	-1.40	< 0.01 *
	Vertebra	Muscle	-3.11	< 0.01 *
<i>Engraulis encrasicolus</i>	Eyes	Dorsal fin	-1.08	0.08
	Gonads	Dorsal fin	-0.09	1.00
	Liver	Dorsal fin	-0.07	1.00
	Muscle	Dorsal fin	1.34	0.02 *
	Vertebra	Dorsal fin	-0.49	0.79
	Gonads	Eyes	0.98	0.13
	Liver	Eyes	1.01	0.12
	Muscle	Eyes	2.42	< 0.01 *
	Vertebra	Eyes	0.59	0.64

	Liver	Gonads	0.02	1.00
	Muscle	Gonads	1.43	< 0.01 *
	Vertebra	Gonads	-0.40	0.90
	Muscle	Liver	1.41	< 0.01 *
	Vertebra	Liver	-0.42	0.87
	Vertebra	Muscle	-1.83	< 0.01 *
<i>Euthynnus alletteratus</i>	Eyes	Dorsal fin	1.42	0.04 *
	Gonads	Dorsal fin	1.38	0.04 *
	Liver	Dorsal fin	1.14	0.14
	Muscle	Dorsal fin	2.25	< 0.01 *
	Vertebra	Dorsal fin	0.01	1.00
	Gonads	Eyes	-0.03	1.00
	Liver	Eyes	-0.27	0.99
	Muscle	Eyes	0.83	0.44
	Vertebra	Eyes	-1.40	0.04 *
	Liver	Gonads	-0.24	0.99
	Muscle	Gonads	0.86	0.40
	Vertebra	Gonads	-1.37	0.05 *
	Muscle	Liver	1.10	0.16
	Vertebra	Liver	-1.13	0.15
	Vertebra	Muscle	-2.23	< 0.01 *
<i>Mola mola</i>	Eyes	Dorsal fin	0.41	0.96
	Gonads	Dorsal fin	-1.24	0.03 *
	Liver	Dorsal fin	-1.55	< 0.01 *
	Muscle	Dorsal fin	-1.84	< 0.01 *
	Vertebra	Dorsal fin	-1.48	< 0.01 *
	Gonads	Eyes	-1.65	0.06

	Liver	Eyes	-1.96	< 0.01 *
	Muscle	Eyes	-2.25	< 0.01 *
	Vertebra	Eyes	-1.89	0.02 *
	Liver	Gonads	-0.31	0.96
	Muscle	Gonads	-0.60	0.62
	Vertebra	Gonads	-0.25	0.99
	Muscle	Liver	-0.29	0.90
	Vertebra	Liver	0.06	1.00
	Vertebra	Muscle	0.36	0.93
<i>Sarda sarda</i>	Eyes	Dorsal fin	0.96	0.07
	Gonads	Dorsal fin	-0.46	0.74
	Liver	Dorsal fin	-0.64	0.39
	Muscle	Dorsal fin	0.57	0.53
	Vertebra	Dorsal fin	-0.48	0.68
	Gonads	Eyes	-1.42	< 0.01 *
	Liver	Eyes	-1.60	< 0.01 *
	Muscle	Eyes	-0.39	0.84
	Vertebra	Eyes	-1.44	< 0.01 *
	Liver	Gonads	-0.19	0.99
	Muscle	Gonads	1.03	0.04 *
	Vertebra	Gonads	-0.03	1.00
	Muscle	Liver	1.21	0.01 *
	Vertebra	Liver	0.16	1.00
	Vertebra	Muscle	-1.05	0.04 *
<i>Sardina pilchardus</i>	Eyes	Dorsal fin	1.34	< 0.01 *
	Gonads	Dorsal fin	0.96	0.03 *
	Liver	Dorsal fin	0.93	0.03 *

	Muscle	Dorsal fin	1.95	< 0.01 *
	Vertebra	Dorsal fin	-0.03	1.00
	Gonads	Eyes	-0.38	0.78
	Liver	Eyes	-0.41	0.72
	Muscle	Eyes	0.61	0.32
	Vertebra	Eyes	-1.37	< 0.01 *
	Liver	Gonads	-0.03	1.00
	Muscle	Gonads	0.98	0.02 *
	Vertebra	Gonads	-0.99	0.02 *
	Muscle	Liver	1.01	0.02 *
	Vertebra	Liver	-0.96	0.03 *
	Vertebra	Muscle	-1.98	< 0.01 *
<i>Scomber colias</i>	Eyes	Dorsal fin	0.79	0.19
	Gonads	Dorsal fin	1.16	0.02 *
	Liver	Dorsal fin	0.58	0.51
	Muscle	Dorsal fin	1.28	< 0.01 *
	Vertebra	Dorsal fin	0.13	1.00
	Gonads	Eyes	0.38	0.86
	Liver	Eyes	-0.21	0.99
	Muscle	Eyes	0.50	0.66
	Vertebra	Eyes	-0.66	0.36
	Liver	Gonads	-0.58	0.50
	Muscle	Gonads	0.12	1.00
	Vertebra	Gonads	-1.04	0.04 *
	Muscle	Liver	0.71	0.30
	Vertebra	Liver	-0.45	0.74
	Vertebra	Muscle	-1.16	0.02 *

<i>Tetrapturus belone</i>	Eyes	Dorsal fin	1.26	< 0.01 *
	Gonads	Dorsal fin	1.76	< 0.01 *
	Liver	Dorsal fin	1.75	< 0.01 *
	Muscle	Dorsal fin	2.19	< 0.01 *
	Vertebra	Dorsal fin	-0.53	0.32
	Gonads	Eyes	0.50	0.39
	Liver	Eyes	0.48	0.42
	Muscle	Eyes	0.93	0.01 *
	Vertebra	Eyes	-1.80	< 0.01 *
	Liver	Gonads	-0.02	1.00
	Muscle	Gonads	0.43	0.55
	Vertebra	Gonads	-2.29	< 0.01 *
	Muscle	Liver	0.44	0.52
	Vertebra	Liver	-2.28	< 0.01 *
	Vertebra	Muscle	-2.72	< 0.01 *

Table D.2: Multi-tissue isotopic niche metrics for the Azohía pelagic community. Data are grouped by tissue type and functional category. Abbreviations: n = sample size; TA = Total Area (Convex Hull); SEA = Standard ellipse area; SEA_C = Standard ellipse area corrected for small sample sizes; SEA_B (95% CI) = Bayesian Standard ellipse area, reported as the posterior mean followed by the 95% Credible Interval in brackets.

Tissue	Functional Group	Species	n	TA (‰ ²)	SEA (‰ ²)	SEA _C (‰ ²)	SEA _B (95% CI)
Dorsal fin	Small pelagic	<i>Engraulis encrasicolus</i>	6	0,93	0,72	0,9	0.77 [0.34 - 1.76]
		<i>Sardina pilchardus</i>	6	0,72	0,66	0,82	0.71 [0.30 - 1.63]
	Medium pelagic	<i>Auxis thazard</i>	5	0,48	0,41	0,55	0.44 [0.18 - 1.09]
		<i>Euthynnus alletteratus</i>	6	2,02	1,67	2,09	2.15 [0.92 - 4.93]
		<i>Sarda sarda</i>	6	0,43	0,34	0,42	0.40 [0.17 - 0.97]

		<i>Scomber colias</i>	6	1,86	1,61	2,01	1.79 [0.78 - 4.08]
	Large pelagic	<i>Mola mola</i>	11	4,07	1,92	2,13	2.01 [1.11 - 3.63]
		<i>Tetrapturus belone</i>	6	0,73	0,58	0,73	0.63 [0.26 - 1.40]
Eyes	Small pelagic	<i>Engraulis encrasicolus</i>	6	1,56	1,13	1,41	1.33 [0.58 - 2.89]
		<i>Sardina pilchardus</i>	6	2,89	2,43	3,03	2.97 [1.25 - 6.84]
	Medium pelagic	<i>Auxis thazard</i>	6	0,63	0,51	0,63	0.59 [0.26 - 1.30]
		<i>Euthynnus alletteratus</i>	6	1,62	1,29	1,61	1.55 [0.68 - 3.39]
		<i>Sarda sarda</i>	6	0,32	0,26	0,33	0.31 [0.13 - 0.71]
		<i>Scomber colias</i>	6	0,68	0,48	0,6	0.56 [0.25 - 1.26]
	Large pelagic	<i>Tetrapturus belone</i>	6	0,31	0,25	0,31	0.29 [0.13 - 0.65]
Gonads	Small pelagic	<i>Engraulis encrasicolus</i>	6	0,28	0,23	0,29	0.24 [0.10 - 0.54]
		<i>Sardina pilchardus</i>	6	0,29	0,2	0,25	0.22 [0.09 - 0.50]
	Medium pelagic	<i>Auxis thazard</i>	6	0,5	0,36	0,46	0.38 [0.17 - 0.87]
		<i>Euthynnus alletteratus</i>	6	1,52	1,07	1,34	1.13 [0.50 - 2.60]
		<i>Sarda sarda</i>	6	0,49	0,4	0,5	0.43 [0.18 - 0.98]
		<i>Scomber colias</i>	6	1,84	1,63	2,03	1.80 [0.77 - 4.28]
	Large pelagic	<i>Mola mola</i>	4	0,36	0,42	0,63	0.45 [0.16 - 1.18]
		<i>Tetrapturus belone</i>	6	0,48	0,35	0,44	0.42 [0.18 - 0.94]
Liver	Small pelagic	<i>Engraulis encrasicolus</i>	6	4,94	4,06	5,07	4.26 [1.79 - 9.20]
	Small pelagic	<i>Sardina pilchardus</i>	6	0,53	0,39	0,49	0.42 [0.17 - 0.98]
	Medium pelagic	<i>Auxis thazard</i>	6	0,25	0,22	0,27	0.22 [0.10 - 0.50]
		<i>Euthynnus alletteratus</i>	6	0,67	0,52	0,64	0.55 [0.23 - 1.21]

		<i>Sarda sarda</i>	6	3,41	2,37	2,96	2.44 [1.06 - 5.62]
		<i>Scomber colias</i>	6	0,9	0,66	0,83	0.69 [0.30 - 1.69]
	Large pelagic	<i>Mola mola</i>	11	2,22	1,31	1,46	1.34 [0.74 - 2.44]
		<i>Tetrapturus belone</i>	6	0,79	0,62	0,77	0.65 [0.27 - 1.52]
Muscle	Small pelagic	<i>Engraulis encrasicolus</i>	6	0,29	0,28	0,35	0.30 [0.13 - 0.68]
		<i>Sardina pilchardus</i>	6	0,29	0,22	0,27	0.23 [0.10 - 0.51]
	Medium pelagic	<i>Auxis thazard</i>	6	0,41	0,34	0,42	0.37 [0.16 - 0.84]
		<i>Euthynnus alletteratus</i>	6	1,84	1,53	1,91	1.63 [0.69 - 3.83]
		<i>Sarda sarda</i>	6	0,36	0,3	0,37	0.35 [0.15 - 0.79]
		<i>Scomber colias</i>	6	0,32	0,28	0,34	0.29 [0.12 - 0.66]
	Large pelagic	<i>Mola mola</i>	11	2,32	1,08	1,2	1.11 [0.60 - 2.03]
		<i>Prionace glauca</i>	58	16,1	3,82	3,89	3.84 [2.95 - 4.98]
		<i>Tetrapturus belone</i>	6	0,47	0,35	0,43	0.39 [0.17 - 0.92]
Vertebra	Small pelagic	<i>Engraulis encrasicolus</i>	6	0,49	0,38	0,47	0.47 [0.21 - 1.07]
		<i>Sardina pilchardus</i>	6	0,84	0,6	0,75	0.66 [0.29 - 1.52]
	Medium pelagic	<i>Auxis thazard</i>	6	0,57	0,46	0,58	0.49 [0.21 - 1.07]
		<i>Euthynnus alletteratus</i>	6	1,96	1,43	1,79	1.52 [0.66 - 3.37]
		<i>Sarda sarda</i>	6	0,66	0,58	0,72	0.64 [0.28 - 1.49]
		<i>Scomber colias</i>	6	0,36	0,31	0,39	0.69 [0.29 - 1.49]
	Large pelagic	<i>Mola mola</i>	4	0,77	0,95	1,42	1.08 [0.36 - 2.87]
		<i>Tetrapturus belone</i>	6	0,76	0,56	0,69	0.60 [0.26 - 1.37]

Table D.3: Statistical comparison of isotopic niche widths (SEA_B) among tissues for Azohía pelagic species. Global differences were tested using Kruskal-Wallis (KW) on posterior ellipse draws. Pairwise differences were resolved using Dunn's tests with Holm-Bonferroni correction. Significance levels: *p<0.05, **p<0.01, ***p<0.001. Only significant pairwise comparisons are listed.

Category	Species	Global KW (p)	Significant Pairwise Diff. (Holm-Corrected)	
Small pelagic	<i>Engraulis encrasicolus</i>	<0.001	Liver vs.	Muscle (***), Vert (***), Gonads (***)
			Eyes vs.	Muscle (***), Vert (***)
	<i>Sardina pilchardus</i>	<0.001	Eyes vs.	Muscle (***), Vert (***), Liver (***)
			Liver vs.	Muscle (***), Vert (***)
Medium pelagic	<i>Auxis thazard</i>	<0.001	Muscle vs.	Liver (***), Gonads (***), Fin (***), Vert (***)
			Liver vs.	Gonads (***), Vert (***)
	<i>Euthynnus alletteratus</i>	<0.001	Muscle vs.	Liver (***), Eyes (***), Fin (***)
			Vert vs.	Muscle (***), Liver (**)
	<i>Sarda sarda</i>	<0.001	Muscle vs.	Liver (***), Fin (***), Eyes (***)
			Liver vs.	Vert (***), Fin (***)
	<i>Scomber colias</i>	<0.001	Muscle vs.	Liver (***), Gonads (***), Fin (***)
			Vert vs.	Liver (***), Gonads (***)
Large pelagic	<i>Mola mola</i>	<0.001	Muscle vs.	Liver (***), Fin (***), Gonads (***)
			Vert vs.	Liver (***), Fin (***)

	<i>Tetrapturus belone</i>	<0.001	Muscle vs.	Liver (***), Fin (***), Eyes (***)
			Vert vs.	Muscle (***), Liver (***)

Table D.4.: Pairwise isotopic niche overlap (%) matrices for the Azohía pelagic community across different tissues. Values represent the percentage of the row species' niche (SEAC) that is overlapped by the column species' niche. Matrices are presented for (a) Liver, (b) Gonads, (c) Muscle, (d) Dorsal fin, (e) Vertebrae and (f) Eyes.

(a) Liver

	<i>Auxis thazard</i>	<i>Engraulis encrasicolus</i>	<i>Euthynnus alletteratus</i>	<i>Mola mola</i>	<i>Sarda sarda</i>	<i>Sardina pilchardus</i>	<i>Scomber colias</i>	<i>Tetrapturus belone</i>
<i>Auxis thazard</i>	100	-	52.38	-	17.85	-	-	77.12
<i>Engraulis encrasicolus</i>	-	100	-	-	-	7.04	-	-
<i>Euthynnus alletteratus</i>	21.94	-	100	-	56.67	-	10.39	71.94
<i>Mola mola</i>	-	-	-	100	-	-	-	-
<i>Sarda sarda</i>	1.63	-	12.33	-	100	-	-	9.18
<i>Sardina pilchardus</i>	-	72.31	-	-	-	100	-	-
<i>Scomber colias</i>	-	-	8.08	-	-	-	100	0.1
<i>Tetrapturus belone</i>	26.94	-	59.99	-	35.18	-	0.11	100

(b) Gonads

	<i>Auxis thazard</i>	<i>Engraulis encrasicolus</i>	<i>Euthynnus alletteratus</i>	<i>Mola mola</i>	<i>Sarda sarda</i>	<i>Sardina pilchardus</i>	<i>Scomber colias</i>	<i>Tetrapturus belone</i>
<i>Auxis thazard</i>	100	0	16,39	0	0	0	38,88	0
<i>Engraulis encrasicolus</i>	0	100	0	0	0	0	0	0
<i>Euthynnus alletteratus</i>	5,57	0	100	5,62	15,5	0	0	23,42
<i>Mola mola</i>	0	0	11,93	100	16,34	0	2,39	0
<i>Sarda sarda</i>	0	0	41,46	20,59	100	0	0	0
<i>Sardina pilchardus</i>	0	0	0	0	0	100	0	0
<i>Scomber colias</i>	8,72	0	0	0,74	0	0	100	0
<i>Tetrapturus belone</i>	0	0	71,32	0	0	0	0	100

(c) Muscle

	<i>Auxis thazard</i>	<i>Engraulis encrasicolus</i>	<i>Euthynnus alletteratus</i>	<i>Mola mola</i>	<i>Prionace glauca</i>	<i>Sarda sarda</i>	<i>Sardina pilchardus</i>	<i>Scomber colias</i>	<i>Tetrapturus belone</i>
<i>Auxis thazard</i>	100	0	68,36	0	9,53	0	0	0	0
<i>Engraulis encrasicolus</i>	0	100	0	22	0	0	4,62	0	0
<i>Euthynnus alletteratus</i>	15,11	0	100	0	26,95	0	0	0	11,86
<i>Mola mola</i>	0	6,44	0	100	0	0	15,39	0	0
<i>Prionace glauca</i>	1,03	0	13,21	0	100	9,55	0	0	0
<i>Sarda sarda</i>	0	0	0	0	100	100	0	0	0
<i>Sardina pilchardus</i>	0	5,94	0	67,58	0	0	100	0	0
<i>Scomber colias</i>	0	0	0	0	0	0	0	100	0
<i>Tetrapturus belone</i>	0	0	52,19	0	0	0	0	0	100

(d) Dorsal fin

	<i>Auxis thazard</i>	<i>Engraulis encrasicolus</i>	<i>Euthynnus alletteratus</i>	<i>Mola mola</i>	<i>Sarda sarda</i>	<i>Sardina pilchardus</i>	<i>Scomber colias</i>	<i>Tetrapturus belone</i>
<i>Auxis thazard</i>	100	0	30,64	0	0	0	0	45,73
<i>Engraulis encrasicolus</i>	0	100	0	0	0	53,6	15,7	0
<i>Euthynnus alletteratus</i>	8,03	0	100	0	0	0	0	12,1
<i>Mola mola</i>	0	0	0	100	1,44	0	0	0
<i>Sarda sarda</i>	0	0	0	7,29	100	0	0	0
<i>Sardina pilchardus</i>	0	58,51	0	0	0	100	2,69	0
<i>Scomber colias</i>	0	7	0	0	0	1,1	100	0
<i>Tetrapturus belone</i>	34,33	0	34,65	0	0	0	0	100

(e) Vertebrae

	<i>Auxis thazard</i>	<i>Engraulis encrasicolus</i>	<i>Euthynnus alletteratus</i>	<i>Mola mola</i>	<i>Sarda sarda</i>	<i>Sardina pilchardus</i>	<i>Scomber colias</i>	<i>Tetrapturus belone</i>
<i>Auxis thazard</i>	100	23,81	0	0	0	50,5	3,3	0
<i>Engraulis encrasicolus</i>	29,21	100	0	0	0	71,99	0	0
<i>Euthynnus alletteratus</i>	0	0	100	0	0	0	5,7	0
<i>Mola mola</i>	0	0	0	100	0	0	0	0
<i>Sarda sarda</i>	0	0	0	0	100	0	0	0
<i>Sardina pilchardus</i>	38,55	44,78	0	0	0	100	0	0
<i>Scomber colias</i>	4,87	0	26,14	0	0	0	100	0
<i>Tetrapturus belone</i>	0	0	0	0	0	0	0	100

(f) Eyes

	<i>Auxis thazard</i>	<i>Engraulis encrasicolus</i>	<i>Euthynnus alletteratus</i>	<i>Sarda sarda</i>	<i>Sardina pilchardus</i>	<i>Scomber colias</i>	<i>Tetrapturus belone</i>
<i>Auxis thazard</i>	100	0	69,3	0	0	4,78	0
<i>Engraulis encrasicolus</i>	0	100	0	0	0	0	0
<i>Euthynnus alletteratus</i>	27,17	0	100	0	2,6	8,7	0
<i>Sarda sarda</i>	0	0	0	100	0	0	0
<i>Sardina pilchardus</i>	0	0	1,38	0	100	10,51	0
<i>Scomber colias</i>	5,02	0	23,31	0	52,98	100	0
<i>Tetrapturus belone</i>	0	0	0	0	0	0	100

Table D.5. Linear regression statistics for ontogenetic shifts ($\delta^{15}\text{N}$ vs. Body Length) in the Azohía pelagic community. Abbreviations: β = regression slope; R^2 = coefficient of determination; P_{val} = significance level; N = sample size. Significant relationships ($p < 0.05$) are highlighted in bold.

Category	Species	Tissue	Slope (β)	R^2	P_{val}
Small pelagic	<i>Engraulis encrasicolus</i>	Dorsal Fin	-0.271	0.070	0.613
		Eyes	0.291	0.060	0.640
		Gonads	0.152	0.086	0.572
		Liver	0.564	0.030	0.744
		Muscle	0.459	0.527	0.102
		Vertebra	0.539	0.121	0.500
	<i>Sardina pilchardus</i>	Dorsal Fin	-0.524	0.371	0.199
		Eyes	0.240	0.017	0.803
		Gonads	-0.174	0.191	0.386
		Liver	-0.129	0.058	0.647
		Muscle	-0.328	0.476	0.129
		Vertebra	-0.227	0.048	0.678
Medium pelagic	<i>Auxis thazard</i>	Dorsal Fin	0.088	0.076	0.598
		Eyes	0.530	0.648	0.053
		Gonads	-0.108	0.367	0.202
		Liver	0.085	0.073	0.603
		Muscle	0.087	0.127	0.488
		Vertebra	0.102	0.086	0.572
	<i>Euthynnus alletteratus</i>	Dorsal Fin	0.022	0.074	0.603
		Eyes	0.022	0.071	0.609
		Gonads	-0.003	0.003	0.917
		Liver	0.016	0.296	0.264
		Muscle	0.021	0.055	0.653
		Vertebra	-0.010	0.017	0.806
	<i>Sarda sarda</i>	Dorsal Fin	-0.135	0.160	0.433
		Eyes	0.041	0.011	0.843
		Gonads	-0.150	0.234	0.331
		Liver	-0.112	0.077	0.594
		Muscle	-0.008	0.001	0.961
		Vertebra	-0.057	0.036	0.718
	<i>Scomber colias</i>	Dorsal Fin	0.011	0.005	0.895
		Eyes	0.132	0.549	0.092
		Gonads	0.158	0.246	0.317
		Liver	-0.014	0.008	0.869
		Muscle	0.064	0.251	0.311
		Vertebra	-0.180	0.800	0.016
Large pelagic	<i>Mola mola</i>	Dorsal Fin (11)	-0.074	0.093	0.362
		Eyes (2)	0.195	1.000	-
		Gonads (4)	-0.100	0.475	0.311
		Liver (11)	-0.029	0.027	0.627
		Muscle (11)	-0.046	0.014	0.730
		Vertebra (4)	-0.213	0.444	0.334
	<i>Prionace glauca</i>	Dorsal Fin (2)	0.033	1.000	-
		Eyes (2)	0.022	1.000	-

		Gonads (2)	0.030	1.000	-
		Liver (2)	0.017	1.000	-
		Muscle (58)	0.013	0.120	0.008
		Vertebra (2)	0.033	1.000	-
	<i>Tetrapturus belone</i>	Dorsal Fin	-0.003	0.010	0.850
		Eyes	-0.040	0.259	0.303
		Gonads	0.011	0.028	0.750
		Liver	0.015	0.075	0.600
		Muscle	-0.001	0.000	0.986
		Vertebra	-0.010	0.013	0.833

Table D.6.: Comparison of Trophic Positions estimated from stable isotope analysis (TP_{SIA}) and stomach content analysis (TP_{SCA}) for the Azohía pelagic community. Isotopic estimates (TP_{SIA}) are derived from somatic muscle tissue relative to the local baseline (*Venus nux*, TP=2.0). Diff: represents the numerical discrepancy between the two methods (TP_{SIA} – TP_{SCA}) N_{SIA} and N_{SCA} represent the number of individuals used for each method. A dash (—) indicates species where stomach content resolution or sample size was insufficient to calculate a comparable trophic level.

Category	Species	Tissue	N _{SIA}	TP _{SIA}	N _{SCA}	TP _{SCA}	Diff
Small pelagic	<i>Engraulis encrasicolus</i>	Dorsal fin	6	2,38	-	-	-
		Eyes	6	2,07	-	-	-
		Gonads	6	2,36	-	-	-
		Liver	6	2,36	-	-	-
		Muscle	6	2,78	-	-	-
		Vertebra	6	2,24	-	-	-
	<i>Sardina pilchardus</i>	Dorsal fin	6	2,34	5	3,74	-1,4
		Eyes	6	2,74	5	3,74	-1
		Gonads	6	2,63	5	3,74	-1,11
		Liver	6	2,62	5	3,74	-1,12
		Muscle	6	2,92	5	3,74	-0,82
		Vertebra	6	2,33	5	3,74	-1,41
Medium pelagic	<i>Auxis thazard</i>	Dorsal fin	6	2,37	-	-	-

		Eyes	6	2,76	-	-	-
		Gonads	6	2,76	-	-	-
		Liver	6	2,8	-	-	-
		Muscle	6	3,3	-	-	-
		Vertebra	6	2,38	-	-	-
	<i>Euthynnus alletteratus</i>	Dorsal fin	6	2,53	-	-	-
		Eyes	6	2,95	-	-	-
		Gonads	6	2,94	-	-	-
		Liver	6	2,87	-	-	-
		Muscle	6	3,19	-	-	-
		Vertebra	6	2,54	-	-	-
	<i>Sarda sarda</i>	Dorsal fin	6	3,25	3	4,15	-0,9
		Eyes	6	3,53	3	4,15	-0,62
		Gonads	6	3,11	3	4,15	-1,04
		Liver	6	3,06	3	4,15	-1,09
		Muscle	6	3,41	3	4,15	-0,74
		Vertebra	6	3,1	3	4,15	-1,05
	<i>Scomber colias</i>	Dorsal fin	6	2,51	6	3,58	-1,07
		Eyes	6	2,74	6	3,58	-0,84
		Gonads	6	2,85	6	3,58	-0,73
		Liver	6	2,68	6	3,58	-0,9
		Muscle	6	2,89	6	3,58	-0,69
		Vertebra	6	2,55	6	3,58	-1,03
Large pelagic	<i>Mola mola</i>	Dorsal fin	11	3,48	4	3,35	0,13
		Eyes	2	3,6	4	3,35	0,25

		Gonads	4	3,12	4	3,35	-0,23
		Liver	11	3,03	4	3,35	-0,32
		Muscle	11	2,94	4	3,35	-0,41
		Vertebra	4	3,05	4	3,35	-0,3
	<i>Prionace glauca</i>	Dorsal fin	2	3,16	-	-	-
		Eyes	2	2,86	-	-	-
		Gonads	2	3,19	-	-	-
		Liver	2	3,24	-	-	-
		Muscle	58	3,56	-	-	-
		Vertebra	2	2,81	-	-	-
	<i>Tetrapturus belone</i>	Dorsal fin	6	2,33	1	4,16	-1,83
		Eyes	6	2,7	1	4,16	-1,46
		Gonads	6	2,85	1	4,16	-1,31
		Liver	6	2,84	1	4,16	-1,32
		Muscle	6	2,97	1	4,16	-1,19
		Vertebra	6	2,17	1	4,16	-1,99
